

LOGARITHMIC KODAIRA DIMENSION AND WHOLE-FIBER VARIATION

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ABSTRACT. We prove the logarithmic Iitaka–Viehweg inequality for projective surjective morphisms with connected fibers between smooth complex quasi-projective varieties whose base has nonnegative logarithmic Kodaira dimension. The variation measures the birational field of definition of the whole geometric generic fiber. This resolves Popa’s logarithmic variation conjecture positively.

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1. INTRODUCTION

A fibration carries two sources of pluricanonical forms: the geometry of its base and the variation of its fibers. Iitaka’s subadditivity problem asks for the contribution of the first. The Iitaka–Viehweg refinement asks how many further directions are forced by a family whose fibers change birationally. On an open base the correct measure of the base is its logarithmic Kodaira dimension. We prove this refinement for projective fibrations over such bases.

For a rational divisor L , write $\kappa(L)$ for the Iitaka dimension of its sufficiently divisible integral multiples. For a smooth quasi-projective variety U , choose a smooth projective compactification X with reduced simple normal crossing boundary $D_X = X \setminus U$. A rational boundary is an effective rational divisor with coefficients in $[0, 1]$; SNC abbreviates simple normal crossing support. The logarithmic Kodaira dimension is

$$\bar{\kappa}(U) = \kappa(X, K_X + D_X);$$

it is independent of this compactification. We give a point Kodaira dimension zero and use $(-\infty) + a = -\infty$, including $a = -\infty$.

Let $f : U \rightarrow V$ be a projective surjective morphism with connected fibers between smooth connected complex quasi-projective varieties. Fix an algebraic closure Ω of $\mathbf{C}(V)$, and let F be

the smooth projective geometric generic fiber. Its *birational variation* is

$$(1.1) \quad \text{Var}(f) = \min_L \text{trdeg}_{\mathbf{C}} L,$$

where $\mathbf{C} \subseteq L \subseteq \Omega$ is algebraically closed and F is birational to the base extension of a variety over L . This is the field-of-definition form of the classical invariant [48, 29]. It records the entire function field of F ; finite changes of the original base do not change it.

Theorem 1.1 (Logarithmic variation). *Let $f : U \rightarrow V$ be a projective surjective morphism with connected fibers between smooth connected quasi-projective complex varieties. Assume $\bar{\kappa}(V) \geq 0$, and let F be its geometric generic fiber. Then*

$$(1.2) \quad \bar{\kappa}(U) \geq \kappa(F) + \max\{\bar{\kappa}(V), \text{Var}(f)\}.$$

This proves Popa’s Conjecture 3.8 in the formulation on page 6 of his author version dated 13 June 2023 [42]. The morphism may have singular fibers. The theorem imposes no abundance, good minimal model, or semiample hypothesis on F . It concerns the lower bound in that conjecture; the separate additivity questions for smooth morphisms in [42, Conjectures 3.1 and 3.9] have a different conclusion. The companion *The reverse logarithmic Kodaira inequality and additivity* [39] proves the corresponding logarithmic additivity for projective complex reduced-SNC pairs whose total space and every boundary stratum are smooth over the open base. That setting allows horizontal boundary and all signs of the base and fiber Kodaira dimensions. The proof of the variation inequality here uses the subadditivity and period inputs described below.

The base hypothesis is genuinely logarithmic. For example, $\bar{\kappa}(\mathbf{C}^*) = 0$, whereas its smooth projective compactification $\mathbf{P}^1$ has Kodaira dimension $-\infty$. A compactification argument must therefore retain the coefficient-one inverse boundary; replacing the open base by its ordinary projective Kodaira dimension would lose the hypothesis used in the proof.

Corollary 1.2 (Projective Iitaka–Viehweg inequality). *Let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective complex varieties, with $\kappa(Y) \geq 0$. For its geometric generic fiber F ,*

$$\kappa(X) \geq \kappa(F) + \max\{\kappa(Y), \text{Var}(f)\}.$$

Proof. Apply Theorem 1.1 with empty compactification boundaries. □

Thus the ordinary projective-complex Iitaka–Viehweg C^+ conjecture also holds in every dimension without a good minimal model hypothesis.

1.1. History and the two constancy problems. Iitaka developed the divisor-dimension and logarithmic frameworks for the addition problem [27, 28]. Viehweg’s weak positivity of pluri-canonical direct images made birational variation part of the expected lower bound [48, 47]. Kollár proved the variation refinement for fibers of general type, and Kawamata established the case of fibers admitting good minimal models [31, 29]. These results distinguish a changing family from a product even when their fiber dimensions agree.

Analytic positivity and generic vanishing brought substantial progress for special bases. Cao–Păun proved subadditivity over abelian varieties, and Hacon–Popa–Schnell treated bases of maximal Albanese dimension [13, 23]. Meng’s determinant-based equalities give related refinements over abelian and maximal-Albanese-dimension bases; the comparison with birational variation there still uses the good-minimal-model condition [36, Theorems 1.9–1.10 and the preceding discussion]. For reduced SNC pairs, Hashizume proved logarithmic subadditivity when the log canonical divisor of the general fiber is abundant [26, Theorem 1.2]. Popa’s formulation asks for the variation bound over an arbitrary base of nonnegative logarithmic Kodaira dimension [42, Conjecture 3.8].

Cyclic covers and Hodge theory also relate pluricanonical positivity to variation. Viehweg–Zuo construct Higgs bundles from cyclic covers associated with pluricanonical sections [49, §4], and Popa–Schnell develop this construction using Hodge modules [43, §§9–11]. For smooth projective families of maximal variation whose geometric generic fiber has a good minimal model, Popa–Schnell prove that the base is of log general type [43, Theorem A].

There are also older public claims in the ordinary setting. Tsuji states full ordinary subadditivity and stronger direct-image generation in [46, Theorems 1.9 and 1.13]; Maehara states an ordinary determinant bound involving whole-fiber variation in [35, §7, Theorem 8]. These are preprint claims, and their proofs are not inputs here. The statements just cited do not formulate the reduced-SNC logarithmic-base inequality of Theorem 1.1.

The proof below starts from the reduced-SNC subadditivity theorem and the full-period adjoint comparison proved in the companion *Orbifold and logarithmic Iitaka subadditivity* [37]. Both interfaces are stated precisely below. Subadditivity accounts for $\bar{\kappa}(V)$, but the variation term requires two additional constancy arguments. The first makes a Kodaira-dimension-zero fiber constant when its entire top Hodge line is flat. The second descends the whole original fiber, retaining the coordinates of its relative Iitaka base. Constancy of either the canonical model or the Kodaira-zero fiber alone would not give this last conclusion.

The proof combines several established methods. Canonical bundle formulas separate discriminant and moduli contributions [20, 1, 2]; the root-cover construction here keeps the actual relative pluriform orders. Deligne’s fixed-part and finite-character results, Schmid’s boundary theory, and functorial Hodge extensions control those lines after restriction [15, 44, 4]. The birational constancy argument uses the singular-weight extension theorem of Demailly–Hacon–Păun, within the Ohsawa–Takegoshi extension method, and the relative pluricanonical Bergman metrics of Păun–Takayama [18, 41]. The latter develop the constructions of Berndtsson–Păun [5, 6]. These estimates are followed by a big klt adjoint model [8]. For marked model pairs, we use the stable-family positivity of Kovács–Patakfalvi [32]. Hanamura’s birational group structure and invariance of finite étale covers provide the final descent mechanism [24, 25, 22, 33]. These inputs are used in the stated settings; the original fiber is never assumed to have a good minimal model.

1.2. The proof and its parameter field. Assume $d = \kappa(F) \geq 0$. Choose compatible compactifications $f : (X, D_X) \rightarrow (Y, D_Y)$, retaining $U = f^{-1}(V)$. Logarithmic subadditivity gives $\bar{\kappa}(U) \geq d + \bar{\kappa}(V)$. Let $q : X' \rightarrow Z$ be the absolute Iitaka map of $K_X + D_X$ on a smooth model, so $\dim Z = \bar{\kappa}(U)$. Take the relative algebraic closure of $\mathbf{C}(Y)\mathbf{C}(Z)$ in $\mathbf{C}(X)$, and resolve the normalization of the combined image in $Y \times Z$ in this field. On compatible smooth models this gives

$$X' \xrightarrow{x} W, \quad g : W \rightarrow Y, \quad h : W \rightarrow Z.$$

The generic x -fiber J has ordinary Kodaira dimension zero. The images $g(W_z)$ form an algebraic family of subvarieties of Y . Let T_0 be a smooth projective model of the normalization of its image parameter space in the relative algebraic closure of that parameter field in $\mathbf{C}(Z)$. Contraction of a nonzero logarithmic pluriform on Y controls the differential of this family and proves

$$(1.3) \quad \bar{\kappa}(U) = d + \dim T_0, \quad \mathbf{C}(T_0) \subseteq \mathbf{C}(Y).$$

Section 2 constructs the maps and establishes the inclusion as an inclusion of the actual function fields.

It remains to define the whole fiber F over $b = \overline{\mathbf{C}(T_0)} \subset \Omega$. Choose the least $p > 0$ with a nonzero section of pK_J , adjoin its p th root as a canonical differential, and resolve the resulting cover. This minimal ordinary root cover has an entire top-form space of dimension one. The adjoint comparison, applied anew to the restricted integral variation, makes its Hodge line

rationally trivial along the h -fibers. Finite character then follows on that restricted algebraic locus.

The analytic criterion of Section 4 turns this flatness into birational constancy of the root covers, equivariantly for their deck action. A lifted base vector field may initially have poles. Minimizing an $L^{2/m}$ integral while fixing a leading coefficient along a divisor shows that every pole has positive asymptotic fixed order. A single sufficiently small ample perturbation of the canonical divisor gives a big adjoint divisor whose model contracts these poles; the resulting regular vector field has flows identifying nearby fibers. The argument is carried out with the weight chosen before the final fiber and with the regularization, degree, and cutoff limits in a fixed order.

The last step is algebraic. Marked canonical models recover a finite normalization V' of the relative Iitaka base over b , together with its map and the images of the old boundary. The family J is then a form of a reference variety over $k = b(V')$. A finite splitting cocycle can be made constant. At every divisor which moves with the parameter, the actual relative-form order calculation supplies a multiplicity-one component attaining the threshold. This forces the splitting cover to be unramified there. After removing finitely many fixed divisors, purity and descent of finite étale covers produce a field F_b with

$$k \subseteq F_b, \quad \text{Frac}(\Omega \otimes_b F_b) \simeq \text{Frac}(\Omega \otimes_{\mathbf{C}(Y)} \mathbf{C}(X)).$$

Thus the entire geometric generic fiber is defined birationally over b , and $\text{Var}(f) \leq \dim T_0$. Together with (1.3) and the subadditivity bound, this proves Theorem 1.1.

Figure 1 records the two constancy steps and the final field inclusion in this argument.

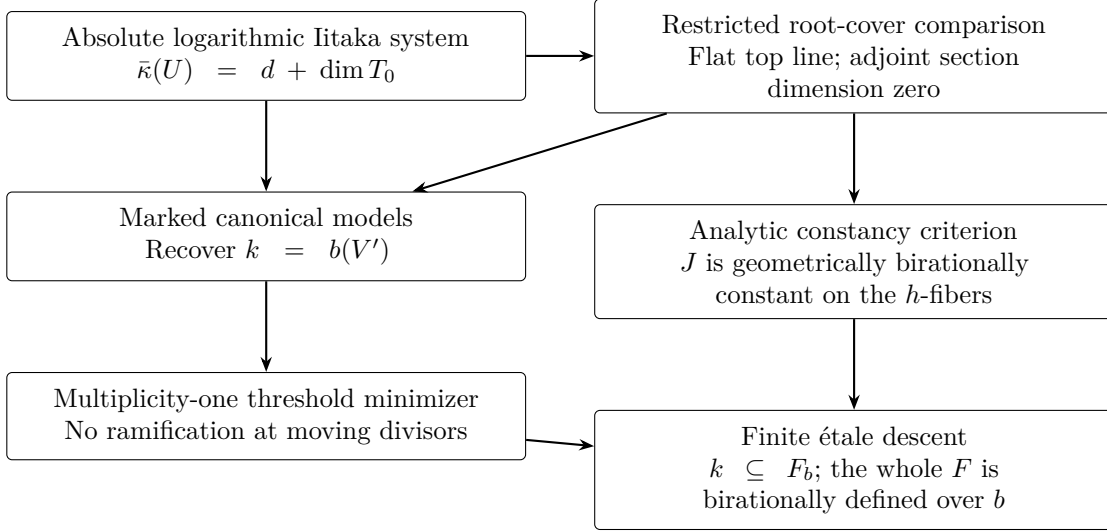


FIGURE 1. The parameter field and the two constancy steps. The restricted root-cover comparison supplies a flat line for analytic constancy and a zero-dimensional adjoint section system for marked-model recovery. The final field retains the relative Iitaka-base coordinates, giving a bound for the variation of the whole F .

1.3. Smooth families. For smooth families, the good-model theorem of the companion *Log abundance in characteristic zero* and Taji's rigidity theorems give a further application. This application is not used in the proof of Theorem 1.1. A smooth quasi-projective base V is *Campana-special* if, for any smooth projective compactification (Y, D) with reduced SNC boundary, every invertible subsheaf $L \subseteq \Omega_Y^p(\log D)$ satisfies $\kappa(L) < p$ for $1 \leq p \leq \dim V$ [45, §1].

Corollary 1.3 (Variation and birational isotriviality). *Let $f : U \rightarrow V$ be a smooth projective surjection with connected fibers between smooth connected complex quasi-projective varieties. Assume that every closed fiber is non-uniruled. Then*

$$\begin{aligned} \bar{\kappa}(V) = -\infty &\implies \text{Var}(f) < \dim V, \\ \bar{\kappa}(V) \geq 0 &\implies \text{Var}(f) \leq \bar{\kappa}(V). \end{aligned}$$

If V is Campana-special, then $\text{Var}(f) = 0$. In particular, if $\bar{\kappa}(V) = 0$, the family is birationally isotrivial.

Proof. Fix a closed fiber F_v . By [11, Corollary 0.3], its non-uniruledness makes K_{F_v} pseudo-effective. The companion [40, Corollary 11.2] gives a normal projective model M_v with semiample $\mathbf{Q}$ -Cartier K_{M_v} and, on a common smooth resolution $p : W \rightarrow F_v$, $q : W \rightarrow M_v$,

$$p^*K_{F_v} = q^*K_{M_v} + E, \quad E \geq 0 \quad \text{and } E \text{ is } q\text{-exceptional.}$$

For any prime divisor P exceptional over M_v , pass to a higher common resolution containing P and denote the effective pullback of E again by E . Since F_v is smooth, the discrepancy identity

$$a(P, M_v) = a(P, F_v) + \text{coeff}_P E \geq 0$$

shows that M_v has canonical singularities. Thus every F_v admits a good minimal model in the sense used by Tajiri. His variation is the same whole-fiber field-of-definition invariant as (1.1), so [45, Theorems 1.1–1.2] gives the dichotomy and the special-base assertion. The case $\bar{\kappa}(V) = 0$ also follows from the second inequality and $\text{Var}(f) \geq 0$. $\square$

Here *birationally isotrivial* means precisely $\text{Var}(f) = 0$. No isomorphism or trivialization of the family, globally or after a finite étale cover, is asserted.

For a smooth family in Theorem 1.1 with $\kappa(F) \geq 0$, the nonnegative-base bound also follows directly from logarithmic additivity [39, Corollary 1.2]. Choose compatible smooth compactifications with $U = f^{-1}(V)$ and reduced complement boundaries; their boundary strata are empty over V . The companion gives $\bar{\kappa}(U) = \kappa(F) + \bar{\kappa}(V)$, where the geometric generic and very general fiber Kodaira dimensions agree by generic base change and upper semicontinuity in each pluricanonical degree. Comparing with (1.2) and cancelling the finite $\kappa(F)$ gives $\text{Var}(f) \leq \bar{\kappa}(V)$. This comparison does not address the negative-base branch or all special bases.

1.4. Organization and an additional numerical result. Section 2 constructs the parameter field, after stating the exact logarithmic lower bound and the section conventions. Section 3 constructs the least-index root cover, computes its parabolic line from actual relative orders, and proves restricted flatness and finite character. Section 4 proves the analytic birational constancy criterion, including contraction, equivariance and algebraic spreading. Section 5 develops the comparison for auxiliary big model pairs at its first use. Section 6 uses recognizable markings to recover the finite Iitaka-base normalization. Section 7 makes the cocycle constant, removes moving ramification after every finite parameter extension, and identifies the whole original function field. This completes the main theorem.

The final two sections give a separate extension. In Section 8, a nef contribution pulled back from an adjoint-positive base replaces the Hodge line: interpolation retains the full second Iitaka-base dimension. Its volume estimate bounds only the effective action on the section image and is independent of the degree of the auxiliary alteration. Section 9 applies that theorem to the ordinary relative Iitaka fibration and recovers the same embedded parameter field. The exact reduced-root and section-comparison inputs from the companion are stated there. These sections are additional results; the logarithmic proof above does not depend on them.

2. THE PARAMETER FIELD DETECTED BY LOGARITHMIC FORMS

We first find a subfield of the original base field whose transcendence degree is exactly the excess of the total logarithmic Kodaira dimension over the fiber Kodaira dimension. This construction uses logarithmic subadditivity and section maps; constancy of the fibers will be proved after that field has been identified.

2.1. The lower bound and section conventions. The precise lower-bound input is the following companion theorem. Only its projective form is used here. A compact manifold belongs to Fujiki class $\mathcal{C}$ if it is bimeromorphic to a compact Kähler manifold; in particular, every projective manifold belongs to this class.

Theorem 2.1 (Reduced-SNC logarithmic subadditivity). *Let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective complex varieties. Let D_X, D_Y be reduced simple normal crossing divisors, with the zero divisor allowed, such that*

$$\text{Supp}(f^*D_Y) \subseteq \text{Supp}(D_X).$$

For a very general fiber F , put $D_F = D_X|_F$. Then

$$\kappa(X, K_X + D_X) \geq \kappa(F, K_F + D_F) + \kappa(Y, K_Y + D_Y).$$

The same assertion holds for compact manifolds in Fujiki class $\mathcal{C}$ and holomorphic fibrations between them.

Proof. Apply [37, Corollary 6.2]. The zero-divisor and $-\infty$ conventions here agree with that statement. Only its projective form is used in the present paper. $\square$

2.2. Models, sections, and fields. In the algebraic arguments below, varieties are integral unless a finite splitting algebra is explicitly considered. For a surjective morphism with connected fibers between normal varieties in characteristic zero, Stein factorization makes the base field relatively algebraically closed in the total function field; the generic field extension is regular and the generic fiber is geometrically integral. We use smooth projective resolutions of varieties and pairs [7]; resolving the closure of a graph also eliminates indeterminacy. Boundaries on these models will always be specified. A rational section of a line bundle on a normal variety is regular exactly when it is regular at every prime divisor.

Lemma 2.2 (Exceptional section comparison). *Let (X, Δ) be a smooth pair with effective SNC rational boundary of coefficients at most one. Let $\mu : X' \rightarrow X$ be a smooth log resolution, and let Δ' be the strict transform of Δ plus the reduced exceptional divisor. For every sufficiently divisible m and every line bundle A on X , pullback and pushdown identify*

$$H^0(X, m(K_X + \Delta) + A) \simeq H^0(X', m(K_{X'} + \Delta') + \mu^*A).$$

More precisely, there is an effective exceptional rational divisor E with $K_{X'} + \Delta' = \mu^(K_X + \Delta) + E$, and*

$$\mu_*\mathcal{O}_{X'}(m(K_{X'} + \Delta')) = \mathcal{O}_X(m(K_X + \Delta)).$$

This sheaf identity respects multiplication and remains valid after pushing to any common base. More generally, adding an effective exceptional divisor to a pulled-back rational divisor does not change its divisible section spaces.

Proof. The discrepancy is exceptional because a birational morphism between smooth varieties is an isomorphism at every generic target divisor. For an exceptional prime E_0 , choose boundary coordinates z_i on X , let $a_i = \text{ord}_{E_0}(\mu^*z_i)$, and let k_{E_0} be its Jacobian order. Pulling back a

logarithmic top wedge gives $k_{E_0} + 1 \geq \sum_i a_i$: a nonzero wedge contains at most one factor du/u at $E_0 = (u = 0)$. If the boundary coefficients are $\delta_i \leq 1$, the discrepancy coefficient is therefore

$$k_{E_0} + 1 - \sum_i \delta_i a_i \geq \sum_i (1 - \delta_i) a_i \geq 0.$$

Thus E is effective. A rational function with poles only on an effective exceptional divisor passes every codimension-one regularity test on the normal target, so $\mu_* \mathcal{O}_{X'}(mE) = \mathcal{O}_X$. The projection formula proves the sheaf and section identities, including their twisted, relative, and multiplicative forms. $\square$

The same comparison applies after restricting to a sufficiently general smooth fiber: the finitely many relevant horizontal strata meet it transversely, and vertical exceptional loci can be avoided. In particular, boundary changes on auxiliary higher models do not alter the original compact-fiber section dimension when only exceptional terms have been introduced on that fiber.

Lemma 2.3 (Finite pullback). *If $\pi : V' \rightarrow V$ is a dominant generically finite morphism of normal projective varieties and D is a rational Cartier divisor on V , then*

$$\kappa(V', \pi^* D) = \kappa(V, D).$$

Proof. Stein factorization reduces to finite π , since a proper birational morphism onto a normal variety has direct image $\mathcal{O}_V$. Clear denominators. Each homogeneous section upstairs is integral over the downstairs section ring: its monic characteristic polynomial in the finite function-field extension has coefficients in the appropriate powers of D . Trivializing D at each prime of V shows that these coefficients are regular there, hence everywhere by normality. The two graded rings therefore have the same transcendence degree. If the downstairs ring has no positive-degree section, the norm excludes one upstairs. These observations prove the assertion in all cases. $\square$

Lemma 2.4 (Fields, sections, and specialization). *For a proper variety and a line bundle, global sections and their multiplication maps commute with extension of the ground field. Consequently Iitaka dimension is unchanged by field extension. For a finite surjective morphism $\pi : Y \rightarrow X$ of normal projective varieties and a rational Cartier divisor L on X ,*

$$\kappa(Y, \pi^* L) = \kappa(X, L).$$

Finite collections of morphisms and birational identifications spread over fields of finite type. Section-system statements on geometric generic fibers may be checked on very general complex fibers, simultaneously in countably many divisible degrees, and conversely.

If $b \subseteq b'$ are algebraically closed fields and I is a geometrically integral b -variety, every nonempty open subset of $I_{b'}$ contains a point of $I(b)$.

Proof. Proper cohomology and flat base change identify the section spaces, compatibly with multiplication, and hence preserve the dimensions of the section maps. The finite-morphism assertion follows from Lemma 2.3. Morphisms, rational maps, and the identities making two rational maps inverse use finitely many coefficients and spread after finitely many denominators are inverted. On the smooth proper locus, in each fixed degree we shrink to an open where the section dimension is constant; cohomology and base change then identify the complete fiber space. Spreading the section maps and excluding the countably many proper closed complements gives the very-general-fiber assertion.

For the last assertion, work on an affine open of I . A regular function after extension to b' is a finite sum $\sum_j c_j f_j$, where the $c_j \in b'$ are linearly independent over b and $f_j \in \mathcal{O}(I)$. If it vanishes on every b -point, then every f_j does too and is zero. Thus $I(b)$ remains Zariski dense after field extension, proving the assertion. $\square$

Lemma 2.5 (Upper addition and interpolation). *For a projective fibration $u : T \rightarrow R$ and a rational divisor D with nonempty divisible section systems, the dimension of its section-system image is at most the dimension of R plus the dimension of the restricted section-system image on the generic fiber. In particular it is at most*

$$\dim R + \kappa(T_{\bar{\eta}}, D|_{T_{\bar{\eta}}}).$$

If D is rationally effective and $b \geq 1$ is rational, then for every rational divisor M ,

$$\kappa(D + M) \geq \kappa(D + bM).$$

Proof. For a sufficiently divisible system, compare its image with the base using the graph of the two rational maps. The fiber image has dimension bounded by the restricted complete system; the fiber dimension theorem gives the assertion, and taking the maximum over divisible systems proves the first inequality. This is the usual upper-addition estimate; see [27, Theorem 4, p. 357].

For the second claim use

$$D + M = b^{-1}(D + bM) + (1 - b^{-1})D.$$

After clearing denominators, multiplication by a fixed effective section of the second summand embeds the corresponding section systems of the first. Their ratios are unchanged. Taking their image dimensions proves the assertion. $\square$

We use the Iitaka fibration theorem for a divisor with nonnegative section dimension [27, Theorems 3–4]: on a suitable smooth model its connected generic fiber has restricted divisor of Kodaira dimension zero, and the base has dimension equal to the original section dimension. Its application to a log adjoint divisor is explained in the two-base construction, including the comparison under subsequent resolutions.

For analytic computations, we normalize $dd^c = \frac{i}{2\pi} \partial \bar{\partial}$, so that $dd^c \log |s_D|^2 = [D]$ in a local holomorphic frame. A weight is a local potential for a singular Hermitian metric; differences of weights on the same line bundle are globally defined functions. Smooth reference metrics are understood whenever a potential or a norm is compared globally. Positive universal normalization constants do not affect the integral estimates below.

Let $f : U \rightarrow V$ be the morphism in Theorem 1.1, and put $d = \kappa(F)$. If $d = -\infty$, the asserted inequality follows from the convention for $-\infty$. We therefore assume throughout this section that $d \geq 0$. We first obtain the contribution of $\bar{\kappa}(V)$ and then construct the bases used to control variation. The subadditivity used in this section is Theorem 2.1.

2.3. Compactification and the first Iitaka base.

Lemma 2.6. *There are smooth connected projective compactifications X and Y of U and V , respectively, and a surjective morphism $X \rightarrow Y$ with connected fibers extending f , such that the reduced divisors*

$$D_X = X \setminus U, \quad D_Y = Y \setminus V$$

have simple normal crossings and the inverse image of V is exactly U . In particular, D_X has no component dominating Y , and

$$(2.1) \quad \kappa(X, K_X + D_X) = \bar{\kappa}(U) \geq d + \bar{\kappa}(V).$$

Proof. Take a smooth projective compactification of V and resolve its boundary. Since f is projective, embed U as a closed subscheme of a projective space over V and take its closure over the resulting projective base Y . Equivalently, one can take the closure of the graph of f in projective compactifications. Over V this closure is already U : the original family is proper over V , so no additional limit point is needed above that open set. Resolve the closure and its boundary, with all centers outside the smooth open U . Resolution and elimination of indeterminacy in

characteristic zero give the required morphism between smooth projective varieties and the SNC boundary; see [7].

The resulting morphism has a geometrically integral generic fiber. Indeed, the original proper morphism has connected fibers and normal base, so its Stein factorization gives $f_*\mathcal{O}_U = \mathcal{O}_V$. Thus $\mathbf{C}(V)$ is relatively algebraically closed in $\mathbf{C}(U)$; in characteristic zero this extension is regular. The finite part of the Stein factorization of the compactified morphism is consequently birational over Y , and hence is an isomorphism because Y is normal. This proves connectedness of its fibers.

As $f^{-1}(V) = U$, every point of D_X lies over D_Y , and $\text{Supp}(f^*D_Y) \subseteq \text{Supp}(D_X)$. A very general fiber lies over V , has no boundary, and has ordinary Kodaira dimension d . Theorem 2.1 now applies to these smooth projective varieties, connected fibers, and reduced SNC boundaries. Its conclusion is (2.1), by the compactification definition of logarithmic Kodaira dimension. $\square$

Fix, once and for all, a nonzero section

$$(2.2) \quad \xi \in H^0(Y, m(K_Y + D_Y)), \quad m > 0.$$

Such a section exists because $\bar{\kappa}(V) \geq 0$. It is essential here that ξ is logarithmic: no nonnegative ordinary Kodaira dimension of Y is assumed.

We recall the boundary convention for all subsequent source models. If $\pi : X_1 \rightarrow X$ is a smooth birational model, set

$$D_1 = \pi_*^{-1}D_X + \text{Exc}(\pi)_{\text{red}}.$$

The SNC discrepancy formula gives

$$(2.3) \quad K_{X_1} + D_1 = \pi^*(K_X + D_X) + \sum_E A(E; X, D_X)E, \quad A(E; X, D_X) \geq 0,$$

where the sum is exceptional. For every positive integer a , divisorial testing on the normal target therefore gives

$$\pi_*\mathcal{O}_{X_1}(a(K_{X_1} + D_1)) = \mathcal{O}_X(a(K_X + D_X)).$$

Thus these modifications preserve the entire adjoint section ring, not just its Iitaka dimension. The same argument permits a pulled-back rational twist in divisible degrees. After each modification we resolve the boundary again, using the same convention.

Proposition 2.7. *There are smooth connected projective varieties X_*, X', W, Z , a birational morphism $X' \rightarrow X_*$, and morphisms*

$$X' \xrightarrow{x} W \xrightarrow{g} Y, \quad h : W \rightarrow Z, \quad q = h \circ x,$$

with the following properties.

- (i) *The variety X_* is a smooth model of X carrying a resolved absolute Iitaka map for $K_X + D_X$. Its boundary D_* , and the boundary D' on X' , follow the strict-transform-plus-reduced-exceptional convention. The map q is the pullback of this Iitaka map, and*

$$(2.4) \quad \dim Z = \bar{\kappa}(U).$$

- (ii) *The morphisms x, g, h are surjective and have connected fibers and geometrically integral generic fibers. The induced map from W to its image in $Y \times Z$ is generically finite. In particular, $W_z \rightarrow g(W_z)$ is generically finite for very general z .*
- (iii) *The divisor $D_W^0 = (g^*D_Y)_{\text{red}}$ is SNC. On very general fibers over Z , all the relevant varieties and boundary strata are smooth, and*

$$(2.5) \quad \kappa(X_{*,z}, K_{X_{*,z}} + D_*|_{X_{*,z}}) = \kappa(X'_z, K_{X'_z} + D'|_{X'_z}) = 0.$$

One may make further compatible smooth birational modifications, retaining these assertions. After the finite preparations in this section, X_ can be fixed as an earlier model, and every subsequent source model is taken to dominate it.*

Proof. By (2.1), $K_X + D_X$ has nonnegative section dimension. The divisor Iitaka theorem gives a resolved map with base dimension $\bar{\kappa}(U)$ and with restriction of the divisor having section dimension zero on a very general fiber. The needed statement is [27, Theorem 3, p. 357, and its proof in §4, pp. 361–363]. The stabilized section field is relatively algebraically closed in $\mathbf{C}(X)$ by [27, §3, Proposition 1, p. 359]. In the case $\bar{\kappa}(U) = 0$, the map is the map to a point.

For clarity, the fiber conclusion concerns the full restricted section system. On a resolution, a sufficiently divisible system has moving part pulled back from a big system on the Iitaka base. In a divisible degree there is consequently a section after subtracting a pulled-back ample divisor. Any extra independent plurisections on the generic fiber extend after an ample pullback twist. Multiplying by the preceding section would then enlarge the stabilized section field, which is impossible. Doing this in each degree gives the assertion on very general fibers. Formula (2.3) allows the theorem to be applied to the log divisor on each higher source model itself. Adjunction on a smooth fiber identifies its restriction with that fiber's log canonical divisor. This proves (2.5), including after further compatible resolutions and birational changes of the Iitaka base.

Let $K = \mathbf{C}(X)$, $L_Y = \mathbf{C}(Y)$, and $L_Z = \mathbf{C}(Z)$, viewed as subfields of K . The combined image in $Y \times Z$ has function field $L_Y L_Z$. Let L_W be the relative algebraic closure of this compositum in K , a finite extension of $L_Y L_Z$. Normalize the combined image in L_W , resolve it, and resolve the resulting rational map from a higher source model. These operations give the displayed morphisms.

The field L_W is relatively algebraically closed in K by construction. Moreover, L_Y and L_Z are relatively algebraically closed in K , so each is relatively algebraically closed in L_W . In characteristic zero the three corresponding generic field extensions are regular. This proves geometric integrality of the generic fibers of x, g, h ; properness and the normality of their targets give connectedness by Stein factorization. Finiteness of $L_W/(L_Y L_Z)$ gives generic finiteness to the combined image and hence generic finiteness of the indicated fiber maps.

Finally resolve D_W^0 and the source boundary and lift all the maps. Outside a proper closed subset of Z , the morphisms and every dominating boundary stratum are smooth, and the non-dominating strata are absent. This follows from generic smoothness applied to the finitely many strata. It gives the asserted very general SNC fiber data. The section-ring and field arguments just given are unchanged by these birational operations. $\square$

The two maps just constructed are shown in Figure 2. The lower arrows in the figure are the next objective; they will identify the required subfield of $\mathbf{C}(Y)$.

2.4. Logarithmic restrictions to the Iitaka fibers. Write

$$n = \dim Y, \quad r = \dim W_z, \quad p_1 = n - r.$$

Generic finiteness of $W_z \rightarrow g(W_z)$ gives $0 \leq r \leq n$, so $p_1 \geq 0$. The next argument explains how the logarithmic form on Y produces a form of the correct degree on W_z .

Lemma 2.8. *For very general $z \in Z$, the form ξ in (2.2) induces nonzero sections of $m(K_{W_z} + D_W^0|_{W_z})$ by contraction with suitable p_1 tangent directions at z . In particular,*

$$\kappa(W_z, K_{W_z} + D_W^0|_{W_z}) \geq 0.$$

Proof. Work over an open subset of Z on which h and all the boundary strata are smooth. On its inverse image the logarithmic cotangent sequence is exact:

$$0 \longrightarrow h^* \Omega_Z^1 \longrightarrow \Omega_W^1(\log D_W^0) \longrightarrow \Omega_{W/Z}^1(\log D_W^0) \longrightarrow 0.$$

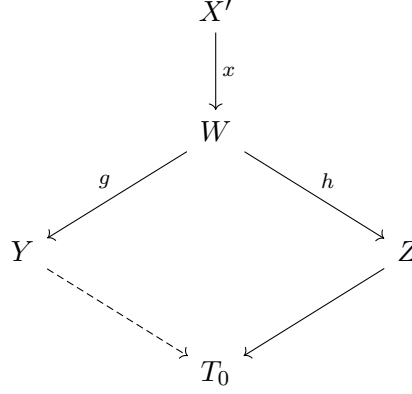


FIGURE 2. The two maps from the total space: gx is the original fibration and hx is the absolute log Iitaka map. The construction of W makes its map to $Y \times Z$ generically finite onto its image. The proof then constructs T_0 by parametrizing $g(W_z)$ and shows that both base fields contain $\mathbf{C}(T_0)$. The lower arrows are conclusions of the argument, not assumptions. Resolving the dashed map later gives $S \rightarrow T_0$ with S birational to Y .

The last bundle has rank r . Since $n = r + p_1$, the exterior-power filtration has no term of base degree below p_1 , and its first quotient is the canonical map

$$(2.6) \quad \bigwedge^n \Omega_W^1(\log D_W^0) \longrightarrow h^* \bigwedge^{p_1} \Omega_Z^1 \otimes \det \Omega_{W/Z}^1(\log D_W^0).$$

Pullback of logarithmic differentials under g is regular here: locally, the pullback of an equation for D_Y is a monomial in equations for D_W^0 times a unit. Thus $g^*\xi$ is a section of the m th tensor power of the left side of (2.6). Apply the quotient tensorwise and, on W_z , evaluate on any p_1 fixed vectors in $T_z Z$. The result is a global section of

$$(\det \Omega_{W/Z}^1(\log D_W^0)|_{W_z})^{\otimes m} \simeq \mathcal{O}_{W_z}(m(K_{W_z} + D_W^0|_{W_z})).$$

This construction is independent of local lifts of the chosen tangent vectors: replacing one lift by a relative tangent vector inserts $r + 1$ relative vectors into an alternating form, so contributes zero. It is therefore valid along the entire boundary of the fiber, rather than only on its interior.

At a generic point of W_z , the differential of g has rank n , while its restriction to the tangent space of W_z has rank r . The induced normal map from $T_z Z$ consequently has rank p_1 . Some choice of the p_1 vectors gives a nonzero determinant, and hence a nonzero contracted section. When $p_1 = 0$, the same construction simply uses the relative top-degree quotient, with no tangent vectors to choose. $\square$

Lemma 2.9. *Let J be the smooth geometric generic fiber of x , or a simultaneous very general complex fiber. Then*

$$(2.7) \quad \kappa(W_z, K_{W_z} + D_W^0|_{W_z}) = 0, \quad \kappa(J) = \kappa(J, K_J + D'|_J) = 0.$$

The source-fiber equalities (2.5) hold on the same very general locus.

Proof. First, $\kappa(J) \geq 0$. To see this using the ordinary canonical divisor, take a very general compact Y -fiber of $X' \rightarrow Y$. It is a smooth birational model of the original fiber F and therefore carries a nonzero ordinary pluricanonical section because $d \geq 0$. The restriction of this section to a general fiber of its map to W_y is nonzero: its zero divisor cannot contain all those fibers. Adjunction identifies this restriction, up to the one-dimensional determinant of the base cotangent

space, with an ordinary pluricanonical section on J . Since $D'|_J$ is effective, its log canonical divisor also has nonnegative section dimension.

Now restrict x to a very general fiber over Z :

$$x_z : (X'_z, D'|_{X'_z}) \longrightarrow (W_z, D_W^0|_{W_z}).$$

The source and base are smooth connected projective varieties, the fibers are connected, and both displayed boundaries are reduced SNC. Every component of the inverse base boundary is in the source boundary, because both lie over D_Y . Thus Theorem 2.1 applies. Its conclusion, together with (2.5) and Lemma 2.8, is

$$0 \geq \kappa(J, K_J + D'|_J) + \kappa(W_z, K_{W_z} + D_W^0|_{W_z}) \geq 0.$$

Both terms vanish. The already proved ordinary nonnegativity, and monotonicity under addition of an effective divisor, give $\kappa(J) = 0$ as well. The simultaneous geometric-generic and very general interpretations follow by base change in the countably many section degrees. $\square$

Although the original compactification has no horizontal boundary, a later exceptional divisor can be horizontal for x . Thus the proof does not assert $D'|_J = 0$; it proves both dimensions in (2.7). On a very general Y -fiber, however, every exceptional divisor that meets that fiber remains exceptional over the original F : its center has codimension at least two after restriction, by the dimension formula for a center dominating Y . Accordingly, (2.3) on this compact fiber identifies its logarithmic section dimension with its ordinary dimension d .

2.5. The image family and the second base.

Proposition 2.10. *The models of Proposition 2.7 may be chosen together with smooth connected projective varieties S, T_0 and morphisms in the commutative diagram*

$$(2.8) \quad \begin{array}{ccccc} & & q=h \circ x & & \\ & & \curvearrowright & & \\ X' & \xrightarrow{x} & W & \xrightarrow{h} & Z \\ & \searrow g & \downarrow g_S & & \downarrow \\ Y & \xleftarrow{\rho} & S & \longrightarrow & T_0 \end{array}$$

such that ρ is birational, $g = \rho \circ g_S$, and

$$(2.9) \quad W \longrightarrow (S \times_{T_0} Z)_{\text{main}} \quad \text{is dominant and generically finite.}$$

Both $Z \rightarrow T_0$ and $S \rightarrow T_0$ have geometrically integral generic fibers. With $D_S = (\rho^* D_Y)_{\text{red}}$ made SNC, one has

$$(2.10) \quad \begin{aligned} \dim(W/Y) &= d, & \dim T_0 &= p_1, & \dim Z &= d + \dim T_0, \\ \kappa(S_t, K_{S_t} + D_S|_{S_t}) &\geq 0 \end{aligned}$$

for very general $t \in T_0$.

Proof. The relative dimension over Y . The dimension of W_y is the image dimension of the absolute Iitaka coordinates restricted to a very general Y -fiber; the finite extension in the definition of W does not change this dimension. Those coordinates form a subsystem of a log pluricanonical system on that fiber. By the exceptional section comparison just explained, their image dimension is at most d . Conversely, upper addition for the ordinary canonical divisor on that compact fiber, whose general fiber over W_y is J , gives

$$d \leq \dim W_y + \kappa(J) = \dim W_y.$$

Here we use the upper addition inequality for section dimensions with irreducible general fibers, as in [27, Theorem 4, p. 357]; no nonnegativity or subadditivity hypothesis on W_y is involved. Consequently $\dim(W/Y) = d$.

The differential of the image family. Let $C_z = g(W_z) \subseteq Y$ with its reduced structure. The scheme image of $W \rightarrow Y \times Z$ has geometrically integral generic fiber over Z : its fiber function field lies between $\mathbf{C}(Z)$ and the regular extension $\mathbf{C}(W)$. Generic flatness, followed by restriction to a smaller open, therefore yields a flat family of reduced integral subschemes C_z of Y . Hilbert representability [21, Theorems 3.1–3.2 and the discussion following Proposition 3.8] therefore gives a parameter map

$$\tau : Z \dashrightarrow \mathrm{Hilb}(Y).$$

We claim that the dimension of its image is exactly p_1 .

Fix a very general z and a basis of $T_z Z$. Perform the contraction in Lemma 2.8 for every p_1 -tuple of basis vectors. By Lemma 2.9, all nonzero resulting sections in degree m are proportional. Indeed, two independent sections in one degree would give a nonconstant ratio and positive Iitaka dimension.

At a general smooth point $y \in C_z$, the normal deformation map is

$$(2.11) \quad T_z Z \longrightarrow N_{C_z/Y, y}.$$

It is computed by lifting a base direction to W , applying dg , and taking its class modulo $T_y C_z$. This is independent of the lift and has rank p_1 , since g is dominant and $W_z \rightarrow C_z$ is generically finite. Locally trivialize the tangential determinant and the form ξ . Each contracted section is the m th power of the corresponding p_1 -minor of (2.11), multiplied by a common nonzero factor. Proportionality of the sections says that the m th powers of all ratios of nonzero minors are constant as the point of W_z varies. The ratios themselves are constant: a rational function whose positive power is constant is algebraic over $\mathbf{C}$, and hence is constant on an integral complex variety. The Plücker coordinates of (2.11) thus have constant ratios. Its kernel, a subspace of the fixed vector space $T_z Z$, is consequently independent of the general point y .

To pass from this pointwise calculation to the parameter map, write $\mathcal{I}_z$ for the ideal of C_z in Y . A Hilbert tangent vector is a homomorphism

$$\mathcal{I}_z / \mathcal{I}_z^2 \longrightarrow \mathcal{O}_{C_z}.$$

Such a homomorphism is zero if it is zero at the generic point: the target is the structure sheaf of an integral scheme and has no torsion. On the general smooth locus of C_z , its normal value is precisely (2.11). It follows that $d\tau_z$ has the same kernel as that normal map and therefore has rank p_1 . In characteristic zero, generic differential rank equals the dimension of the reduced image. This proves the claim.

This argument includes the extremes. If $p_1 = 0$, every C_z has dimension $\dim Y$ and is therefore Y , so the family is constant. If $r = 0$, the smooth connected W_z is a point and its parameter map has rank $\dim Y = p_1$.

Construction of S and T_0 . Let the function field of T_0 be the relative algebraic closure of the image parameter field in $\mathbf{C}(Z)$, and take a smooth projective model. Then $\dim T_0 = p_1$, and $Z \rightarrow T_0$ has geometrically integral generic fiber. Pull the universal image family back to T_0 and take the component dominated by W , with a smooth projective resolution denoted by S . After compatible resolutions all the maps in (2.8) are morphisms. Its right square commutes, and the induced map (2.9) is dominant and generically finite: over a general z it is the generically finite map to the corresponding image member.

The generic fiber of $S \rightarrow T_0$ has dimension r . Thus $\dim S = r + p_1 = \dim Y$, and its dominant map $\rho : S \rightarrow Y$ is generically finite. In fact it is birational. The maps give the field inclusions

$$(2.12) \quad \mathbf{C}(Y) \subseteq \mathbf{C}(S) \subseteq \mathbf{C}(W) \subseteq \mathbf{C}(X).$$

The first extension is algebraic, whereas $\mathbf{C}(Y)$ is relatively algebraically closed in $\mathbf{C}(X)$. Therefore $\mathbf{C}(S) = \mathbf{C}(Y)$. In particular, the construction embeds $\mathbf{C}(T_0)$ in the original base field $\mathbf{C}(Y)$.

For completeness, geometric integrality of the generic fiber of $S \rightarrow T_0$ can be checked in the same field tower. The field $\mathbf{C}(T_0)$ is relatively algebraically closed in $\mathbf{C}(Z)$ by construction, and $\mathbf{C}(Z)$ is relatively algebraically closed in $\mathbf{C}(W)$. It follows that $\mathbf{C}(T_0)$ is relatively algebraically closed in $\mathbf{C}(W)$, and hence in its subfield $\mathbf{C}(S)$. The extension $\mathbf{C}(S)/\mathbf{C}(T_0)$ is therefore regular. This also verifies the geometric connectedness independently of the universal-family description.

Dimensions and the logarithmic base fiber. The equality already proved over Y gives

$$\dim W = \dim Y + d = \dim Z + r.$$

Together with $\dim T_0 = \dim Y - r$, this yields $\dim Z = d + \dim T_0$.

Resolve the support of $D_S = (\rho^* D_Y)_{\text{red}}$, lifting the other maps once more. Logarithmic pullback gives a nonzero section $\rho^* \xi$ of $m(K_S + D_S)$. Restrict it to a very general smooth fiber S_t ; this restriction is nonzero, since its zero divisor cannot contain all fibers. Adjunction, with the determinant of $T_t^* T_0$ a constant one-dimensional factor, gives a nonzero section of $m(K_{S_t} + D_S|_{S_t})$. This proves the last assertion of (2.10). All these final resolutions preserve the section and fiber properties in Proposition 2.7. We now fix an earlier resolved Iitaka source X_* and take the common source X' above it; later source modifications will continue to dominate this fixed model. $\square$

Remark 2.11. The field identifications in (2.12) will also control the constant field in the final descent. Inside the prescribed algebraic closure Ω of $\mathbf{C}(Y)$, set $b = \overline{\mathbf{C}(T_0)}$. The regular extensions $\mathbf{C}(Y)/\mathbf{C}(T_0)$ and $\mathbf{C}(W)/\mathbf{C}(Y)$ remain regular after the corresponding base changes. Thus, if I and H_0 are the geometric generic fibers of S and W over T_0 , their map $H_0 \rightarrow I$ has geometrically integral generic fiber. Moreover, $b(I) = b\mathbf{C}(Y) \subseteq \Omega$ is algebraic over $\mathbf{C}(Y)$, so Ω is also an algebraic closure of $b(I)$. These are statements about the fields in the original diagram; they do not assert constancy of the x -fiber or of the whole original fiber.

3. THE ROOT COVER AND ITS HODGE LINE

We retain the diagram and the boundary conventions of Proposition 2.7. Thus

$$X' \xrightarrow{x} W, \quad h : W \longrightarrow Z, \quad q = hx,$$

and X' dominates a fixed earlier log Iitaka model X_* . Its boundary D' is obtained from the boundary on X_* by strict transform and addition of the reduced exceptional divisor. Put $D_W^0 = (g^* D_Y)_{\text{red}}$. Lemma 2.9 gives, for very general $z \in Z$ and for the geometric generic fiber J of x ,

$$\begin{aligned} (3.1) \quad & \kappa(X_{*,z}, K_{X_{*,z}} + D_*|_{X_{*,z}}) = 0, \\ & \kappa(W_z, K_{W_z} + D_W^0|_{W_z}) = 0, \\ & \kappa(J) = \kappa(J, K_J + D'|_J) = 0. \end{aligned}$$

All subsequent modifications are smooth projective birational modifications, with this same source-boundary convention. We shall construct an effective rational boundary B and a nef rational Hodge line M on W , prove a section comparison on W_z , and then show that the Hodge line is rationally trivial there.

3.1. An ordinary cyclic cover.

Lemma 3.1 (The root cover). *Let p be the least positive integer for which $H^0(J, pK_J) \neq 0$. A generator ω_J determines a geometrically connected cyclic cover of J . Every smooth projective*

resolution $\tilde{J}$ of its finite normalization satisfies

$$\kappa(\tilde{J}) = 0, \quad h^0(\tilde{J}, mK_{\tilde{J}}) = 1 \quad \text{for every } m > 0.$$

The root form spans the entire ordinary top-form space. After shrinking a nonempty open $W^\circ \subset W$, these covers admit an equivariant resolution forming a smooth projective family over W° .

Proof. The spaces of pluricanonical sections commute with extension to an algebraic closure of the ground field. We first make that extension and work with the smooth projective variety J over an algebraically closed field of characteristic zero. Since $\kappa(J) = 0$, every nonzero pluricanonical system has dimension one.

Choose a rational canonical frame τ and write $\omega_J = f\tau^p$. Normalize J in the field obtained by adjoining a p th root of f . This field extension has degree p . Indeed, since the ground field contains the p th roots of unity, reducibility of the Kummer polynomial would make f a q th power for some prime $q \mid p$. If $f = a^q$, then the rational (p/q) -canonical form $a\tau^{p/q}$ has regular q th power. Its order at every prime is therefore nonnegative, so it is regular, contrary to the minimality of p . This also proves geometric connectedness before the extension of the ground field.

Denote the finite normalization by $\pi_0 : J^{\text{nor}} \rightarrow J$ and a smooth resolution by $\mu : \tilde{J} \rightarrow J^{\text{nor}}$. The rational root form η on $\tilde{J}$ satisfies

$$(3.2) \quad \eta^{\otimes p} = (\pi_0\mu)^*\omega_J$$

as pluricanonical differentials. Pullback of a regular pluricanonical differential under a morphism of smooth varieties is regular. Thus (3.2) proves that η is regular at every prime of $\tilde{J}$, including the resolution-exceptional primes.

Set $E = \text{div}(\omega_J)$. At a prime of J^{nor} over a prime of J at which E has coefficient $a \geq 0$, let e be the ramification index. The tame differential formula gives

$$(3.3) \quad \text{ord}(\eta) = \frac{ea}{p} + e - 1.$$

If $a = 0$, adjoining the root of a unit is unramified at that prime and $e = 1$. If $a \geq 1$, the right side of (3.3) is at most $2ea$. This bound is a statement at the primes of the finite normalization; we do not assert it at every prime of its resolution.

For $s \in H^0(\tilde{J}, mK_{\tilde{J}})$, the rational function s/η^m consequently has poles at the primes of J^{nor} bounded by $2m\pi_0^*E$. Normality therefore gives an injection

$$(3.4) \quad H^0(\tilde{J}, mK_{\tilde{J}}) \longrightarrow H^0(J^{\text{nor}}, \mathcal{O}_{J^{\text{nor}}}(2m\pi_0^*E)), \quad s \longmapsto s/\eta^m.$$

Only codimension-one tests on J^{nor} enter this injection.

Lemma 2.3 gives $\kappa(J^{\text{nor}}, \pi_0^*E) = \kappa(J, E) = 0$. Since π_0^*E is effective, each space on the right of (3.4) has dimension one: two independent sections would give a nonconstant section ratio. The nonzero section η^m gives the reverse inequality on the left, proving the assertion.

The integer p and the one-dimensional generator space are unchanged geometrically. A rational generator can consequently be chosen as a section of $pK_{X'/W}$. The Kummer construction is algebraic over a nonempty open of W . An equivariant resolution of its projective normalization, followed by shrinking the base and generic smoothness, gives the asserted smooth projective family, with its fiberwise cyclic deck action. $\square$

3.2. The entire top line and its extension. A highest Hodge line means the highest nonzero filtration piece F^p of a pure variation, with $F^{p+1} = 0$ and $\text{rank } F^p = 1$. The rank condition concerns the entire highest step; a rank-one eigenspace in a larger highest step does not satisfy it. Its projective period map is the map on the connected universal cover to the projective space of a flat reference vector space determined by that line. Its differential rank need not equal

the rank of the full period map. Our applications use rational polarizations and a monodromy-invariant lattice in the rational local system. The full-period comparison in Theorem 3.8 also allows a real polarization, with a lattice in the underlying real local system. The parabolic extension is the rational line obtained by making boundary monodromy unipotent on local root covers, taking its canonical Hodge extension, and descending with the resulting rational weights. Finite-monodromy weights are retained in this convention.

Lemma 3.2 (Parabolic extension). *Let P be smooth projective, let D be a reduced SNC divisor, and let L° be the highest line of a polarized integral pure variation on $P \setminus D$. Its rational parabolic extension L is nef. This extension commutes with positive tensor powers and with pullback by morphisms of smooth projective log pairs whose interiors map into $P \setminus D$.*

Proof. For unipotent local monodromy, the extension is the top filtration subbundle of the Deligne–Schmid extension. Nefness and compatibility with powers and log-pair pullbacks are [4, Lemmas 2.16 and 5.6]. The pullback statement follows from functoriality of the unipotent extension and the extended Hodge filtration.

Integral polarized monodromy is quasi-unipotent at the boundary. In a crossing chart, take sufficiently divisible roots of the boundary coordinates to make it unipotent. The resulting line is equivariant for the finite deck group. A common positive power kills the stabilizer characters on its fibers and descends to a line bundle in the original chart. These descents agree on overlaps: compare them on a common root cover and use uniqueness of the unipotent extension. They define the rational parabolic line, including its finite-monodromy weights. The same comparison proves compatibility with powers and further log-pair pullbacks. This proves the quasi-unipotent assertions by descent from the unipotent ones.

These comparisons can be made on a global smooth adapted alteration. For example, a finite neat level of the integral monodromy, followed by compactification and resolution, makes the boundary monodromies unipotent. Extra boundary components on which the original variation already extends have weight zero. The pullback of L is the unipotent extension there and is nef. Nefness descends under a proper surjective map: every curve downstairs is dominated by a curve upstairs, and the projection formula compares their degrees. Thus L is nef. $\square$

Fix the rational relative form ω_J of Lemma 3.1. Enlarge an SNC divisor on W so that the root-cover family and its pure integral polarized middle-cohomology variation are defined on its complement. This enlargement specifies the domain of the variation; it does not add logarithmic poles to the target section systems.

Let $\mathcal{L}$ be the highest nonzero Hodge filtration piece of the middle-cohomology variation of the resolved root-cover family. Lemma 3.1 says that $\mathcal{L}$ has rank one. Write M for its parabolic rational extension to the prepared SNC compactification W . Lemma 3.2 makes M nef.

3.3. Actual divisorial orders and the boundary. Two divisorial tests enter the comparison. The first records which poles a coefficient on W may have while its product with the relative pluriform remains logarithmic at every actual source component. It gives a minimum over those components. The second test ranges over divisorial valuations on all higher models; the metric calculation below will identify it with an integrability threshold. For each prime P of W , we denote these two quantities by t_P and λ_P , respectively.

Proposition 3.3 (Divisorial normalization). *For a prime divisor P on W , and for a prime divisor Q on the actual source X' dominating P , set*

$$(3.5) \quad \begin{aligned} m_Q &= \text{ord}_Q(x^*P), & r_Q &= \frac{1}{p} \text{ord}_Q(\omega_J), & d_Q &= \text{coeff}_Q(D'), \\ t_P &= \min_{\substack{Q \mapsto P \\ Q \text{ on } X'}} \frac{r_Q + d_Q}{m_Q}, \\ \lambda_P &= \inf_{Q \mapsto P} \frac{1 + r_Q}{m_Q}, & B_P &= 1 - \lambda_P + t_P. \end{aligned}$$

In the definition of λ_P , Q runs over divisorial valuations on all higher smooth source models. In both formulas r_Q is measured in the actual relative canonical bundle with respect to the smooth base W .

The two rational divisors

$$T = \sum_P t_P P, \quad B = \sum_P B_P P$$

have finite support. The divisor B is an effective boundary and

$$(3.6) \quad 0 \leq B_P \leq 1, \quad B \geq D_W^0 = (g^*D_Y)_{\text{red}}.$$

The formulas at unchanged codimension-one base points are invariant under further source resolutions with the prescribed boundary convention. They are also preserved, allowing splitting of components, by an étale extension of the base DVR and the corresponding source base change. After a further birational preparation of W , B can be assumed to have SNC support.

Proof. At the generic point of P , correct the relative form by the factor u^{-pt_P} , where u is a local equation for P ; take a tensor power to make the exponent integral. At every actual component over P its normalized order satisfies

$$(3.7) \quad r_Q - m_Q t_P + d_Q \geq 0.$$

Orders at divisors horizontal over W are nonnegative, since ω_J is an ordinary regular pluricanonical form on the generic fiber. Near the generic point of P , the corrected form is therefore a logarithmic pluriform for the actual SNC pair (X', D') .

The discrepancy inequality for a reduced SNC pair says that pullback to a higher smooth model is still logarithmic for strict transform plus reduced exceptional boundary. In particular, the normalized ordinary order of the corrected form at any higher prime is at least -1 . It follows that

$$\frac{1 + r_Q}{m_Q} \geq t_P \quad \text{for every } Q \mapsto P, \quad \text{and hence} \quad \lambda_P \geq t_P.$$

If Q attains the actual minimum defining t_P , then

$$(3.8) \quad 0 \leq \lambda_P - t_P \leq \frac{1 + r_Q}{m_Q} - \frac{r_Q + d_Q}{m_Q} = \frac{1 - d_Q}{m_Q} \leq 1.$$

This proves $0 \leq B_P \leq 1$. If P is contained in $g^{-1}D_Y$, every actual component over P lies in D' and has $d_Q = 1$. The middle term of (3.8) is then zero, so $B_P = 1$.

Resolve, over the generic point of P , the support of the form's zero and pole divisor together with x^*P . On this resolution the relevant conditions are monomial inequalities. The SNC discrepancy inequality shows that inequalities at its components imply the inequalities on every higher model. Thus a finite log resolution computes λ_P ; in particular it is rational. Outside a finite divisorial subset of W , the fibers are smooth and integral, the relative form has no vertical zero or pole, and there is no vertical source boundary. There the actual component has $m_Q = 1$,

$r_Q = d_Q = 0$, and the smooth-fiber calculation gives $t_P = 0$ and $\lambda_P = 1$. This proves finite support.

On a further source resolution, the old actual divisors remain. They still attain the old minimum. Equation (3.7) holds at every new actual divisor by logarithmic pullback, so the minimum cannot decrease. The all-valuation infimum is unchanged. An étale base-DVR extension preserves canonical frames up to units, multiplicities, and all the displayed orders; the same assertion holds on a fixed log resolution, including when a divisor splits. Finally resolve the finite support on the base, resolve the source map, and recompute (3.5). All previously unchanged codimension-one points keep their coefficients, and new support lies on the exceptional base divisor. The inequalities just proved apply to the recomputed data. $\square$

Corollary 3.4 (A zero boundary coefficient). *If an actual coefficient B_P in Proposition 3.3 is zero, some actual source component Q over P satisfies*

$$(3.9) \quad d_Q = 0, \quad m_Q = 1, \quad \frac{1 + r_Q}{m_Q} = \inf_{Q' \mapsto P} \frac{1 + r_{Q'}}{m_{Q'}}.$$

The same conclusion applies after the étale DVR extensions described there. It refers to the actual coefficient before any operation that assigns a new reduced-exceptional boundary coefficient.

Proof. Take Q attaining the first minimum in (3.5). The equality $B_P = 0$ makes $\lambda_P - t_P$ in (3.8) equal to one. Thus

$$1 \leq \frac{1 - d_Q}{m_Q} \leq 1.$$

Since m_Q is a positive integer and $d_Q \in \{0, 1\}$, this forces $m_Q = 1$ and $d_Q = 0$. Equality throughout also gives the final assertion in (3.9). $\square$

The component singled out by Corollary 3.4 will enter the ramification argument in Sections 6–7. That argument uses both its multiplicity one and its attainment of the all-valuation threshold. We first use the two order tests to identify the Hodge line and compare sections.

3.4. The metric identifies the Hodge contribution.

Proposition 3.5 (The Hodge order). *At every prime P of W , the parabolic order of the root form η is $\lambda_P - 1$. Consequently*

$$(3.10) \quad M \sim_{\mathbf{Q}} \sum_P (\lambda_P - 1)P, \quad T \sim_{\mathbf{Q}} B + M.$$

Proof. On the smooth family locus, finite change of variables gives

$$(3.11) \quad \|\eta_w\|_{\text{Hdg}}^2 = c_p \int_{J_w} |\omega_{J,w}|^{2/p},$$

where $c_p > 0$ is constant and the expression on the right is the canonical density associated with a pluricanonical form. In a local canonical frame, this density is the $2/p$ power of its coefficient times the usual canonical volume density.

Take a general point of P , with base coordinates $(u, s_2, \dots, s_l)$ and $P = (u = 0)$. Integrate (3.11) against base Lebesgue measure and the factor $|u|^{-2\gamma}$. Fubini and change of variables reduce local integrability to an integral on a log resolution of the total space. The factors from the absolute canonical density and from the base canonical frame cancel precisely as prescribed by $K_{X'/W} = K_{X'} - x^*K_W$. Thus, at a vertical prime Q , the monomial exponent in the resulting density is

$$r_Q - \gamma m_Q.$$

The transverse integral is finite exactly when $r_Q - \gamma m_Q > -1$. Horizontal primes have nonnegative ordinary order and cause no further condition. A finite SNC resolution computes all these conditions, so the supremal integrability exponent is

$$(3.12) \quad \sup \left\{ \gamma : |u|^{-2\gamma} \|\eta\|_{\text{Hdg}}^2 \in L_{\text{loc}}^1 \right\} = \inf_{Q \rightarrow P} \frac{1 + r_Q}{m_Q} = \lambda_P.$$

Here local integrability is measured against base Lebesgue measure. The value at the endpoint itself is irrelevant to this supremum.

We compare this with the parabolic extension. After a transverse root coordinate killing the finite part of the monodromy, a nonvanishing canonical extension frame has squared Hodge norm bounded above and below by powers of $-\log |u|$. This is the one-variable nilpotent-orbit growth estimate for a polarized pure variation [44, §§4 and 6]. Accordingly, a rational section of parabolic order a has squared norm

$$|u|^{2a} (-\log |u|)^{O(1)},$$

where the notation means two-sided bounds by fixed powers of the logarithm. Its supremal exponent in (3.12) is $1 + a$.

There is a meromorphic coefficient to which this statement applies. The form η is an algebraic section of the top line on the family locus. One may use the regular-singular extension of the geometric Gauss–Manin system. Equivalently, the finiteness of the threshold in (3.12) gives a weighted L^2 bound for its holomorphic coefficient in a nonvanishing extension frame. Absorb the logarithmic factors into an arbitrarily small power of the radius. The Laurent-series integral test then bounds the negative exponents of that coefficient, excluding an essential singularity and proving meromorphic extension. The finite root coordinate gives a rational parabolic order downstairs. Clearing its denominator by a tensor power, and applying the codimension-one extension test, gives the divisor of the rational line M .

Comparing with (3.12) proves $a = \lambda_P - 1$. There is no additional factor of p : the normalized order r_Q already contains $1/p$, and the density in (3.11) uses the same normalization. Finally $t_P = B_P + (\lambda_P - 1)$ at every prime, which proves (3.10). The transverse root coordinate has only been used to compute the metric order; no section system has been descended from a ramified alteration. $\square$

3.5. Preparation and transfer of sections. We first record the birational preparation needed to avoid an uncontrolled codimension-two image in a section comparison.

Lemma 3.6 (Extracting source divisorial valuations). *Let T_* be a fixed smooth projective model of the source of a dominant rational map to a smooth projective variety R . After resolving the map, making a birational modification of R , and resolving the source again, we can arrange that every source divisor whose image in the new base has codimension at least two is exceptional over T_* . Further smooth birational preparations of the base and compatible source resolutions preserve this assertion.*

If the maps also commute with maps to a fixed base Z , the same assertion holds on their very general fibers over Z .

Proof. Resolve the rational map, obtaining a morphism $T_1 \rightarrow R$. Put $e = \dim T_1 - \dim R$. A prime divisor of T_1 with image of codimension at least two lies in the closed locus where the fiber dimension is at least $e + 1$. There are only finitely many divisorial components in that locus. Consider those not exceptional over T_* .

For each such prime, its divisorial valuation v on $K = \mathbf{C}(T_1)$ has nontrivial restriction w to $L = \mathbf{C}(R)$. The latter is a discrete rank-one valuation. We justify that it is divisorial. Residue elements algebraically independent over $\kappa(w)$ lift to elements algebraically independent over L :

in a proposed relation, divide by a coefficient of least valuation and reduce to the residue field. Hence

$$\mathrm{trdeg}_{\kappa(w)} \kappa(v) \leq \mathrm{trdeg}_L K.$$

Since v is divisorial, this gives

$$\mathrm{trdeg}_{\mathbf{C}} \kappa(w) \geq (\dim T_1 - 1) - (\dim T_1 - \dim R) = \dim R - 1.$$

Conversely, lifts of algebraically independent residue elements for w , together with an element of positive valuation, are algebraically independent over $\mathbf{C}$. Thus $\mathrm{trdeg}_{\mathbf{C}} \kappa(w) \leq \dim R - 1$. Equality follows.

Choose a residue transcendence basis and lift it to units of L , together with an element generating the value group of w . They give a purely transcendental subfield of L over which L is finite. Normalize the corresponding model in L and take the center of w above the DVR cut out by that positive-valuation parameter. This realizes w as a prime divisor on a birational model of R . Extract all the finitely many restrictions in this way and dominate the result by a smooth projective model. On a compatible smooth source model, each of the original nonexceptional divisors now has divisorial image. Any new source prime is exceptional over T_* . A further birational base modification preserves the strict transform of each extracted valuation divisor, proving persistence.

For the last assertion, shrink a nonempty open in Z so that the finitely many relevant loci, their images, and their boundary strata have the expected fiber dimensions. A prime divisor contracted to codimension at least two on a fiber lies in the restriction of the fixed excess-fiber-dimension locus. A global component of codimension at least two dominating Z still has codimension at least two on a general fiber. Thus a divisorial component on a general source fiber comes from a global divisorial component of that locus, possibly after splitting. Its image over T_* has codimension at least two, and the same dimension calculation makes its image exceptional on the general fiber of T_* . This proves the fiberwise assertion. $\square$

Proposition 3.7 (The Iitaka-fiber section bound). *The models in Proposition 2.7 can be chosen, with the divisors of Proposition 3.3 recomputed on them, so that*

$$(3.13) \quad \kappa(W_z, K_{W_z} + B|_{W_z} + M|_{W_z}) \leq 0$$

for very general $z \in Z$. The comparison is a comparison with the section system on the fixed fiber $X_{*,z}$ in (3.1).

Proof. Apply Lemma 3.6 with fixed source X_* and base W . Retain the maps to Y and Z , resolve the finite boundary supports, and recompute (3.5). By the lemma, every divisor of a very general fiber X'_z whose image under x_z has codimension at least two is exceptional over $X_{*,z}$.

We also shrink the good open in Z for the finitely many supports and resolutions used in the divisorial calculation. The tests defining T then restrict to W_z . To see this explicitly, over the smooth good locus of W those tests are zero. Outside it, only restrictions of the finitely many divisors dominating Z can meet a general W_z in a divisor. Generic smoothness and adjunction preserve the multiplicities and relative-canonical orders there, with splitting of components allowed. Loci of higher codimension stay of higher codimension after the shrinking just made. The same reasoning on a fixed log resolution computes all the higher-valuation inequalities.

Write $T_z = T|_{W_z}$. For a sufficiently divisible integer m , take

$$\sigma \in H^0(W_z, m(K_{W_z} + T_z)).$$

Using the relative canonical identification, multiply its pullback by $\omega_J^{m/p}$. This is a rational m -canonical form on X'_z . If Q is an actual source divisor with divisor image P , its normalized

order, after allowing the source boundary, is at least

$$m(r_Q - m_Q t_P + d_Q) \geq 0$$

by (3.7). A horizontal divisor passes the ordinary test since the generic fiber generator is regular. All other untested divisors are exceptional over $X_{*,z}$.

Push the rational form to $X_{*,z}$. At every prime of that smooth fixed fiber, its strict transform has just passed the required logarithmic test; hence the pushdown belongs to

$$H^0(X_{*,z}, m(K_{X_{*,z}} + D_*|_{X_{*,z}})).$$

Possible poles on a divisor exceptional over $X_{*,z}$ have no bearing on this codimension-one test downstairs. The construction is injective. More precisely, the common factor $\omega_J^{m/p}$ cancels in ratios, so ratios of two input sections pull back by the dominant map x_z and their rational-map dimension is preserved. The target system has Iitaka dimension zero by (3.1). The identity $T_z \sim_{\mathbf{Q}} B|_{W_z} + M|_{W_z}$ from Proposition 3.5 proves (3.13). $\square$

3.6. Adjoint positivity for the restricted variation. The section bound is now available on a very general h -fiber. On that fiber, $K+B$ has a nonzero divisible section because $B \geq D_W^0$. We shall use adjoint positivity to show that any positive-dimensional adjoint base for the restricted Hodge line would contradict the bound for $K+B+M$. The following is the full companion input; its projective form follows immediately after.

Theorem 3.8 (Full-period adjoint comparison). *Let B be a smooth compact connected manifold in Fujiki class $\mathcal{C}$, and let $B^\circ \subset B$ be a dense Zariski open whose complement is an SNC divisor. Let $\mathbb{V}$ be a real-polarizable pure variation of Hodge structures on B° with an integral lattice. Suppose a complex direct summand of $\mathbb{V}_{\mathbf{C}}$ has a highest nonzero Hodge filtration step of rank one, with rational parabolic extension L . There exist a modification $\tau : B' \rightarrow B$, a connected-fiber surjective morphism $p : B' \rightarrow R$ to a smooth projective variety, and a nef rational line bundle A on R such that*

$$\tau^*L = p^*A \quad \text{in } \text{Pic}(B') \otimes \mathbf{Q}, \quad K_R + cA \text{ is big for some } c \in \mathbf{Q}_{>0}.$$

*Positive rational rescaling of L is allowed, and the line identity persists under further modifications. A point is allowed for R ; then τ^*L is rationally trivial.*

Proof. This is [37, Theorem 3.1], whose proof uses the full period image and boundary numerical-rank loss. The statement permits a complex summand of an integral ambient variation; it is stronger in this respect than a theorem requiring that summand itself to carry an integral lattice. $\square$

Proposition 3.9 (Projective adjoint comparison). *Let P be smooth connected projective, let D be reduced SNC, and let L be the rational parabolic extension of the entire highest Hodge line of a polarizable integral pure variation on $P \setminus D$. There are a birational morphism $\mu : P' \rightarrow P$ with P' smooth projective, a surjective morphism $a : P' \rightarrow R$ with connected fibers to a smooth projective variety, and a nef rational line bundle A on R such that*

$$(3.14) \quad \mu^*L \sim_{\mathbf{Q}} a^*A, \quad K_R + bA \text{ is big for all sufficiently large rational } b.$$

*The point quotient is allowed, in which case μ^*L is rationally trivial.*

Proof. The hypotheses meet Theorem 3.8 directly. A projective manifold belongs to class $\mathcal{C}$, a rational polarization is a real polarization, and the entire complexification of the given variation is a complex direct summand of itself. Its highest nonzero filtration step is the specified rank-one line. Thus the companion supplies $\tau : B' \rightarrow P$, $p : B' \rightarrow R$, A and $c > 0$ with

$$(3.15) \quad \tau^*L \sim_{\mathbf{Q}} p^*A, \quad K_R + cA \text{ big.}$$

It remains to obtain a projective source model without changing this identity or the connected-fiber property. The map $P \dashrightarrow R$ induced by p is meromorphic between projective varieties and hence rational: its graph closure is algebraic. Resolve that graph projectively to obtain a smooth projective P' and morphisms

$$\mu : P' \rightarrow P, \quad a : P' \rightarrow R.$$

On a common smooth modification of B' and P' , the two maps to R agree. Pulling back (3.15) there therefore identifies the pullbacks of μ^*L and a^*A .

Pullback by a proper birational morphism $v : T \rightarrow P'$ is injective on rational line bundles. Indeed, if a line bundle N pulls back to a torsion line bundle, some positive power v^*N^m is trivial. Normality of P' gives $v_*\mathcal{O}_T = \mathcal{O}_{P'}$, and the projection formula then gives $N^m \simeq \mathcal{O}_{P'}$. Applying this observation to the common modification proves

$$\mu^*L \sim_{\mathbf{Q}} a^*A.$$

The generic fibers of a are connected as well. The original map p has connected fibers and normal target, so $\mathbf{C}(R)$ is relatively algebraically closed in $\mathbf{C}(B') = \mathbf{C}(P)$. The same field extension occurs for a . Its Stein factor is consequently finite birational over the normal variety R , and hence is R itself. Thus all fibers of a are connected.

Finally A is nef. For every rational $b \geq c$, $K_R + bA = (K_R + cA) + (b - c)A$ is big. If R is a point, the line identity already gives rational triviality; the zero-dimensional bigness statement uses the usual point convention. This proves the proposition. $\square$

Remark 3.10. The conclusion is stable under further smooth projective birational modification $\pi : R' \rightarrow R$ and a compatible modification of the source. The parabolic pullback property preserves the line-bundle identity and nefness, while

$$K_{R'} + b\pi^*A = \pi^*(K_R + bA) + E_\pi, \quad E_\pi \geq 0,$$

preserves bigness. This is the form used when preparing the later section comparisons.

3.7. Rational triviality on the Iitaka fiber. The following weak-positivity consequence specifies the section arithmetic used in the argument.

Lemma 3.11 (Paying for a weak-positivity twist). *Let $a : T_1 \rightarrow R$ be a surjective morphism between smooth connected projective varieties. Let Δ be an effective rational SNC boundary with coefficients in $[0, 1]$, and let $N \sim_{\mathbf{Q}} a^*A$ for a nef rational divisor A on R . Suppose*

$$\kappa(T_1, K_{T_1} + \Delta) \geq 0, \quad K_R + cA \text{ is big}$$

for a positive rational number c . Then

$$(3.16) \quad \kappa(T_1, K_{T_1} + \Delta + cN) \geq \dim R.$$

Proof. Use Lemma 3.6 with fixed source T_1 and base R . On the resulting smooth source T_2 , let Δ_2 be the strict transform of Δ plus the reduced exceptional divisor, and pull back N and A ; write $N_2 = \pi^*N$ for the former pullback. The SNC discrepancy inequality gives

$$K_{T_2} + \Delta_2 = \pi^*(K_{T_1} + \Delta) + E_{\text{exc}}, \quad E_{\text{exc}} \geq 0 \text{ exceptional over } T_1.$$

The section systems in the assertion therefore have the usual pullback and pushdown comparison. For a smooth birational base map $\rho : R' \rightarrow R$,

$$K_{R'} + c\rho^*A = \rho^*(K_R + cA) + K_{R'/R}$$

is still big, because $K_{R'/R}$ is effective. Relabel the prepared base as R , its divisor as A , and its source morphism as a_2 .

Choose a sufficiently divisible positive integer m for which the relative direct image

$$\mathcal{E}_m = (a_2)_*\mathcal{O}_{T_2}(m(K_{T_2/R} + \Delta_2))$$

has positive rank. Such a choice is possible by restriction of a nonzero section of a multiple of $K_{T_1} + \Delta$ to the generic fiber, followed by pullback. The pair (T_2, Δ_2) is log canonical, T_2 is projective, and the base is smooth projective. Thus [19, Theorem 1.1] makes $\mathcal{E}_m$ weakly positive.

Put $L = K_R + cA$ and fix an ample integral divisor H on R . Increase m to clear the denominators needed below. Since L is big, choose a positive integer α such that

$$\alpha mL - H \text{ is big.}$$

The definition of weak positivity [19, Definition 7.2] supplies a positive integer β for which

$$(3.17) \quad (\text{Sym}^{\alpha\beta} \mathcal{E}_m)^{**} \otimes \mathcal{O}_R(\beta H) \text{ is generically generated by global sections.}$$

At the generic point of R , multiplication of relative sections is not the zero map: the power of any nonzero section on the integral generic fiber is nonzero. Generic generation in (3.17) therefore yields a global section whose evaluation is a nonzero rational section u of

$$\alpha\beta m(K_{T_2/R} + \Delta_2) + \beta a_2^* H.$$

At every codimension-one point of R , the direct image is locally free and the reflexive symmetric power agrees with its ordinary symmetric power. Evaluation is regular above those points. Any remaining pole divisor of u thus has image of codimension at least two and is exceptional over the fixed source T_1 .

For a sufficiently divisible positive integer q , multiply u^q by the pullbacks of all sections of

$$q\beta(\alpha mL - H).$$

The resulting rational sections lie in the class

$$\begin{aligned} & q\alpha\beta m(K_{T_2/R} + \Delta_2) + q\beta a_2^* H + q\beta a_2^*(\alpha mL - H) \\ & = q\alpha\beta m(K_{T_2} + \Delta_2 + cN_2). \end{aligned}$$

They are regular at every prime not exceptional over T_1 , and so push down to the required absolute section system on T_1 . Their ratios are the pullbacks of the ratios of sections of the big divisor $\alpha mL - H$. These ratios give a rational map of dimension $\dim R$, proving (3.16). $\square$

Proposition 3.12 (Triviality on the Iitaka fiber). *For very general $z \in Z$,*

$$(3.18) \quad M|_{W_z} \sim_{\mathbf{Q}} 0, \quad \kappa(W_z, K_{W_z} + B|_{W_z}) = 0.$$

These assertions also hold on the geometric generic h -fiber.

Proof. Set $T_1 = W_z$, $\Delta = B|_{W_z}$, and $N = M|_{W_z}$. For the chosen z , the fiber is smooth and all relevant boundary strata meet it transversely, including those of the enlarged divisor defining the domain of the variation. The log-pair pullback property in Lemma 3.2 therefore identifies N with the parabolic extension of the restricted variation. Moreover

$$\kappa(T_1, K_{T_1} + \Delta) \geq 0$$

by (3.1) and $B \geq D_W^0$. In particular, $D = K_{T_1} + \Delta$ is rationally equivalent to an effective rational divisor.

Apply Proposition 3.9 to the restricted integral polarized variation whose *entire* highest piece is this line. The comparison is applied directly to this restricted variation; its entire complexification is an allowed summand in the companion theorem. On a higher smooth model it gives

$$N \sim_{\mathbf{Q}} a^* A, \quad a : T_1 \longrightarrow R, \quad A \text{ nef}, \quad K_R + cA \text{ big for } c \gg 1.$$

On the higher source model use strict transform plus reduced exceptional boundary for Δ . The logarithmic section comparison preserves (3.13) and the rational effectivity of D ; the Hodge line is pulled back. If $\dim R > 0$, choose a sufficiently large rational $c \geq 1$. Lemma 3.11 gives

$$\kappa(T_1, D + cN) \geq \dim R > 0.$$

But

$$(3.19) \quad D + N = \frac{1}{c}(D + cN) + \left(1 - \frac{1}{c}\right) D.$$

After clearing denominators, multiplication by the section of the effective last summand preserves rational-map dimensions. Thus the same positive lower bound holds for $D + N$, contradicting (3.13). We conclude that R is a point and $N \sim_{\mathbf{Q}} 0$. Rational triviality descends through the intervening birational morphisms, since their direct image of the structure sheaf is the structure sheaf. Now (3.13) and $\kappa(D) \geq 0$ give $\kappa(D) = 0$.

Finally, these algebraic assertions pass to the geometric generic fiber. For the Kodaira dimension assertion, use generic base change in the countably many divisible degrees. For rational triviality, choose an integer clearing the denominator of M . For each positive integer j , simultaneous nonvanishing of sections of the j th power of that line and of its inverse is a closed condition on a smooth proper family, and is equivalent to triviality on a connected projective fiber. The countable union of these loci contains every very general z by what we proved. Hence some one of them contains a dense open of Z ; otherwise a very general point would avoid them all. Generic base change gives the required rational triviality on the geometric generic fiber. $\square$

3.8. Flatness and finite character. Rational triviality gives degree zero on every complete curve in an Iitaka fiber. The curvature calculation below converts those zero degrees into a flat highest line; finite character is then a statement about the integral variation restricted to that algebraic locus.

Corollary 3.13 (Degree zero on a curve). *In Lemma 3.2, suppose that P is a smooth projective curve. If $\deg L = 0$, the projective highest-line map is constant on the connected universal cover of $P \setminus D$. Equivalently, L° is a flat line subbundle.*

Proof. Pass to a finite cover making the boundary monodromies unipotent. The extension pulls back, and its degree remains zero. Let α be the highest-line Hodge curvature. Griffiths' curvature formula makes it semipositive and identifies its kernel with the kernel of the projective line differential.

We check that its integral is the degree of the extension. For a smooth reference curvature β , write $\alpha - \beta = dd^c u$. Near a puncture, with $\rho = -\log |z|^2$, the Hodge norm and horizontal Schwarz estimates give

$$|u| \leq C(1 + \log \rho), \quad 0 \leq \alpha \leq C \frac{i dz \wedge d\bar{z}}{|z|^2 \rho^2};$$

see [14, Theorem 5.1] and [10, Proposition 2.4 and Lemma 2.5]. Use products of cutoffs $\chi((\log \rho)/N)$, equal to one away from a shrinking puncture neighborhood. Their second derivatives are uniformly bounded in the cusp metric and supported on escaping tails. Integration by parts bounds the error by the tail integral of $(1 + \log \rho)\rho^{-2} d\rho$, which tends to zero. Thus $\int_{P \setminus D} \alpha = \deg L = 0$. A continuous nonnegative curvature form with zero integral vanishes. Its kernel identity proves constancy on the cover and therefore on the original universal cover. $\square$

Corollary 3.14 (Finite character of a line). *Let $\mathbb{V}$ be a polarized integral variation of Hodge structures on a connected smooth complex algebraic variety. Every rank-one complex local subsystem of $\mathbb{V}_{\mathbf{C}}$ has finite monodromy.*

Proof. Deligne's determinant theorem [15, Corollary 4.2.8(iii)(b)] states that a positive tensor power of the determinant of any complex local subsystem is trivial. For a rank-one subsystem, the determinant is the subsystem itself. Its character therefore takes values in a fixed finite group of roots of unity. The theorem applies anew to the pulled-back variation on any connected smooth algebraic locus. $\square$

Corollary 3.15 (Restriction and finite character). *Let T be a connected smooth complex algebraic variety mapping into the domain of a polarized integral pure variation whose entire highest Hodge step is a line. If the pulled-back projective highest-line map is constant, its highest line has finite monodromy character.*

Proof. The pulled-back variation is again polarized, integral and pure. Constancy of its projective highest line makes that line a rank-one complex local subsystem. Apply Corollary 3.14 to this restricted variation. No common exponent for different algebraic loci is needed. $\square$

Corollary 3.16 (Flatness and finite character on the Iitaka fiber). *For very general $z \in Z$, the projective entire top line of the restricted root-cover variation is constant on the connected good locus of W_z , and its top-line character has finite image.*

Proof. Put $T_1 = W_z$ and $N = M|_{W_z}$. Proposition 3.12 gives $N \sim_{\mathbf{Q}} 0$, and the pullback property in Lemma 3.2 identifies N with the parabolic extension of the restricted entire top line. The degree of N on every complete smooth test curve is zero. Corollary 3.13 makes the projective top line constant along such curves. Complete-intersection curves through general points and tangent directions give projective constancy on the connected good locus of T_1 . Corollary 3.15, applied to this restricted integral polarized variation, gives finite monodromy for its original top line. $\square$

3.9. Alternative arguments for the Hodge line.

Remark 3.17 (Tensor comparison on restrictions). The direct comparison above was applied to the restricted variation. A separate integral tensor replacement, which makes projective monodromy faithful, is proved in [38, Lemma 8.4]; it gives the independent adjoint comparison in [38, Theorem 8.1]. On a connected algebraic restriction, that tensor construction must be applied anew to the pulled-back integral variation. A replacement faithful for the ambient moving line need not remain faithful after restriction, so the monodromy and rational isotypic calculation must be repeated on the restricted locus. Its positive tensor exponent may depend on that locus; no common exponent is required. Thus it gives an alternative adjoint comparison on each restriction. The proof here uses the direct comparison and Corollary 3.15.

There is also a direct metric proof of the finite-character conclusion in Corollary 3.16. It uses the rational triviality from Proposition 3.12, in addition to zero curvature. Put $T_1 = W_z$ and $N = M|_{W_z}$ on the prepared smooth projective model. The constant projective top line on its connected good open is a flat line. Its polarization, with the fixed Hodge sign, is a positive flat metric, so its monodromy character χ is unitary. Quasi-unipotence of the integral ambient variation makes the characters around the finitely many SNC boundary components roots of unity [44].

Choose $e > 0$ killing these local characters and clearing the parabolic weights. Then χ^e extends across the boundary to a unitary flat line on T_1 ; its holomorphic line bundle is the parabolic tensor power eN . Since $N \sim_{\mathbf{Q}} 0$, a further positive tensor power is holomorphically trivial. For a nowhere-zero holomorphic section of that power, the logarithm of its norm is a globally defined pluriharmonic function. Compactness makes this function constant. In a local flat unitary frame the section is holomorphic of constant absolute value, hence constant, so it is parallel. A positive power of χ is therefore trivial. This recovers finiteness on this restricted locus without requiring a common exponent for different loci.

4. A BIRATIONAL CONSTANCY CRITERION

Section 3 made the entire highest line flat, with finite character, along the relevant Iitaka fibers. We now turn this Hodge-theoretic information into birational constancy. The argument is first

carried out over a curve, where a single lifted vector field can be analyzed, and then spread to arbitrary algebraic parameter loci.

Theorem 4.1 (Birational constancy over a curve). *Let $\pi^\circ : P^\circ \rightarrow C^\circ$ be a smooth projective morphism with connected fibers, where C° is a smooth connected complex algebraic curve. Suppose that its very general fiber D satisfies*

$$\kappa(D) = 0, \quad h^0(D, K_D) = 1.$$

Put $n = \dim D$ and assume that the entire top Hodge bundle $F^n(R^n\pi_^\circ \mathbf{C} \otimes \mathcal{O}_{C^\circ})$ is a flat line with finite monodromy character. Then there are a finite extension $L/\mathbf{C}(C^\circ)$ and a smooth connected projective variety $D_0/\mathbf{C}$ such that the generic fiber after extension to L is birational to $(D_0)_L$.*

If a finite group Γ acts on P° over C° , one can choose D_0 with a Γ -action and choose this birational identification Γ -equivariantly. The same conclusion holds for a finite fiberwise birational action, after replacing the family by an equivariant smooth projective model.

The proof lifts a base direction to a meromorphic vector field on the total space. We first control this field along the zero divisor of a global top form. A singular-metric estimate then shows that every pole is a fixed divisor of a small ample perturbation of the canonical system. Contracting those divisors makes the vector field regular; its flow identifies nearby fibers. Finite changes of the curve and birational changes of the total space over a dense open do not change the asserted conclusion. A zero-dimensional connected smooth fiber is a point, so throughout the construction we assume $n > 0$.

In the log-smooth Kähler setting of Cao–Guenancia–Păun, with klt boundary and numerically trivial fiberwise log-canonical class, flatness of the pluricanonical direct image in its Narasimhan–Simha metric implies analytic local triviality of the pairs over the smooth locus [12, §1.1, Theorem 1.2]. Here the flat line is the entire top Hodge line, with finite character, and the fibers satisfy $\kappa(D) = 0$ and $h^0(D, K_D) = 1$. The top form may have zeros, allowing poles in the lifted base vector field; contracting its possible poles yields the birational conclusion.

4.1. A global form and its polarization.

Lemma 4.2. *After a finite change of the curve there are a smooth projective curve C , a smooth projective variety P of dimension $n + 1$, and a projective morphism $\pi : P \rightarrow C$ whose restriction over a nonempty open is birational to the changed family, with the following properties. There is a closed holomorphic n -form Θ on P whose restriction to each fiber over that open spans its ordinary top-form space. Write*

$$s \in H^0(P, K_{P/C}), \quad \mathcal{E} = \operatorname{div}(s)$$

for its relative canonical section and zero divisor. The horizontal part of $\mathcal{E}$ is relatively simple normal crossing over a smaller open of C : the morphism and every horizontal zero-divisor stratum are smooth there. We call fibers over this open good fibers.

One can choose an ample line bundle H with a positive smooth metric, whose curvature form is denoted by h , so that, for a positive area class ℓ pulled back from C , there is a holomorphic n -form α satisfying

$$(4.1) \quad [h \wedge \Theta] = [\ell \wedge \alpha] \quad \text{in } H^{n+1,1}(P).$$

In the equivariant case P and H can be chosen equivariant, h is invariant, and Θ transforms by the character of the fiber top line. Moreover, P dominates a smooth projective model $P_0 \rightarrow C$ with reduced fibers, and Θ is the pullback of a holomorphic form on P_0 .

Identity (4.1) has two uses: it removes the obstruction to lifting the base vector field constructed below to H , and later converts a weighted integral on P into an integral over good

fibers. We obtain it by choosing the global lift of the flat top form; that choice will leave the relative form s and its zero divisor unchanged.

Proof. Kill the finite character by a finite cover of C° and complete the resulting curve. Semistable reduction over a curve, followed by the usual toroidal resolution, gives, after a further finite change, a smooth projective model $P_0 \rightarrow C$ with reduced fibers; see [30, Chapter II, semistable reduction over a curve]. Take a smooth projective common dominating model of this family and the original changed family. When Γ is present, include the finitely many translates in the graph construction and use an equivariant resolution. This also regularizes a finite birational action: take the closure of the graph of all its maps in the product of their projective models, on which the group permutes the factors, and resolve equivariantly. Over a dense smooth open, birational pullback preserves ordinary top forms and is compatible with the cohomology local systems. The chosen top class is therefore still invariant.

The flat generator already gives a relative top form on this smooth open. Choose a log resolution of its geometric generic zero divisor. That resolution and its finitely many geometric components are defined after a finite extension of the curve's function field. Make this change now and spread the resolution. Repeat semistable preparation at the boundary and take a smooth common dominating model, preserving the chosen resolution over a dense open. Thus the geometric divisor components used below are defined before the final P , H , and Θ are fixed.

The global invariant cycle theorem gives a surjection from the cohomology of the smooth compactification onto the invariant part of the fiber cohomology. The restriction map is a morphism of Hodge structures, so its strictness gives a preimage of the invariant $(n, 0)$ -class in $H^{n,0}(P)$; these are precisely holomorphic n -forms. Here we use [15, Theorem 4.1.1(ii) and Corollary 4.1.2], with smooth proper family over the good curve and smooth projective total compactification. Holomorphic forms on a smooth projective variety are closed.

To describe the relative section, choose a local coordinate t on C . The absolute form $dt \wedge \Theta$, divided by the base frame dt , defines s in $K_{P/C}$. These expressions agree under a change of coordinate on the curve. The section is holomorphic globally and is nonzero on every good fiber. Resolve its horizontal zero divisor and shrink the good open so that the resulting horizontal components and their intersections are smooth over the curve. This gives the stated relative normal crossings. Any two lifts of the fixed flat top class have the same restriction to every good fiber, and therefore give the same relative section. Thus the following change of lift does not alter s or this preparation of its zero divisor.

Put $V = H^0(P, \Omega_P^n)$ and let N be the kernel of restriction to a good fiber. This kernel is independent of the chosen good fiber: restriction of a global closed form gives a flat cohomology section, and a holomorphic top form represents the zero cohomology class only when it is zero. Choose an ample H and a positive metric, and set

$$B_h(\beta, \gamma) = i^{n^2} \int_P h \wedge \beta \wedge \bar{\gamma}.$$

This is a positive definite Hermitian pairing on V . Indeed its integrand is a positive multiple of the pointwise norm squared when $\beta = \gamma$, and a nonzero holomorphic form is nonzero on an open set. For a positive smooth area form ℓ from the curve, the pairing B_ℓ is semidefinite. Fiber integration identifies its kernel with N . Consequently the image of the map

$$V \longrightarrow \bar{V}^*, \quad \beta \longmapsto B_\ell(\beta, -)$$

is the annihilator of N .

Choose the lift Θ orthogonal to N for B_h . The functional $B_h(\Theta, -)$ annihilates N , so it equals $B_\ell(\alpha, -)$ for some $\alpha \in V$. The duality pairing of $H^{n+1,1}(P)$ with $H^{0,n}(P)$ is nondegenerate. Applying it to the difference proves (4.1).

In the equivariant case a tensor product of the translates of an ample bundle admits a natural Γ -linearization; its metric can be made invariant while keeping positive curvature. Both N and the pairing are then invariant. The unique orthogonal lift of the chosen top class transforms by its top-line character. The same character can be imposed on α by projection to its character space.

Finally, a holomorphic form on a smooth projective birational model is the pullback of a holomorphic form on every smooth projective model it dominates. Applying this to $P \rightarrow P_0$ gives the asserted descent of Θ . A later positive tensor power of H multiplies h by a positive integer and preserves all these conclusions, with α scaled accordingly. $\square$

On the complement of the zeros and the critical locus, the kernel of Θ is a holomorphic line transverse to the fibers. On a good base coordinate disk it has a unique generator v satisfying

$$(4.2) \quad d\pi(v) = \partial_t, \quad \iota_v \Theta = 0.$$

Here and below we use $\mathcal{E}$ for a divisor on the total space and $E = \mathcal{E}|_D$ for its divisor on a good fiber.

Lemma 4.3 (A meromorphic differential operator). *The vector field v extends meromorphically to a neighborhood of each general fiber, with poles bounded by $\mathcal{E}$. After shrinking the base disk, it is the symbol of a meromorphic first-order differential operator ∇ on H with the same pole bound.*

Proof. In a local relative canonical frame, write ω for the relative n -form represented by s . The normalization in (4.2) gives

$$(4.3) \quad \Theta = \iota_v(dt \wedge \omega).$$

Thus sv is a holomorphic section of $T_P \otimes K_{P/C}$: contraction with an absolute top form identifies this bundle, after choosing dt , with Ω_P^n . Dividing by s proves the pole bound.

Use the absolute scalar-symbol sequence

$$(4.4) \quad 0 \longrightarrow \mathcal{O}_P \longrightarrow \text{At}(H) \longrightarrow T_P \longrightarrow 0.$$

Its middle sheaf consists of first-order differential operators on H with scalar symbol. Its extension class is $c_1(H)$, up to the fixed normalizing constant; this is the connection obstruction of [3]. For the specific assertion needed here, take frames $e_i = g_{ij}e_j$. Local lifts of the same vector symbol differ by its contraction with $d \log g_{ij}$. This is the cocycle representing the stated extension class.

Tensor (4.4) by $K_{P/C}$. The obstruction to lifting sv is its contraction with $c_1(H)$ in $H^1(K_{P/C})$ on the neighborhood in question. In Dolbeault notation and using (4.3), its restriction to D_t is, up to a nonzero constant and sign, the class obtained from $h \wedge \Theta$ by dividing its top holomorphic degree by dt and restricting to D_t . This operation respects Dolbeault coboundaries. In fact the total space has dimension $n+1$, so a primitive of type $(n+1, 0)$ necessarily contains dt in submersion coordinates; division by dt commutes with $\bar{\partial}$ before restriction. Equivalently this is the canonical restriction $K_P|_{D_t} \simeq K_{D_t} \otimes T_t^*C$ with a base frame chosen.

The right side of (4.1) gives zero under this operation: a smooth area representative of ℓ contains $d\bar{t}$. Hence the fiber obstruction is zero for every good t . Shrink to a Stein disk on which cohomology and base change hold for $K_{P/C}$. Leray and the vanishing of higher coherent cohomology on the disk identify the obstruction with a section of $R^1\pi_*K_{P/C}$. It is zero because its fiber values are zero. Therefore sv lifts to a holomorphic section of $\text{At}(H) \otimes K_{P/C}$ on the whole neighborhood. Dividing this lift by s produces ∇ , with the claimed symbol and pole bound. Replacing H by a tensor power uses the induced operator and leaves its symbol and its pole bound unchanged. $\square$

4.2. Vanishing on the zero divisor. The lifted field may have poles along $\mathcal{E}$. We first prove that the absolute form vanishes on every component of this divisor. This will make its pairing with every positive curvature current of $K_{P/C}$ vanish, and supply the integral estimate used below.

Lemma 4.4. *The pullback of Θ to a resolution of every irreducible component of $\mathcal{E}$ is zero.*

Proof. We first treat a horizontal component. Choose a simultaneous very general good fiber D , so that all its positive pluricanonical section spaces have dimension one. Write

$$(4.5) \quad E = \sum_j e_j D_j, \quad R_0 = E + E_{\text{red}}, \quad S = D_i, \quad B' = \sum_{j \neq i} D_j.$$

The coefficients e_j are positive integers, $E \sim K_D$, and the support is simple normal crossing. The components extend relatively over a small disk. In the local computations we use the same letters for these relative extensions and their local equations.

The contradiction will take place in the section spaces of R_0 . Since

$$E \leq R_0 \leq 2E, \quad E \sim K_D,$$

for every positive integer m we have

$$1 \leq h^0(D, mR_0) \leq h^0(D, 2mE) = h^0(D, 2mK_D) = 1.$$

Thus every section of mR_0 is a multiple of its divisor section and vanishes on S . We will construct a section whose restriction to S is nonzero.

Suppose that the restriction of Θ to the horizontal component through S is nonzero. Let z and G be local equations for S and R_0 , and put

$$W = Gv, \quad u = (W(z)/z)|_S.$$

The pole bound in Lemma 4.3 shows that W is holomorphic and vanishes on the reduced zero divisor. It is consequently logarithmic along that divisor, so that both $W(z)/z$ and $W(G)/G$ are holomorphic. Under changes of the equations, u transforms as a section of $R_0|_S$. Formula (4.3) identifies its nonvanishing at the general point of S with the supposed nonvanishing of Θ on the relative component. Thus

$$(4.6) \quad 0 \neq u \in H^0(S, \mathcal{O}_S(R_0)).$$

Moreover u is divisible by the full reduced divisor $B'|_S$. To see this at a crossing, the coefficients of sv are holomorphic, while G/s has one factor for each component of E_{red} . Division by z removes the factor belonging to S and leaves all other reduced factors.

Write the operator in a regular frame of H as $\nabla = v + c$. Its pole bound also gives $Gc|_S = 0$. Replace H by a sufficiently high tensor power. Relative generation supplies finitely many sections a_λ on a neighborhood of the whole fiber, vanishing on the relative S , such that a_λ/z generate $H(-S)$ along S . For each λ set

$$q_{k,\lambda} = \frac{\nabla^k a_\lambda}{k!}, \quad b_{k,\lambda} = \frac{G^k \nabla^k a_\lambda}{z}.$$

The following recurrence proves in particular that the second expression is holomorphic:

$$(4.7) \quad b_{k+1,\lambda} = W(b_{k,\lambda}) + \left(\frac{W(z)}{z} - k \frac{W(G)}{G} + Gc \right) b_{k,\lambda},$$

$$(4.8) \quad b_{k,\lambda}|_S = u^k (a_\lambda/z)|_S \prod_{j=0}^{k-1} (1 - j(e_i + 1)).$$

For the second equality, W vanishes on S , $Gc|_S = 0$, and $(W(G)/G)|_S = (e_i + 1)u$. The other components of G contribute zero on S ; this can be checked away from their intersections and

then extended holomorphically. These facts prove both formulas by induction. They also show that the restrictions of $q_{k,\lambda}$ are sections of $\mathcal{O}_D(kR_0 + H - S)$. More explicitly, in a local frame of H ,

$$q_{k,\lambda} = \frac{z}{G^k} \frac{b_{k,\lambda}}{k!}.$$

The meromorphic expression on the left is therefore a holomorphic section of the indicated twisted bundle, with local coefficient $b_{k,\lambda}/k!$.

We need estimates as k tends to infinity. Cover a compact neighborhood of D by finitely many pairs of nested coordinate charts. For a fixed k , divide each nesting margin into k equal parts. On these charts the coefficients of W , $W(z)/z$, $W(G)/G$, and Gc are bounded. At the j th step of (4.7), Cauchy's estimate for the derivative costs at most a constant times k , and the zeroth-order coefficient costs at most a constant times $1 + j \leq k + 1$. It follows that the holomorphic coefficients $b_{k,\lambda}/k!$, in smooth metrics for $kR_0 + H - S$, are bounded by

$$(4.9) \quad \frac{(Ck)^k}{k!} \leq C_1^k.$$

This is a bound in the indicated twisted bundle, including across the zero divisor.

Put $a = e_i + 1 \geq 2$. No factor in (4.8) is zero, and

$$\lim_{k \rightarrow \infty} \left| \frac{\prod_{j=0}^{k-1} (1 - ja)}{k!} \right|^{1/k} = a > 0;$$

for example, the ratio of successive absolute values tends to a . On D , take the weights

$$\frac{1}{k} \log \sum_{\lambda} \left| \frac{b_{k,\lambda}}{k!} \right|^2 - \frac{1}{k} \varphi_{H-S},$$

where φ_{H-S} is a smooth weight on $H - S$ in the same local frames. They are weights on R_0 with curvature loss tending to zero and uniform upper bounds by (4.9). Their regularized upper limit is a semipositive singular weight. Formula (4.8), together with generation of $H(-S)|_S$, gives its restriction a lower bound by $\log |u|^2$ up to a constant. Finally take the maximum with the divisor weight of R_0 . We obtain a semipositive weight φ on R_0 satisfying

$$(4.10) \quad \varphi \geq \log |s_{R_0}|^2, \quad \varphi|_S \geq \log |u|^2 - C.$$

All weights here are written in compatible local frames; the inequalities are invariant statements about the corresponding metrics.

We now extend a nonzero power of u using [18, Theorem 4.1, conditions (19)–(23)]. We record every hypothesis in this application. The ambient manifold is the smooth projective D , and its smooth hypersurface is S . For any integer $m \geq 2$ put

$$(4.11) \quad \begin{aligned} F' &= mR_0 - (K_D + S) \sim (m-1)R_0 + B', \\ \varphi_{F'} &= (m-1)\varphi + \log |s_{B'}|^2, \\ A_0 &= R_0 - aS = \sum_{j \neq i} (e_j + 1)D_j, \\ \varphi_S &= 1 + \frac{\varphi - \log |s_{A_0}|^2}{a}. \end{aligned}$$

The first line uses $K_D \sim E$ and $E_{\text{red}} = S + B'$. Thus F' is an actual line bundle with the indicated weight, after the indicated line-bundle identification.

Condition (19) of the extension theorem holds with its constant equal to one, since (4.10) implies

$$|s_S|^2 e^{-\varphi_S} \leq e^{-1}.$$

For condition (20), the weights $1 + \varphi/a$ and $a^{-1} \log |s_{A_0}|^2$ are semipositive weights on rational line bundles and have difference φ_S . If integral bundles are used, choose an integral G_2 sufficiently ample that $G_2 - A_0/a$ has a positive smooth rational-bundle metric, and add this metric to both weights. The difference is unchanged and both underlying bundles become integral.

For condition (21), let $T_R = dd^c \varphi$. Directly,

$$(4.12) \quad \begin{aligned} dd^c \varphi_{F'} &= (m-1)T_R + [B'] \geq 0, \\ dd^c \varphi_{F'} - dd^c \varphi_S &= \left(m-1 - \frac{1}{a}\right) T_R + [B'] + \frac{1}{a} [A_0] \geq 0. \end{aligned}$$

Here $a \geq 2$ and $m \geq 2$. For condition (22), choose a positive number

$$\varepsilon_0 \leq \min \left\{ 1, \min_{j \neq i} \frac{a}{e_j + 1} \right\},$$

omitting the inner minimum if there are no other components. The weight φ is locally bounded above. After normalizing local defining functions to have norm at most one, each coefficient of $\log |s_{B'}|^2$ is at least the corresponding coefficient in $\varepsilon_0 a^{-1} \log |s_{A_0}|^2$. Therefore

$$\varphi_{F'} \leq \frac{\varepsilon_0}{a} \log |s_{A_0}|^2 + C.$$

A common smooth metric added to the two weights above changes this inequality only by a bounded local term. This proves precisely the required domination of the subtracted weight's singularities.

The section to extend is

$$u^m \in H^0(S, \mathcal{O}_S(mR_0)) = H^0(S, K_S + F'|_S).$$

The lower bound for $\varphi|_S$ and $u \neq 0$ also ensure that the restricted weights are not identically minus infinity. Condition (23) follows from the second bound in (4.10):

$$(4.13) \quad |u|^{2m} e^{-\varphi_{F'}|_S} \leq C \frac{|u|^2}{|s_{B'}|_S^2}.$$

The quotient on the right is locally bounded because u is divisible by the full reduced divisor $B'|_S$. There is in fact integrability with a slightly larger weight exponent. Writing $u = s_{B'} w$, the density with exponent $1 + \delta$ on $\varphi_{F'}$ is bounded by

$$C |s_{B'}|^{-2m\delta} |w|^{2-2(m-1)\delta}.$$

It is integrable for $0 < \delta < 1/m$, by the normal crossings of $B'|_S$ and the positive exponent on $|w|$. This argument requires no normal-crossing assertion about the other zeros of w .

The extension theorem now gives a section of mR_0 with restriction $u^m \neq 0$, contradicting the one-dimensionality of this section space established above: its divisor section vanishes on S . This proves the horizontal vanishing.

For vertical components, use the reduced-fiber model $P_0 \rightarrow C$ in Lemma 4.2, to which Θ descends. At a general point of a reduced fiber component the map has local equation $t = z$. There the vanishing of the relative coefficient s is exactly the vanishing of the restriction of Θ to that component. Every vertical component on P which is not exceptional over P_0 is tested at the same generic point. An exceptional divisor maps into a subset of P_0 of dimension at most $n-1$, and the pullback of an n -form to its resolution is zero. These observations prove the remaining vanishing assertions. They also explain why the reduced-fiber model is used: for $t = z^r$ with $r > 1$, an additional differential factor z^{r-1} would not itself imply vanishing of the absolute form. $\square$

4.3. Curvature pairing and an integrability estimate. We have proved that Θ vanishes on every component of $\mathcal{E} = \text{div}(s)$. Put $\mu = i^{n^2} \Theta \wedge \bar{\Theta}$. The resulting identity $[\mathcal{E}] \wedge \mu = 0$ will control singular metrics on $K_{P/C}$, and hence the poles of the normalized kernel field v .

For a semipositive weight T on $K_{P/C}$, the curvature pairing with μ has the same integral as $[\mathcal{E}] \wedge \mu$ and therefore vanishes. When $T \geq \log |s|^2$, applying this observation to truncations shows that $\psi = T - \log |s|^2$ is constant along almost every local kernel leaf. This permits the weighted use of (4.1) and gives an integrability exponent strictly greater than one. We prove the estimate for arbitrary T before constructing the weight supplied by pluricanonical section spaces.

We use weights for metrics written as $e^{-\varphi}$, and the convention

$$dd^c = \frac{i}{2\pi} \partial \bar{\partial}, \quad dd^c \log |z|^2 = [z = 0].$$

In particular, if φ_H is a local weight for the metric on H , then $dd^c \varphi_H = h$. Expressions such as $\log |s|^2$ below denote local weights in holomorphic frames; the difference of two weights on the same bundle is a globally defined function wherever both are finite.

Lemma 4.5. *Let T be a semipositive singular weight on $K_{P/C}$ such that $T \geq \log |s|^2$, and put $\psi = T - \log |s|^2 \geq 0$ off $\mathcal{E}$. There exists $\delta > 0$ such that*

$$(4.14) \quad \int_P e^{(1+\delta)\psi} i^{n^2} h \wedge \Theta \wedge \bar{\Theta} < \infty.$$

Proof. The form μ is smooth, closed, and positive of type (n, n) . Lemma 4.4 implies that $[\mathcal{E}] \wedge \mu = 0$: the integral over each component can be computed on a resolution of that component, where the pullback of Θ vanishes. Since the curvature current of $\log |s|^2$ is $[\mathcal{E}]$, cohomology and closedness give

$$\int_P (dd^c T) \wedge \mu = \int_P [\mathcal{E}] \wedge \mu = 0.$$

Here and subsequently $dd^c T$ denotes the globally defined curvature current of the bundle weight. The measure on the left is positive, so it vanishes. For every real b , the weight

$$T_b = \max\{T, \log |s|^2 + b\}$$

is another semipositive weight on the same bundle. The same argument shows that $(dd^c T_b) \wedge \mu = 0$.

Delete $\mathcal{E}$, the zeros of Θ , and the bad-fibration locus. In a holomorphic flow box for the kernel of Θ , use coordinates $(w_1, \dots, w_n, \tau)$, where the leaves vary only in τ . Closedness gives

$$\Theta = a(w) dw_1 \wedge \dots \wedge dw_n, \quad a(w) \neq 0.$$

The weight $\log |s|^2$ is pluriharmonic here. Thus wedging the curvature of T , or of T_b , with μ selects the τ -Laplacian of ψ , or of $\max\{\psi, b\}$, against a nonvanishing transverse volume. Slicing these zero positive measures shows that both functions are harmonic on almost every local leaf. Take the intersection of these full-measure sets for rational b . If a harmonic function is nonconstant, there is a small leaf neighborhood on which its differential is nonzero and whose image contains an open real interval. Choose a rational b in that interval. Taking the maximum with b produces a nonzero positive Laplacian on the level crossing in that neighborhood. Hence ψ is constant on almost every local leaf.

In these coordinates $\bar{\Theta}$ contains all the transverse differentials $d\bar{w}_j$. Thus only a derivative in the leaf direction can contribute to $\bar{\partial}(\chi(\psi)\bar{\Theta})$, and leafwise constancy gives, for every bounded continuous function χ ,

$$\bar{\partial}(\chi(\psi)\bar{\Theta}) = 0$$

off the deleted analytic set, in the sense of distributions. This identity extends across that set. Indeed, the coefficients of this current are locally bounded. Across a smooth analytic stratum

of complex codimension $q \geq 1$, a cutoff at distance ϵ has derivative $O(\epsilon^{-1})$ on a tube of volume $O(\epsilon^{2q})$, so its contribution tends to zero. Applying this successively to the smooth strata, and choosing the tubes about the remaining lower-dimensional strata sufficiently small, proves the distributional extension. This argument uses boundedness of χ ; it makes no differentiability assumption on ψ .

We may now pair (4.1) with $\chi(\psi)\bar{\Theta}$. Choose the representative ℓ there to be the pullback of a nonnegative smooth area form supported in relatively compact good base charts. This does not change its cohomology class. On those charts, after shrinking within the good locus, fiberwise rank one gives

$$\alpha|_{D_t} = a(t)s|_{D_t}$$

with $a(t)$ bounded on the support in question. Dolbeault integration by parts against the closed current just constructed therefore gives

$$(4.15) \quad \int_P \chi(\psi) h \wedge \Theta \wedge \bar{\Theta} = \int_P \chi(\psi) \ell \wedge \alpha \wedge \bar{\Theta}.$$

On a good fiber write $\omega = s|_{D_t}$. In local frames the absolute value of the right-hand integrand for $\chi(x) = \min\{M, e^{(1+\delta)x}\}$ is bounded by a fixed multiple of

$$|\omega|^2 e^{(1+\delta)(T - \log |\omega|^2)} = e^{(1+\delta)T} |\omega|^{-2\delta}.$$

Here $|\omega|^2$ denotes the canonical volume density. The weights T have local upper bounds. The zero divisor of ω is relatively simple normal crossing on this compact set, so the last expression is integrable if $\delta > 0$ is small enough that $\delta e_j < 1$ for its finitely many multiplicities. The bound is independent of M . Multiply (4.15) by i^{n^2} , take absolute values on the right, and let $M \rightarrow \infty$. Monotone convergence proves (4.14). $\square$

4.4. Fixed orders and Bergman weights. Calculate the following orders first on the geometric generic fiber. The section-space and vanishing-filtration comparisons below will give the same values on every simultaneous very general fiber. For a prime divisor S on such a fiber D and positive rational ϵ , define its actual asymptotic fixed order by

$$a_S(\epsilon) = \inf_{\substack{m > 0 \\ m \in \mathbf{Z}}} \frac{1}{m} \min_{0 \neq U \in H^0(D, mK_D + m\epsilon H|_D)} \text{ord}_S(U),$$

where degrees with zero section space are omitted. These spaces are nonzero in sufficiently divisible degree, since $K_D \sim E$ and $\epsilon H|_D$ is ample. Multiplication by an ample section not vanishing generically on S shows that $a_S(\epsilon)$ is nondecreasing as ϵ decreases. In making this comparison, first pass to common divisible degrees and raise any chosen section to a power. Thus the limit

$$\sigma_S(K_D) = \lim_{\epsilon \downarrow 0} a_S(\epsilon)$$

exists. We need only this definition and its elementary monotonicity.

Our objective is to prove that $\sigma_S(K_D) > 0$ whenever S comes from a horizontal pole divisor of v . A zero value would give section degrees k and positive ample twists r_k for which both r_k/k and the normalized minimal vanishing order j_k/k tend to zero. The next two lemmas associate to such degrees a semipositive weight with zero generic divisorial coefficient along S . The pole-density estimate will then show that this weight cannot satisfy Lemma 4.5 at a pole of v .

The weight must be constructed on P before we select the fiber on which its global integral is finite. We therefore use relative section spaces on the fixed model of Lemma 4.2. The weight will work on every fiber satisfying the corresponding base-change conditions. Only afterwards, in Proposition 4.8, will we make the final fiber choice.

For any good fiber D under consideration, write

$$\omega = s|_D, \quad E = \mathcal{E}|_D = \sum_j e_j D_j, \quad \varphi_0 = \log |\omega|^2.$$

The geometric divisor components are already defined on the fixed model of Lemma 4.2. Choose the integer degrees and relative section spaces on this model; no individual sequence of generic sections needs to be spread. For each degree, cohomology and base change also apply to the section spaces with prescribed vanishing along these components. Excluding the resulting countable union of proper closed subsets fixes all section dimensions and minimal vanishing orders simultaneously.

Lemma 4.6. *Choose integers $k \rightarrow \infty$ and $r_k > 0$ with $r_k/k \rightarrow 0$, such that $H^0(D, kK_D + r_k H|_D) \neq 0$ on the generic fiber. There is a semipositive weight T on $K_{P/C}$, depending only on these degrees and the fixed family and metrics, with $T \geq \log |s|^2$. On every good fiber D satisfying the simultaneous base-change conditions for these unfiltered section spaces, for every sequence $U_k \in H^0(D, kK_D + r_k H|_D)$ normalized by*

$$(4.16) \quad \int_D |U_k|^{2/k} e^{-r_k \varphi_H/k} = 1,$$

the weights and functions

$$(4.17) \quad \varphi_k = \frac{\log |U_k|^2 - r_k \varphi_H}{k}, \quad x_k = \varphi_k - \varphi_0, \quad \psi = T|_D - \varphi_0$$

satisfy uniform local upper bounds for φ_k and $\limsup_k x_k \leq \psi$ off E . The integral in (4.16) is understood in canonical coordinate densities, and is independent of those coordinates and bundle frames.

Proof. Apply the relative k -Bergman construction to $\pi : P \rightarrow C$ and the twisting bundle $r_k H$ with its smooth positive metric. The source and base are smooth, the morphism is projective and surjective with connected fibers, and a nonzero fiber section exists generically. Generic base change supplies a nonempty open on which all such sections extend locally. The smooth twisting metric makes every fiber section $2/k$ -integrable. Thus Theorem 4.2.7 of [41], in the cited arXiv version, supplies a semipositive weight β_k on $kK_{P/C} + r_k H$. Over the smooth extendibility locus, its exponential is the extremum of $|U|^2$ among sections normalized as in (4.16). No extremal interpretation on exceptional fibers is needed.

Set

$$\tau_k = \frac{\beta_k - r_k \varphi_H}{k}.$$

These are weights on $K_{P/C}$, and $dd^c \tau_k \geq -(r_k/k)h$. On a fixed smaller submersion chart, the mean inequality for the subharmonic function given by the absolute value of a section coefficient to the power $2/k$, together with its normalized integral and the bounded factor $e^{-r_k \varphi_H/k}$, gives an upper bound independent of k . Taking the extremum gives that bound for τ_k . The bound persists across degree-dependent exceptional base values by plurisubharmonic extension. Equivalently, the ambient estimate in Remark 4.3.1 of [41] gives precisely this uniformity when the divided twisting weights are bounded above.

For completeness, local boundedness on one fixed nonempty open also globalizes as follows. Subtract a fixed smooth reference weight on $K_{P/C}$. The resulting globally defined potentials have a uniform lower bound for their complex Hessians. If their suprema were unbounded, subtracting those suprema and using compactness of sup-normalized quasi-plurisubharmonic potentials would give a subsequence converging in L^1 to a finite quasi-plurisubharmonic function. On the fixed open, however, the normalized functions would tend uniformly to minus infinity.

This is impossible. Hence the weights τ_k are uniformly bounded above in local smooth frames on P .

These constructions use the relative section spaces, without selecting a final fiber or an individual section. Take their regularized upper limit on P and then its maximum with the divisor weight:

$$(4.18) \quad T = \max \left\{ \left(\limsup_{k \rightarrow \infty} \tau_k \right)^*, \log |s|^2 \right\}.$$

The curvature losses tend to zero, so this is semipositive; if the first term is identically minus infinity, the second term still defines the weight. Its restriction to a good fiber is well defined because it dominates φ_0 . Every normalized section is bounded above by the corresponding Bergman extremum. Consequently $\varphi_k \leq \tau_k|_D$, which proves all the assertions. $\square$

The weight T now bounds the upper limit of the normalized section weights in the chosen degrees. We next compare its divisorial coefficient with the minimal vanishing orders of those complete section spaces. This comparison holds on every fiber satisfying the stated generic conditions; the final choice of a fiber on which the integral in (4.14) is finite is not needed for the next lemma.

Lemma 4.7. *Let D satisfy the simultaneous generic conditions above, and fix a component $S = D_i$ of E . Suppose the degrees chosen in Lemma 4.6 have minimal actual vanishing orders*

$$j_k = \min\{\text{ord}_S(U) : 0 \neq U \in H^0(D, kK_D + r_k H|_D)\}$$

satisfying $j_k/k \rightarrow 0$. The weight T constructed in (4.18) has zero generic divisorial Lelong coefficient along S after restriction to D .

If $T|_D$ had positive divisorial coefficient along S , an adjoint estimate would produce a section with the same leading coefficient and a strictly smaller $L^{2/k}$ integral. We prove both the preservation of that coefficient and a gain proportional to $1/k$; the dependence on k matters because the exponent $2/k$ tends to zero.

Proof. Suppose that coefficient is $\nu > 0$, with Lelong coefficients normalized by $\log |z|^2$. For each k , choose a nonzero leading section in the image of

$$H^0(D, kK_D + r_k H|_D - j_k S) \longrightarrow H^0(S, (kK_D + r_k H|_D - j_k S)|_S).$$

Among sections with this prescribed image, minimize the $2/k$ -integral in (4.16). This minimum exists: the constraint is a nonempty closed affine subset of a finite-dimensional vector space, and the integral is continuous, positive away from zero, and homogeneous of positive degree. It is therefore coercive. Denote a minimizer by U_k , and rescale both it and the prescribed image so that (4.16) holds. All lower coefficients along S are zero by the definition of j_k .

Choose $c > 0$ so large that, for all sufficiently large k ,

$$(4.19) \quad c\nu - (1+c)\frac{j_k}{k} > 1.$$

Choose $p_0 > 0$ sufficiently small that

$$(4.20) \quad \int_D |\omega|^2 e^{(1+p_0)\psi} < \infty.$$

This follows already from the local upper bound for $T|_D$: the integrand is bounded by a constant times $|\omega|^{-2p_0}$, which is integrable when $p_0 e_j < 1$ for every component of the simple normal crossing divisor E .

Fix a smooth convex function $g : \mathbf{R} \rightarrow \mathbf{R}_{\geq 0}$ with $0 \leq g' \leq 1$, equal to 0 sufficiently far to the left and to its argument sufficiently far to the right, and with $g'' > 0$ on $[-1, 1]$. Choose a smooth

function $0 \leq \theta \leq 1$ that is zero on $(-\infty, -1]$, one on $[1, \infty)$, and whose derivative is supported in $[-1, 1]$. For $b > 0$, put

$$(4.21) \quad \Phi = (k-1-c-p_0)\varphi_k + cT|_D + p_0(\varphi_0 + g(x_k - b)) + r_k\varphi_H.$$

This is a weight on the integral bundle $(k-1)K_D + r_kH|_D$: the coefficients of its canonical weights sum to $k-1$, and x_k is a global function off the divisors. We henceforth take $k > 1+c+p_0$. The constants c, p_0 are fixed. For each fixed k and cutoff b , we first remove the regularization used below. We then estimate the upper limit as $k \rightarrow \infty$ with b fixed, and finally let $b \rightarrow \infty$. The contradiction will use one sufficiently large b followed by sufficiently large k .

The curvature and the adjoint estimate. Off E and the zero divisor of U_k , one has $dd^c\varphi_k = -(r_k/k)h$, $dd^c\varphi_0 = 0$, and $dd^c x_k = -(r_k/k)h$. Direct differentiation of (4.21) gives

$$(4.22) \quad \begin{aligned} dd^c\Phi &= \frac{r_k}{k}(1+c+p_0-p_0g'(x_k-b))h + cdd^c(T|_D) \\ &\quad + p_0g''(x_k-b)\frac{i}{2\pi}\partial x_k \wedge \bar{\partial} x_k. \end{aligned}$$

In particular it dominates $(1+c)r_kh/k$ and retains the displayed gradient term. The resulting adjoint estimate produces $V_k \in H^0(D, kK_D + r_kH|_D)$ such that, on the complement of the divisors,

$$u_k = \theta(x_k - b)U_k - V_k$$

satisfies

$$(4.23) \quad \int_D \left| \frac{u_k}{U_k} \right|^2 w_k \leq C_{p_0} \int_{\{|x_k-b|\leq 1\}} w_k, \quad w_k = |\omega|^2 e^{(1+c+p_0)x_k - c\psi - p_0g(x_k-b)}.$$

The constant is independent of b and k : the retained gradient term controls the datum on the cutoff shell, even as the ample coefficient tends to zero. We give the singular-weight justification, including the choice of complete metric, before using this inequality.

Subtract a fixed smooth reference metric from $T|_D$ and apply Theorem 14.12(a),(c) and Corollary 14.13(c) of [16]. We obtain weights T_ℓ converging almost everywhere to $T|_D$, uniformly bounded above locally, smooth outside an analytic set Z_ℓ , and satisfying $dd^c T_\ell \geq -\epsilon_\ell h$, where $\epsilon_\ell \rightarrow 0$. The cited construction gives pointwise approximation above the original potential. In any event, taking a smooth regularized maximum with φ_0 , of width tending to zero, ensures $T_\ell \geq \varphi_0$. The regularized maximum is invariant under simultaneous translation of its arguments, so it gives a weight on the same bundle. Its monotonicity and convexity preserve the lower curvature bound. It is smooth off $Z_\ell \cup E$; preservation of analytic singularities of the maximum itself is not required.

Keep k, b fixed and replace $T|_D$ by T_ℓ in (4.21), obtaining Φ_ℓ . For large ℓ , the loss $c\epsilon_\ell h$ is less than half the ample term in (4.22). On

$$D \setminus (Z_\ell \cup E \cup \text{div}(U_k))$$

the weight is smooth with positive curvature and the same gradient lower bound. This complement admits a complete Kähler metric. Indeed its deleted analytic set is the zero set of a section of a Hermitian vector bundle: twist its coherent ideal by a sufficiently ample bundle and use finitely many generators. Lemma 12.9 of [16] applies, because D is compact Kähler and the curvature of that smooth bundle has a uniform upper bound.

Write $A_\ell = i\partial\bar{\partial}\Phi_\ell$ for the full positive curvature form on the complement. Add A_ℓ to a complete Kähler metric there, obtaining a complete metric $\hat{h}_\ell \geq A_\ell$. This particular choice also verifies the ordinary L^2 hypothesis for the datum

$$f_k = \theta'(x_k - b)\bar{\partial} x_k U_k.$$

To see both this and the uniform estimate explicitly, let G be the matrix of an auxiliary Kähler metric and A the positive curvature matrix. For a bundle-valued $(n, 1)$ -form with local coefficient $U\eta$, matrix cancellation gives

$$(4.24) \quad \langle [A, \Lambda_G]^{-1}(U\eta), U\eta \rangle_G e^{-\Phi} dV_G = |U|^2 e^{-\Phi} \eta^* A^{-1} \eta d\lambda,$$

up to the fixed convention for the canonical volume form. The determinants from the top holomorphic degree and the volume cancel, and so do the metric factors in the antiholomorphic degree and in the curvature operator. The solution norm in bidegree $(n, 0)$ is likewise independent of G . When $G \geq A$, the ordinary datum norm is bounded by the right side of (4.24).

The gradient term in (4.22) implies, for $\eta = \partial x_k$, the matrix bound

$$\eta^* A_\ell^{-1} \eta \leq \frac{1}{p_0 g''(x_k - b)}.$$

Since θ' is supported where g'' has a positive minimum, the inverse-curvature density of f_k is at most a constant times the shell density. That density is integrable. The datum is ∂ -closed, being the derivative of $\theta(x_k - b)U_k$ on the complement. The complete Kähler estimate, Theorem 5.1 of [16], consequently gives $\bar{\partial}u_{k,\ell} = f_k$ with the desired bound for Φ_ℓ .

Here the local adjoint density has precisely the required exponents:

$$\begin{aligned} |U_k|^2 e^{-\Phi_\ell} &= \exp((1 + c + p_0)\varphi_k - cT_\ell - p_0(\varphi_0 + g(x_k - b))) d\lambda \\ &= |\omega|^2 e^{(1+c+p_0)x_k - c(T_\ell - \varphi_0) - p_0g(x_k - b)}. \end{aligned}$$

On the fixed shell, it is bounded above by $e^{(1+c+p_0)(b+1)}|\omega|^2$, because $T_\ell - \varphi_0 \geq 0$. Moreover Φ_ℓ is bounded above locally uniformly in ℓ , using the positive coefficient $k - 1 - c - p_0$ and

$$\varphi_0 + g(\varphi_k - \varphi_0 - b) \leq \max\{\varphi_0, \varphi_k - b\} + C_g.$$

The weighted solution estimate therefore bounds the ordinary local L^2 norms of $u_{k,\ell}$. The holomorphic section $V_{k,\ell} = \theta(x_k - b)U_k - u_{k,\ell}$ extends across its deleted analytic set by the distributional L^2 removable-singularity lemma [16, Lemma 12.10]. In a local bundle frame the right side of the distributional $\bar{\partial}$ equation is zero, so the extended coefficients have holomorphic representatives. The extensions lie in the fixed finite-dimensional space $H^0(D, kK_D + r_k H|_D)$, with bounded ordinary norm. Take a subsequence converging as global sections. Fatou's lemma for the left side and dominated convergence on the fixed shell give (4.23). For this passage one may discard the countable union of the sets Z_ℓ ; no nesting of those sets is needed.

The shell becomes small. Put $B = 1 + c + p_0$. For fixed b , the shell densities in (4.23) are bounded by $e^{B(b+1)}|\omega|^2$. At a point off E , a subsequence contributing to their limsup has a further subsequence with $x_k \rightarrow x_* \in [b - 1, b + 1]$. Lemma 4.6 gives $x_* \leq \psi$; thus $\psi \geq b - 1$ and

$$Bx_* - c\psi - p_0g(x_* - b) \leq (1 + p_0)\psi.$$

Reverse Fatou therefore yields

$$(4.25) \quad \limsup_{k \rightarrow \infty} \int_{\{|x_k - b| \leq 1\}} w_k \leq \int_{\{\psi \geq b - 1\}} |\omega|^2 e^{(1+p_0)\psi}.$$

By (4.20), the right side tends to zero as $b \rightarrow \infty$.

The prescribed leading section is preserved. Near a general point of $S = (z = 0)$, write $U_k = z^{j_k} a_k$ and $\omega = z^{e_i} a_0$, with the coefficients nonvanishing in the chosen frames. Since $j_k/k < e_i$, the function x_k tends to plus infinity as $z \rightarrow 0$. Thus $\theta(x_k - b) = 1$ and $g(x_k - b) = x_k - b$ sufficiently near that point. The divisorial decomposition of the positive curvature current of $T|_D$

gives $T|_D \leq \nu \log |z|^2 + O(1)$: subtracting $\nu[S]$ leaves a positive current, whose local potential is bounded above. See [17, Chapter III, Proposition 8.16]. Consequently

$$(4.26) \quad \Phi \leq \left((k-1-c) \frac{j_k}{k} + c\nu \right) \log |z|^2 + O(1).$$

The coefficient exceeds $j_k + 1$ by (4.19). In this neighborhood $u_k = U_k - V_k$ is holomorphic. If its vanishing order were at most j_k , the transverse integral against $e^{-\Phi}$ would diverge, contradicting (4.23). Hence $U_k - V_k$ vanishes along S to order at least $j_k + 1$. This divisorial statement says exactly that V_k has the prescribed leading section and all the prescribed zero lower coefficients.

A gain of order $1/k$. Let $\rho_k dV$ be the normalized density in (4.16) relative to a fixed smooth volume on D , and put $q_k = |u_k/U_k|$ off the zero divisor. The functions ρ_k have a uniform upper bound by Lemma 4.6. In local canonical frames the density is

$$w_k = \exp(B\varphi_k - cT|_D - p_0(\varphi_0 + g(x_k - b))) d\lambda.$$

The local upper bounds for $T|_D$, and the earlier bound $\varphi_0 + g(x_k - b) \leq \max\{\varphi_0, \varphi_k - b\} + C_g$, therefore give, on a finite cover by smooth frames,

$$(4.27) \quad w_k \geq C^{-1} \rho_k^B dV$$

with C independent of k and of b bounded below. Fix $0 < \eta < \min\{1, 2/B\}$. Hölder's inequality gives

$$(4.28) \quad \begin{aligned} \int_D q_k^\eta \rho_k dV &\leq \left(\int_D q_k^2 \rho_k^B dV \right)^{\eta/2} \left(\int_D \rho_k^{(2-B\eta)/(2-\eta)} dV \right)^{1-\eta/2} \\ &\leq C_\eta \left(\int_D q_k^2 w_k \right)^{\eta/2}. \end{aligned}$$

The exponent in the second factor is positive, so its uniform bound follows from compactness and the upper bound for ρ_k . Equations (4.23) and (4.25) show that this moment can be made arbitrarily small by first fixing b sufficiently large and then taking k sufficiently large.

The set where $\theta(x_k - b) \neq 0$ is contained in $\{\varphi_0 \leq C - b + 1\}$, by the upper bound for φ_k . Its smooth volume tends to zero with b , and therefore its ρ_k -mass tends to zero uniformly in k . Markov's inequality applied to (4.28) now gives a set of ρ_k -mass at least $1/2$ on which $\theta(x_k - b) = 0$ and $q_k \leq 1/2$. On this set $|V_k/U_k| = q_k \leq 1/2$; everywhere else, $|V_k/U_k| \leq 1 + q_k$.

For $0 < p \leq \eta/2$ and $q \geq 0$,

$$(4.29) \quad (1+q)^p \leq 1 + C_\eta p q^\eta.$$

For $q \leq 1$, this follows from concavity. For $q \geq 1$, integrate the derivative with respect to the exponent and use $(1+q)^{\eta/2} \log(1+q) \leq C_\eta q^\eta$. Taking $p = 2/k$, we therefore obtain

$$(4.30) \quad \begin{aligned} \int_D |V_k|^{2/k} e^{-r_k \varphi_H/k} &= \int_D |V_k/U_k|^p \rho_k dV \\ &\leq 1 - \frac{1}{2}(1 - 2^{-p}) + C_\eta p \int_D q_k^\eta \rho_k dV < 1 \end{aligned}$$

for the indicated choices of b and then k . Indeed $(1 - 2^{-p})/p \rightarrow \log 2$, whereas the moment can be made arbitrarily small independently of large k . This is a strict improvement of the minimizing integral for a section with the same prescription, a contradiction. Therefore $\nu = 0$. $\square$

4.5. Excluding poles with zero limiting fixed order. We now apply the estimates to a horizontal pole of v . The degrees and the limiting weight will be chosen from the geometric generic section spaces. Only after applying Lemma 4.5 to that fixed weight will we choose a complex fiber and the minimizing sections on it.

Proposition 4.8. *Every horizontal pole divisor of v has positive limiting fixed order on a simultaneous very general fiber. In particular, for one sufficiently small positive rational ϵ , every such divisor has positive actual asymptotic fixed order for $K_D + \epsilon H|_D$.*

Proof. Suppose a horizontal pole divisor induces $S = D_i$ and has zero limiting fixed order on the geometric generic fiber. Choose rational perturbations tending to zero and degrees in which the normalized minimal orders approach their infima. Passing to powers as necessary gives integers $k \rightarrow \infty$, $r_k > 0$, with

$$\frac{r_k}{k} \rightarrow 0, \quad \frac{j_k}{k} \rightarrow 0,$$

where j_k is the minimal actual order of the whole section space. Use the corresponding relative section spaces and vanishing filtrations on the fixed prepared family, and impose the countably many base-change conditions that preserve these orders. No further field extension or spreading of individual sections is required. Lemma 4.6 now constructs one global weight T from the chosen degrees, before any final fiber is selected.

Apply Lemma 4.5 to this T . Fubini gives a full-measure set of good fibers on which the integral is finite. The excluded algebraic conditions form a countable subset of the complex curve and hence have measure zero. Choose D satisfying both requirements. Lemma 4.7 applies on this same fiber and proves that $T|_D$ has zero generic divisorial coefficient at S . The normalized minimizing sections in that Lemma are chosen only now; they do not change T .

In a smooth chart over the base, contraction gives

$$\Theta = \iota_v(dt \wedge \omega).$$

The density of $i^{n^2} h \wedge \Theta \wedge \bar{\Theta}$ relative to base area is comparable, by the smooth positive metric, to $|v|_h^2 |\omega|^2$. This formula includes the mixed components of the metric. At a generic point of a pole of order at least one, shrink the chart so its leading coefficient is bounded below. Since $\text{ord}_S(\omega) = e_i$, the finite fiber integral from (4.14) would then dominate a positive constant times

$$(4.31) \quad \int |z|^{-2-2\delta e_i} e^{(1+\delta)T|_D} d\lambda, \quad S = (z = 0).$$

We show that this last integral is infinite.

The generic Lelong coefficient is zero. The analyticity of Lelong upper-level sets, or equivalently the generic-Lelong-number statement in [17, Chapter III, Lemma 8.15], implies that the point Lelong number is zero at almost every center on a small smooth patch of S . For a local plurisubharmonic weight T , the logarithmic mean characterization [17, Chapter III, Corollary 6.8 and Example 6.9] then gives

$$\frac{1}{\text{vol}(B(a, r))} \int_{B(a, r)} T d\lambda = o(|\log r|)$$

at almost every such center a . The characterization for ball means follows from that for sphere means by radial integration.

Fix $\epsilon > 0$. For almost every center, the ball mean is at least $\epsilon \log r$ at all sufficiently small dyadic radii. The sets on which this holds beyond one fixed dyadic index have a union of full divisor measure. Hence there is a measurable set A of positive divisor measure on a compact patch of S , and one common starting index, for which the lower bound holds uniformly. In dimension one the divisor measure here is counting measure.

On $B(a, r)$, $|z| \leq Cr$. Jensen's inequality therefore gives, for the nonnegative integrand F in (4.31),

$$\int_{B(a, r)} F d\lambda \geq C^{-1} r^{2n-2-2\delta e_i + (1+\delta)\varepsilon} \quad (a \in A).$$

Integrate this inequality in a . For each ambient point, the measure of centers $a \in A$ whose radius- r ball contains it is at most Cr^{2n-2} , by projection to the smooth tangential coordinates of S . Fubini consequently yields

$$\int F d\lambda \geq C^{-1} r^{-2\delta e_i + (1+\delta)\varepsilon}.$$

The overlap bound is constant when $n = 1$, so the same formula holds in that case. Since $e_i \geq 1$, choose $\varepsilon < 2\delta e_i / (1 + \delta)$ and let the dyadic radius tend to zero. The right side diverges, contradicting (4.31) and the Fubini consequence of (4.14). A pole with zero limiting fixed order is therefore impossible.

There are only finitely many horizontal pole divisors. For each, positivity of its limit and monotonicity give a sufficiently small rational perturbation with positive actual asymptotic fixed order. Taking one common smaller rational perturbation proves the last assertion. $\square$

4.6. Contraction of poles and the geometric generic fiber.

Proof of Theorem 4.1. We may assume $n > 0$, and use the preparation and notation of Lemmas 4.2–4.4. Proposition 4.8 gives one sufficiently small positive rational ε for which every horizontal pole divisor has positive actual asymptotic fixed order for $K_D + \varepsilon H|_D$. We fix this same perturbation for the relative model construction. The order comparisons hold on the geometric generic fiber and on simultaneous very general fibers by base change in the countably many divisible section degrees.

Over the generic curve choose an effective rational divisor $\Delta \sim_{\mathbf{Q}} \varepsilon H$ such that the pair is klt. One obtains it by taking a general sufficiently divisible member of the ample system and a small coefficient. In the equivariant case average its translates; the klt condition is preserved because discrepancies are linear in the boundary and the translates are klt. Spread this choice and shrink the curve so that the total pair is klt. The relative adjoint divisor is big: on a good fiber $K_D \sim E \geq 0$, and an ample positive perturbation of an effective divisor is big.

Apply [8, Theorem 1.2(2)] to this projective morphism over the shrunk curve. Its hypotheses are a klt pair and a big relative adjoint divisor. It gives a normal relative log canonical model

$$\chi : P \dashrightarrow Q$$

with Q projective over the curve. Since the adjoint divisor is relatively big, this map is birational. The ample-model comparison [8, Definitions 3.6.4 and 3.6.6, Lemma 3.6.5(4)] shows that it is a birational contraction, so it extracts no divisors. In a common resolution the pullback adjoint system is the pullback of the ample system on Q plus an effective exceptional fixed part.

Every horizontal pole divisor is contracted by χ . Indeed a prime which survived would be tested at its corresponding prime on the ample model. Sufficiently divisible ample systems have no fixed component there, and the exceptional section comparison changes no order at that prime. Its asymptotic fixed order would therefore be zero, contrary to the choice of ε . The canonical ring is invariant under the finite group when Δ is invariant, and hence the action descends to Q .

Choose a nonvanishing algebraic vector field on a smaller open of the curve. Normalize the kernel line of Θ against that vector field, and denote the resulting rational derivation of $\mathbf{C}(P) = \mathbf{C}(Q)$ by ∂ . It is rational because Θ is algebraic and its kernel and prescribed base projection are given by algebraic linear equations. This normalization multiplies the previous local v by a nonzero base function, so it creates no horizontal pole. Remove the finitely many vertical pole loci and the zeros or poles of the chosen base vector field.

The derivation is regular at every prime divisor of Q . Because χ extracts no divisors, each such prime corresponds to a prime on P , with the same discrete valuation ring; it cannot be one of the contracted pole divisors. It remains to pass through codimension two. For any normal affine open with coordinate ring A , and any $f \in A$, regularity at the height-one points gives

$$\partial f \in \bigcap_{\text{ht } \mathfrak{p}=1} A_{\mathfrak{p}} = A.$$

Thus ∂ is a regular derivation on the entire normal space. Its projection to the base is still the chosen base vector field, since this equality is an identity of derivations on the function field. Locally reparametrize the base analytically so that $\partial t = 1$.

A regular derivation on a complex analytic space has local holomorphic flows, including at singular points. Here is the local justification. Embed the space in a polydisk with coordinate functions $z_1, \dots, z_N$, lift the functions ∂z_j to holomorphic functions on the polydisk, and form the corresponding ambient vector field. It preserves the defining ideal, because its induced derivation on the quotient is ∂ . Its flow preserves the analytic subspace: for finitely many generators of the ideal, their values along an integral curve satisfy a homogeneous linear differential system with zero initial value. Uniqueness makes the restrictions of these local flows independent of the ambient lifts and permits them to glue.

Properness now gives a uniform flow time near a compact fiber. More explicitly, the inverse image of a sufficiently small closed base disk is compact; a finite cover by flow neighborhoods provides a common smaller disk of complex times. Since $\partial t = 1$, forward and backward flows identify the fiber over its center with every fiber over a smaller disk. These identifications are biholomorphic, and hence algebraic because the fibers are projective. They commute with Γ : the kernel of Θ is invariant, and its normalization by a base vector field is unique. Shrink the algebraic open as needed so that the original family and the model have birational fibers and the model fibers under consideration are geometrically integral.

Fix one fiber Q_{t_0} in this disk. Graphs of isomorphisms from Q_{t_0} to fibers of Q belong to countably many finite-type loci of relative Hilbert schemes. Requiring equivariance is a closed condition on these graphs. Their images in the algebraic curve are constructible and together contain the disk, by the algebraic flow identifications. If none were dominant, each image would be a finite subset of the curve; their countable union could not contain a disk. Some graph locus is therefore dominant. Its generic fiber has a closed point defined over a finite extension of the curve's function field. Over that extension the generic fiber of Q is isomorphic to the fixed Q_{t_0} , equivariantly when prescribed.

Take a smooth projective resolution D_0 of Q_{t_0} , equivariant in the finite-group case. The birational comparison with P gives the required birational identification of its generic fiber with D_0 after a finite extension. The preliminary finite changes of the curve compose with this extension. For $n = 0$ the fiber is a point and the conclusion is immediate. This completes the proof. $\square$

4.7. From curves to geometric birational constancy. The curve criterion gives birational identifications after finite extensions of curve function fields. We now use algebraic loci of graphs to obtain the corresponding statement on an arbitrary constant-line locus and on a geometric generic parameter fiber.

Corollary 4.9 (Constant-line loci). *Let $\pi : \mathcal{P} \rightarrow B$ be a smooth projective morphism with connected fibers over a smooth connected complex algebraic variety. Let $T \subseteq B$ be a connected smooth algebraic locus on which the very general fiber D satisfies $\kappa(D) = 0$ and $h^0(D, K_D) = 1$. If the entire ordinary top Hodge line of the restricted family has constant projective period in a*

flat reference, its geometric generic fiber is birational to a smooth projective variety over $\mathbf{C}$. The conclusion is equivariant for any prescribed finite fiberwise birational group action.

More generally, let $B \rightarrow Z$ be a morphism of smooth connected complex algebraic varieties with geometrically integral generic fiber. If the same fiber and constant-line hypotheses hold on a nonempty smooth open of every very general fiber over Z , then the family over the geometric generic fiber of $B \rightarrow Z$ is birationally constant over $\overline{\mathbf{C}(Z)}$, with the prescribed finite action. These birational identifications remain valid after algebraically closed extension of their fields of definition.

Proof. Restrict the integral polarized cohomology variation to T . Its entire highest Hodge step has rank one. Constancy of the projective line makes it a flat rank-one local subsystem, and Corollary 3.15, applied anew to this restricted variation, gives finite character. On an algebraic curve in T , Theorem 4.1 therefore applies, including its equivariant assertion.

We pass from curves to the geometric generic point of T . After shrinking T to an affine open, graphs of birational identifications between pairs of fibers are parametrized by countably many finite-type algebraic loci over $T \times T$. For example, take relative Hilbert schemes of the product family and their constructible loci of geometrically integral graphs on which both projections are dominant of generic degree one. Geometric integrality and these degree conditions are imposed after flat stratification. Equivariance is expressed by invariance of the graph under the diagonal finite action; a birational action is first regularized as in Lemma 4.2. The images of the graph loci in $T \times T$ are constructible.

For $\dim T = 0$ the conclusion is immediate, and for $\dim T = 1$ it is Theorem 4.1. Suppose $\dim T \geq 2$. Choose a smooth projective compactification and a sufficiently ample complete-intersection family of curves. The incidence of a curve with two distinct points on it has an irreducible open dominating $T \times T$: sufficiently ample systems separate the two point conditions, and their general members through those points are smooth and connected. Its general fiber over the curve parameter space is the irreducible surface of pairs of points on that curve.

If none of the graph images were dense in $T \times T$, their closed image closures would pull back to countably many proper closed subsets of this incidence. For a very general curve, each such subset meets its surface of pairs in a proper closed subset; those not dominating the curve parameter space are avoided. The curve can also satisfy the countably many generic section conditions giving $\kappa(D) = 0$ and $h^0(D, K_D) = 1$. Theorem 4.1 and spreading its birational identification give a dense open of pairs of fibers on that curve which are birational, equivariantly when required. A countable union of proper closed subsets of an irreducible complex surface cannot contain this open. This contradiction proves that some graph locus dominates $T \times T$.

A dense constructible image contains a nonempty open. Specialize the first coordinate to a sufficiently general complex point so that the slice of this open is dense in the second factor. The first fiber is now a fixed smooth projective variety. The graph parameter space over this slice dominates T , and a closed point of its generic fiber is defined over a finite extension of $\mathbf{C}(T)$. It gives the desired birational identification, with the finite action.

For the relative assertion use these same graph loci over $B \times_Z B$, taking the component that dominates Z . Its geometric generic fiber is integral by hypothesis. If no graph locus dominated this component, the closures of their images would remain proper on every very general fiber over Z , after excluding countably many proper specialization loci. This contradicts the assertion just proved on those complex fibers. Thus one graph locus dominates over the generic point of Z . Over $b_Z = \mathbf{C}(Z)$, its constructible image contains an open in the space of pairs. Choose a b_Z -point in a suitable open for the first projection. Its fiber is a fixed smooth projective b_Z -variety, and the graph locus on the remaining slice supplies the birational identification after a finite extension of the function field of the geometric generic B -fiber. This proves the relative

conclusion. Finally, all the graphs, inverse maps, and equivariance identities are algebraic data, so these identifications persist under field extension. $\square$

Proposition 4.10 (Constancy in the h -fiber directions). *For very general $z \in Z$, the geometric generic fiber of the J -family over W_z is birational to the base change of a smooth projective variety over $\mathbf{C}$. More intrinsically, after extending the function field of Z to its algebraic closure, the geometric generic fiber of x over the resulting h -fiber is birationally constant over that algebraically closed field.*

Proof. On the smooth family locus W° , Lemma 3.1 gives a smooth projective cyclic-cover family with connected fibers, Kodaira dimension zero, and a one-dimensional *entire* ordinary top-form space. On the connected open of every very general W_z , Corollary 3.16 makes its projective top line constant. Corollary 4.9 therefore makes that restricted cover family geometrically birationally constant, equivariantly for its cyclic deck group. Taking invariant function fields gives the asserted constancy of the original J -family.

For the relative assertion, apply the second part of Corollary 4.9 to $W^\circ \rightarrow Z$, shrinking Z if necessary. Its geometric generic fiber is integral by Proposition 2.7, and the preceding paragraph verifies the fiber and constant-line hypotheses on its very general complex fibers. Over $b_Z = \overline{\mathbf{C}(Z)}$ the reference cover and its finite deck action are consequently defined over b_Z . Their invariant function field has a smooth projective b_Z -model, whose base extension is birational to the original geometric generic x -fiber. This is geometric birational constancy; the descent of a chosen trivialization and of the whole original fiber is addressed in the remaining sections. $\square$

5. GENERAL-TYPE MODELS OVER A LOGARITHMIC BASE

We now compare the variation of a log canonical model with sections on the original total space. This comparison has two later uses. Marked models detect which relative Iitaka bases are constant, while Section 8 uses complete model systems to retain the dimension of a second Iitaka base. In both applications, a finite base change is only an auxiliary step: the sections must be tested on the finite normalization of the original source.

Throughout this section, k is an algebraically closed field of characteristic zero and all varieties are projective. An effective rational boundary has coefficients in $[0, 1]$. For a dominant morphism $p : H \rightarrow I$ between smooth integral varieties, put $r = \dim H - \dim I$ and let $\mathcal{V}_p$ be the saturated kernel of dp . Its canonical sheaf is $(\bigwedge^r \mathcal{V}_p^\vee)^{**}$; at the generic point this is the one-dimensional $k(H)$ -space $\bigwedge^r \Omega_{k(H)/k(I)}^1$. We compare the rank-one lattices supplied by this sheaf and the ordinary relative canonical sheaf at prime divisors of H . If R_p is the divisorial zero of the pulled-back top base differential, then taking determinant lattices gives

$$(5.1) \quad K_{\mathcal{V}_p} = K_{H/I} - R_p.$$

Here $K_{H/I} = K_H - p^*K_I$, whereas $K_{\mathcal{V}_p}$ is the canonical divisor of the saturated relative tangent sheaf. The term R_p records the difference between these two determinant lattices. The saturated lattice of rational relative top forms depends only on $k(I) \subset k(H)$ and the source model. It is therefore unchanged by a birational change of the base. For a rational boundary B on H , write $B = C + B_v$, where C is horizontal and B_v is vertical over I .

Lemma 5.1 (The inverse boundary). *Let D_I be a reduced SNC divisor on I . Then*

$$R_p + (p^*D_I)_{\text{red}} \geq p^*D_I.$$

*If B is an effective rational boundary satisfying $B \geq (p^*D_I)_{\text{red}}$, it follows that*

$$(5.2) \quad K_H + B = K_{\mathcal{V}_p} + C + p^*(K_I + D_I) + V, \quad V := R_p + B_v - p^*D_I \geq 0.$$

Proof. Test a prime E of H , with uniformizer x . In local boundary coordinates on I , write $p^*t_j = x^{a_j}u_j$, with u_j a unit. Each logarithmic differential is $a_j dx/x + du_j/u_j$, and a nonzero wedge contains at most one factor dx/x . If $a = \text{ord}_E(p^*D_I) > 0$, the pulled-back ordinary base volume thus vanishes to order at least $a - 1$. This proves the first inequality, also when $p(E)$ has higher codimension. If $a = 0$, use $R_p \geq 0$. The second assertion follows from Equation (5.1) and the coefficient-one boundary along the reduced inverse image of D_I . $\square$

The boundary inequality supplies a fixed effective contribution from the base. To use relative positivity for the remaining divisor, we first arrange that all original vertical divisors have multiplicity one after an alteration. Divisors introduced by a resolution are allowed to remain exceptional: normality will remove them on the finite normalization of the original source.

Lemma 5.2 (Comparison at the original divisors). *Let C be an effective rational horizontal boundary on H , log smooth over the generic point of I , and fix a finite extension of $k(I)$. After enlarging it to a finite Galois extension K' , there are a finite normalization $\tau : H^a \rightarrow H$ in a chosen compositum $k(H)K'$, a smooth projective birational model $\mu : H' \rightarrow H^a$, and a morphism $p' : H' \rightarrow I'$ to a smooth projective model of K' . Put $\rho = \tau\mu$ and let C' be the strict transform of the horizontal pullback of C . They can be chosen so that:*

- (i) *C' has SNC support and the geometric generic pair is unchanged, up to the field extension and choice of component;*
- (ii) *every prime of H' mapping onto a prime of the original H is horizontal, or lies with multiplicity one over a prime of I' ;*
- (iii) *at every such prime, the relative-form lattices of $K_{H'/I'} + C'$ and $\rho^*(K_{Y_p} + C)$ agree.*

Consequently, in every divisible degree m , a rational section of $m(K_{H'/I'} + C')$ regular at these primes is an actual section of $m\tau^(K_{Y_p} + C)$ on H^a . The same assertion holds after adding the pullback of a rational divisor on H . Every divisor above a subset of codimension at least two in I' is exceptional over H .*

Proof. There are only finitely many prime divisors of H outside a fixed smooth log-family open. They include every multiple fiber component and every divisor whose image on I has codimension at least two. Their restricted valuations can all be extracted on one smooth base model. Indeed, if a divisorial valuation v of $k(H)$ restricts nontrivially to w on $k(I)$, the residue-field inequality gives

$$\text{trdeg}_k \kappa(w) \geq (\dim H - 1) - r = \dim I - 1.$$

The valuation inequality gives the reverse bound. Thus w is a positive multiple of a divisorial valuation. Extract the finitely many such valuations and resolve their common base model. Every original vertical prime now has a divisorial center, which remains divisorial on higher smooth base models.

All multiplicities in the next step are measured over this prepared base. Normalize it in the prescribed field extension. For each valuation above the selected base primes, take a root of a uniformizer of degree equal to the least common multiple of the relevant original multiplicities. Do this for all the finitely many valuations and take a Galois closure. The ramification indices in these towers multiply, so the final base index is divisible by every required multiplicity; compare [32, Lemmas 9.10–9.11].

The local calculation is at discrete valuation rings. If $t = ux^e$, where u is a unit, base ramification of index n changes the normalized fiber multiplicity to $e/\gcd(e, n)$. After an étale unit-root extension this is the normalization of $s^n = x^e$; separable residue extensions do not change the indices. Thus $e \mid n$ makes the multiplicity one, and further base ramification preserves it. This also protects every original vertical prime outside the chosen list, whose multiplicity was already one. Resolve the altered base, the induced source map, and the boundary without

changing the generic smooth pair. The original divisorial centers and their DVRs retain the properties just established.

At a vertical prime, choose separating relative residue parameters $z_1, \dots, z_r$. For the extracted base model, the ordinary relative lattice has generator $x^{-(e-1)}dz_1 \wedge \dots \wedge dz_r$ up to a unit: complete these with base residue parameters and use $dt = x^{e-1}(eudx + xdu)$. Saturation removes the factor $x^{-(e-1)}$. The same residue differentials remain separating after finite extension, and the new multiplicity is one. Hence the ordinary relative lattice upstairs is the pullback of the original saturated lattice. At a horizontal prime the DVR contains $k(I)$ as a field, so finite separable base extension is étale there. Its relative lattice and boundary coefficient are unchanged. This proves (i)–(iii).

Every prime of H^a lies over a prime of H , since τ is finite. Its strict transform is therefore among the primes just tested. Normality of H^a gives the asserted global section. The same tests hold after tensoring by a pulled-back line bundle. Property (ii) proves the final assertion. Poles at the remaining resolution-exceptional primes are harmless because the section is asserted on H^a , not on that resolution. $\square$

The finite-generation and log-canonical-model statements from [8], whose cited version is formulated over $\mathbf{C}$, are used over k by the following field comparison. Descend the projective morphism, the rational boundary, a log resolution, and an ample decomposition witnessing relative bigness to a field k_0 finitely generated over $\mathbf{Q}$. Enlarge k_0 to define all this finite data. An embedding $k_0 \hookrightarrow \mathbf{C}$ preserves the geometric klt and bigness conditions, so the complex theorem applies. Formation of each graded direct-image sheaf in the relative adjoint ring commutes with field extension. Finite generation descends through the faithfully flat extension $k_0 \subset \mathbf{C}$: finitely many homogeneous generators upstairs involve finitely many elements of the original graded algebra, and faithful flatness detects the vanishing of the resulting cokernel. Applying this on an affine cover of the target gives finite generation over k_0 , hence over k . Relative Proj commutes with field extension, so the resulting canonical model and its comparison maps have the same compatibility. We use this argument also over the parameter fields in Section 6.

We can now apply positivity for families of general-type pairs without changing the section problem on H . The variation below is the variation of the log canonical *model together with its boundary*; it is a different invariant from birational descent of the whole geometric generic fiber.

Proposition 5.3 (Model-pair variation over a logarithmic base). *Let B be an effective rational SNC boundary on H , and let D_I be a reduced SNC divisor on I , such that*

$$B \geq (p^*D_I)_{\text{red}}, \quad \kappa(I, K_I + D_I) \geq 0.$$

Suppose the log canonical divisor of a geometric generic fiber component is big. Then $\kappa(H, K_H + B) \geq r$. If that generic pair is klt, then

$$(5.3) \quad \kappa(H, K_H + B) \geq r + \text{Var}(p, C), \quad C = B_{\text{hor}},$$

Here $\text{Var}(p, C)$ is the minimum transcendence degree over k of an algebraically closed field over which the geometric generic log canonical model pair, including its boundary, is isomorphic to a base extension. Thus it measures descent of that marked model; the invariant $\text{Var}(f)$ in Equation (1.1) measures birational descent of the whole geometric generic fiber. In the klt case, if the left side is at most r , this model pair becomes constant after a finite extension of the base function field. For disconnected geometric generic fibers, the model and its variation refer to a chosen component after adjoining the Stein field.

Proof. If $r = 0$, the chosen geometric generic component is a point and its variation is zero. The saturated relative tangent sheaf has rank zero, so $K_{\mathcal{V}_p} = 0$, and there is no horizontal boundary. Equation (5.2) and a nonzero plurisecion of $K_I + D_I$ give the required nonnegative

litaka dimension directly. The model is a point after the Stein extension. We may therefore assume $r > 0$.

First suppose the generic pair is klt. Apply Lemma 5.2, prescribing the algebraic closure of $k(I)$ in $k(H)$ and choosing the corresponding component. The altered geometric generic fiber is integral and has the same klt big pair. The relative form of Kovács–Patakfalvi’s inequality, [32, Theorem 9.9] with its auxiliary base divisor $M = 0$, gives

$$\kappa(H', K_{H'/I'} + C') \geq r + \text{Var}(p, C).$$

Its hypotheses are satisfied by the smooth projective SNC pair and its klt big generic log divisor; no condition on $\kappa(I')$ is required. The model-pair convention is [32, Definition 6.16, Remark 6.17, Corollary 6.20, Definition 9.3 and Remark 9.4]; the canonical model exists for klt big pairs by finite generation [8]. Finite extension does not change this variation.

By Lemma 5.2, these systems give sections of $m\tau^*(K_{\mathcal{V}_p} + C)$ on the finite normalization of the original H . Their ratios are unchanged. Lemma 2.3 therefore gives the same lower bound for $K_{\mathcal{V}_p} + C$ on H . Choose $0 \neq \alpha \in H^0(I, \ell(K_I + D_I))$. In common divisible degrees m , multiply by $p^*\alpha^{m/\ell}$ and the canonical section of mV in Equation (5.2). These common factors embed the systems into those of $m(K_H + B)$ without changing their rational-map dimensions. This proves Equation (5.3).

For a generic lc pair, decrease the horizontal coefficients by a sufficiently small rational factor. Its log divisor stays big, while the generic pair becomes klt. The required vertical boundary is unchanged. The resulting systems are contained in those for B , which proves the lower bound r ; no assertion about the variation of the unperturbed lc pair is needed.

In the zero-excess klt case, the inequality forces model-pair variation zero. A stable general-type pair has finite automorphism group. The constant moduli point has a representative over k , and its generic isomorphism torsor has a point after a finite field extension. This makes the model pair constant, with its boundary retained. $\square$

6. MARKINGS AND A CONSTANT FINITE NORMALIZATION

The geometric constancy in Proposition 4.10 concerns the fibers of x along a geometric fiber of h . We next control the base of that family. The purpose of the markings below is to recover a finite cover of the geometric generic fiber of $Z \rightarrow T_0$, together with the divisorial images of the old boundary. The markings will then be discarded. There are two separate conclusions: the finite normalization and its morphism must descend, and each positive old-boundary image must become fixed. The second conclusion supplies the actual multiplicity-one component needed in the ramification argument of Section 7.

6.1. The geometric generic parameter space. Fix the algebraic closure Ω of $\mathbf{C}(Y)$ used to define $\text{Var}(f)$, and put

$$b = \overline{\mathbf{C}(T_0)} \subset \Omega,$$

where the bar denotes the algebraic closure inside Ω .

Lemma 6.1. *Let H_0, I, P_0 be the geometric generic fibers over T_0 of W, S, Z , respectively. Write B_0 and D_I for the restrictions of B and D_S . These are smooth projective varieties over b , with the indicated SNC boundary supports. The induced morphisms*

$$p_I : H_0 \longrightarrow I, \quad a_0 : H_0 \longrightarrow P_0$$

have geometrically integral generic fibers, and the combined morphism satisfies

$$(6.1) \quad H_0 \longrightarrow I \times_b P_0 \quad \text{dominant and generically finite,} \quad \dim P_0 = \dim(H_0/I) = d.$$

Moreover,

$$(6.2) \quad \kappa(I, K_I + D_I) \geq 0, \quad B_0 \geq (p_I^* D_I)_{\text{red}},$$

and a geometric general fiber $H_{0,t}$ of a_0 satisfies

$$(6.3) \quad \kappa(H_{0,t}, K_{H_{0,t}} + B_0|_{H_{0,t}}) = 0.$$

Proof. The diagram and dimension assertions follow from Propositions 2.7 and 2.10. The latter identifies $\mathbf{C}(S) = \mathbf{C}(Y)$ inside $\mathbf{C}(X)$, so the present parameter field is the embedded field $b(I) = b \mathbf{C}(Y)$ of Remark 2.11. The generic extensions for $W \rightarrow S$ and $W \rightarrow Z$ are regular. Restriction to the generic point of T_0 and extension to b therefore preserve geometric integrality. Stein factorization over the normal bases also gives connected fibers. Generic smoothness, applied also to the finitely many boundary strata, gives the stated smoothness and SNC properties. The first assertion of (6.2) is the geometric generic form of the corresponding assertion for S_t in Proposition 2.10; nonzero divisible sections are preserved by field extension. The second follows from Proposition 3.3. Finally, (6.3) is precisely Proposition 3.12 after this generic restriction. $\square$

Proposition 5.3 applies over the algebraically closed field b of characteristic zero. Its base hypothesis is precisely $\kappa(I, K_I + D_I) \geq 0$, as supplied by Lemma 6.1. In particular, the base pair in the marked-model argument is (I, D_I) .

6.2. Recognizable markings. The generic I -fiber of H_0 is generically finite over P_0 . We will first decrease the positive horizontal boundary coefficients to make its pair klt, and then add pullbacks of divisors from P_0 to make its adjoint divisor big. The marking line bundles restrict trivially to general a_0 -fibers, while decreasing the old boundary can only lower its adjoint dimension from zero. Upper addition therefore bounds the total section dimension by d , and Proposition 5.3 bounds it below by d plus the variation of the marked model pair. The marked model must become constant after a finite parameter extension. We choose its markings so that they also recover the map to P_0 , and hence the normalization of P_0 in the model's function field. Once this normalization is fixed, Proposition 4.10 will supply a reference variety for the J -family over its function field.

Proposition 6.2. *After a finite extension of $b(I)$ there is a normal projective variety V' over b , finite over P_0 , such that the generic I -fiber of H_0 is birational over P_0 to the base extension of V' . Equivalently, after replacing I by a smooth projective model of that finite extension, there is a birational product description over $I \times_b P_0$,*

$$(6.4) \quad H_0 \dashrightarrow I \times_b V'.$$

Here H_0 denotes the main component after the parameter extension. Every component of the old B_0 which has a divisorial image on V' over the generic point of I has an image defined over b .

Put

$$(6.5) \quad k = b(V'), \quad K = k(I).$$

The original generic x -fiber, viewed by (6.4) as a variety J/K , is geometrically birational to a smooth projective geometrically integral variety J_0/k . One has $\kappa(J_{0,\bar{k}}) = 0$, and, for the minimal positive pluricanonical index p of J , there is a nonzero ordinary p -pluricanonical form on J_0 and

$$(6.6) \quad \dim_k H^0(J_0, pK_{J_0}) = \dim_{\bar{k}} H^0(J_{0,\bar{k}}, pK_{J_{0,\bar{k}}}) = 1.$$

Proof. If $d = 0$, the generic fiber of p_I is geometrically integral and zero-dimensional, so its degree is one. The variety P_0 is a point and the product assertion holds with $V' = \text{Spec } b$. The construction of J_0 at the end of the proof applies in this case as well. Assume for the marking construction that $d > 0$.

The relative model. Let $E_0 = b(I)$ and $H_\eta = (H_0)_{E_0}$. Decrease every positive coefficient of B_0 horizontal over I slightly, keeping it positive and strictly less than one; leave the vertical coefficients unchanged. Denote the resulting boundary by B_0^- . Since the generic support is SNC, the pair (H_η, B_η^-) is klt. The morphism

$$a_\eta : H_\eta \longrightarrow (P_0)_{E_0}$$

is generically finite. Consequently $K_{H_\eta} + B_\eta^-$ is big relative to a_η , since its generic fiber is zero-dimensional. There is therefore a relative log canonical model

$$H_\eta \dashrightarrow C_1 \xrightarrow{c} (P_0)_{E_0}$$

by [8, Theorem 1.2(2)]. To apply that theorem over an algebraically closed ground field, one can first shrink I so that the spread of the decreased pair is klt, apply the theorem to the projective morphism to $I \times_b P_0$, and then restrict to its generic I -fiber. The characteristic-zero field comparison in Section 5 justifies this use of the cited complex formulation over b . The model C_1 is normal, and its adjoint divisor is relatively ample. The canonical rational map is a birational contraction, so it extracts no divisors; see [8, Definition 3.6.6]. The map c is proper, surjective and generically finite, and factors through the normalization of $(P_0)_{E_0}$ in $E_0(H_\eta)$.

Choice of the markings. Choose a very ample line bundle L on P_0 and a general basis $x_0, \dots, x_N$ of $H^0(P_0, L)$. The coordinate divisors will recover each ratio x_j/x_0 up to a scalar. The divisor of $x_0 + x_j$ will determine that scalar, so that the marked model remembers the morphism to this fixed P_0 . We therefore use the divisors

$$(6.7) \quad (x_0 = 0), \quad (x_j = 0), \quad (x_0 + x_j = 0) \quad (1 \leq j \leq N).$$

Their pullbacks receive distinct positive rational coefficients, chosen sufficiently small that the resulting generic pair is still klt. They are also chosen distinct from every coefficient of the decreased generic strict boundary B_η^- . The finite number of dependencies in (6.7) is harmless: on a fixed log resolution, sufficiently small coefficients preserve all the strict klt discrepancy inequalities. Add sufficiently many further general members of $|L|$, again with small coefficients less than one and with no coefficient equal to one of the displayed labels. These can have arbitrarily large total degree. General members of the free pullback system meet the strata on a resolution transversely, so the generic pair remains klt.

Make the following avoidance choices at the same time. None of the hyperplanes contains the image of any relevant divisorial exceptional component on the fixed generic models, any old generic boundary component, or any branch divisor. In addition, none contains the P_0 -image of an I -vertical divisor of H_0 having proper image in P_0 . There are only finitely many divisors in the last condition: such a divisor has image of codimension at least two under the generically finite morphism $H_0 \rightarrow I \times_b P_0$, so it is one of its divisorial exceptional components. An I -vertical divisor dominating P_0 cannot be a component of the pullback of a hyperplane. Thus the last condition prevents any added marking from overlapping a vertical old coefficient-one component or acquiring a vertical divisorial multiplicity. It also makes the corresponding total boundary a boundary with coefficients at most one.

These finitely many Bertini, avoidance, and klt conditions define a nonempty open in the parameter space of the chosen sections, initially over E_0 . Lemma 2.4, applied after extending E_0 to its algebraic closure, shows that this open contains a b -point. Thus all the markings can be chosen over b . The same argument applies to the open set of bases of $H^0(P_0, L)$ and to each additional member.

Write A_{mark} for the weighted sum of all the chosen divisors on P_0 , and set

$$\Delta = B_0^- + a_0^* A_{\text{mark}} \quad \text{on } H_0.$$

The total degree of A_{mark} is chosen sufficiently large that

$$K_{C_1} + (B_\eta^-)_{C_1} + c^* A_{\text{mark}}$$

is ample: a relatively ample divisor plus a sufficiently large pullback of an ample divisor is ample. Twisting by pullback from P_0 tensors each divisible graded piece of the relative adjoint ring by the corresponding power of a line bundle on P_0 . It therefore leaves the relative Proj, and hence the model, unchanged. Equivalently, on a common resolution the exceptional comparison for the old adjoint divisor remains identical after adding this pullback on both sides. Hence C_1 is now the absolute log canonical model of (H_η, Δ_η) .

The pullbacks of the displayed divisors on C_1 are their complete pullback divisors, each with multiplicity one at every component. A divisorial exceptional component occurs in a pullback only when its image is contained in the hyperplane; merely meeting its image does not produce that component. The avoidance of the branch divisors gives multiplicity one at the other generic points. No old boundary prime is shared with a marking. Thus their distinct coefficients identify the complete displayed divisors on the canonical model, even when a pullback is reducible.

The upper bound and the model-pair estimate. On a general a_0 -fiber the marking line bundles restrict trivially, and $B_0^- \leq B_0$. Lemma 6.1 therefore gives restricted adjoint dimension at most zero. The upper addition estimate in Lemma 2.5 gives

$$(6.8) \quad \kappa(H_0, K_{H_0} + \Delta) \leq \dim P_0 = d.$$

If the restricted section system is empty, the same bound follows because the total system is empty. Only the upper bound is needed.

Resolve the support of Δ , writing $\mu : \hat{H} \rightarrow H_0$. Over the generic point of I , use the crepant boundary and replace its negative exceptional coefficients by zero; keep all strict coefficients. Call the resulting generic boundary $\hat{\Delta}_\eta$. The generic pair is effective and klt, and

$$(6.9) \quad K_{\hat{H}_\eta} + \hat{\Delta}_\eta = \mu_\eta^*(K_{H_\eta} + \Delta_\eta) + E_{\text{exc}}, \quad E_{\text{exc}} \geq 0 \quad \mu_\eta\text{-exceptional}.$$

For divisible m , normality and testing at codimension-one points give $\mu_{\eta*} \mathcal{O}(mE_{\text{exc}}) = \mathcal{O}$. The generic adjoint rings are therefore identical. The generic model pair remains the marked pair on C_1 : exceptional divisors added in (6.9) are contracted to the original generic source, and the canonical map from that source extracts no divisors.

On $\hat{H}$, assign coefficient one to each new exceptional divisor vertical over I , and retain the specified coefficients on all other components. This gives an effective rational SNC boundary $\hat{\Delta}$, with generic restriction just described. All strict components above D_I already had coefficient one, and any newly created component above it is vertical over I . Consequently

$$(6.10) \quad \hat{\Delta} \geq ((p_I \mu)^* D_I)_{\text{red}}.$$

The total adjoint sections on $\hat{H}$ inject into those on H_0 : a rational pluriform that is a section upstairs passes the unchanged strict-divisor tests at every prime downstairs. This argument does not require the vertical exceptional coefficients to be crepant. In particular, even if one of these exceptional divisors dominates P_0 , (6.8) still yields

$$(6.11) \quad \kappa(\hat{H}, K_{\hat{H}} + \hat{\Delta}) \leq d.$$

The markings now define a big generic model pair, while the total section dimension still has the upper bound d . We apply Proposition 5.3 to

$$p_I \mu : (\hat{H}, \hat{\Delta}) \longrightarrow (I, D_I).$$

Both varieties are smooth projective over b ; the morphism is surjective with connected fibers and relative dimension d ; the boundary is effective rational SNC; (6.10) holds; and the base has

nonnegative log Kodaira dimension by (6.2). Its geometric generic pair is klt and has big adjoint divisor by the construction of the absolute ample model C_1 . Thus Proposition 5.3 and (6.11) give

$$d + \text{Var}(p_I \mu, \hat{\Delta}_{\text{hor}}) \leq \kappa(\hat{H}, K_{\hat{H}} + \hat{\Delta}) \leq d.$$

The variation here is exactly that of the generic log canonical model *pair*. It is zero. The finite-extension conclusion of Proposition 5.3 makes that model pair isomorphic, after a finite extension E/E_0 , to the base extension of a model pair over b . Denote its underlying normal projective variety by C_* . Its coefficient-labeled displayed divisors are now defined over b .

Recovering the map to P_0 . View C_1 after extension to E as $(C_*)_E$. The displayed markings fix both

$$\text{div}(x_j/x_0) \quad \text{and} \quad \text{div}(1 + x_j/x_0).$$

These are differences of the corresponding complete marking divisors. In particular, the first is the base extension of a divisor over b which is principal after extension to E . Cartierness descends under this faithfully flat extension. Triviality of its associated line bundle descends as well: its space of sections commutes with field extension, and a nonzero descended section becomes a scalar multiple of a nowhere-vanishing section upstairs. We may therefore write

$$(6.12) \quad x_j/x_0 = c_j f_j, \quad f_j \in b(C_*), \quad c_j \in E^*.$$

Here the quotient of two functions with the same divisor is a constant, since C_* is proper and geometrically integral.

The sum marking $(x_0 + x_j = 0)$ has a b -point off the two coordinate markings. Indeed, for a general basis this is a nonempty open part of the pullback of that hyperplane, by surjectivity onto P_0 and $d > 0$; as it is defined over the algebraically closed field b , it has a b -point. Evaluation at such a point in $1 + c_j f_j = 0$ gives $c_j \in b$. Thus every coordinate ratio of the map to the fixed projective embedding of P_0 belongs to $b(C_*)$. The map itself descends to a morphism $C_* \rightarrow P_0$, either by descent of its graph or by descent of regularity of the resulting rational map.

Let V' be the normalization of P_0 in the finite extension $b(C_*)/b(P_0)$. It is finite over P_0 . The map $C_* \rightarrow P_0$ factors properly and birationally through V' . Normalization commutes with the present field extensions: b has characteristic zero and is algebraically closed, so V' is geometrically normal; its base extension is finite normal with the required fraction field and hence has the normalization's universal property. The original generic source H_η after extension to E also maps properly and birationally to V'_E . This proves (6.4).

A proper birational morphism to a normal variety is an isomorphism at the generic point of every target divisor. Consequently a prime on V'_E has its unique strict prime both on the original generic source and on $(C_*)_E$. If the old boundary coefficient at that prime is positive, its decreased coefficient is still positive and occurs in the constant model pair. Its image on V' is therefore defined over b . This proves the asserted control of old boundary images. Notice that this reasoning uses the old boundary; a newly assigned exceptional coefficient on $\hat{H}$ cannot hide such a divisor.

Choosing the reference fiber. Replace I by a smooth projective model of E , and use the notation (6.5). All subsequent parameter extensions will be made relative to this choice. The original x -fiber gives a smooth projective geometrically integral variety J/K with geometric Kodaira dimension zero. Proposition 4.10 applies after extending k to an algebraic closure: the I -directions in (6.4) lie over one geometric generic Z -point. Thus this family is geometrically birationally constant in those directions.

Spread J as a smooth projective family over a nonempty open of I_k . Fix one geometric birational identification with a reference variety over $\bar{k}$, together with its inverse. Their finitely many coefficients lie in a finite extension of $\bar{k}(I)$. Normalize an affine open of $I_{\bar{k}}$ in that extension.

After shrinking, this is a finite surjective cover on which the maps and their inverse identities spread. Shrink again so that their fiberwise domains are dense and the two maps are inverse there. This uses one chosen birational identification and finitely many equations, not a finite-type parameter space for all birational maps.

The image of this comparison cover contains a nonempty open of $I_{\bar{k}}$. Intersect it with the smooth family open. By Lemma 2.4, that intersection contains a point of $I(b)$; a point of the finite comparison cover above it exists over $\bar{k}$. Specializing the original k -family at this b -point gives a smooth projective variety J_0/k . Its base extension to $\bar{k}$ is birational to the chosen reference, hence geometrically birational to J . Geometric integrality and geometric Kodaira dimension zero follow. Ordinary birational invariance of pluricanonical sections and their compatibility with field extension give (6.6); in particular a nonzero ordinary p -pluriform is defined over k itself. $\square$

6.3. A multiplicity-one divisor at every moving test. The boundary control has a stronger consequence than the existence of some reduced component in a degeneration. It supplies a reduced component attaining the precise threshold used in the relative-form comparison.

Definition 6.3. In the notation (6.5), let $E/b(I)$ be a finite extension. A *moving test* is a prime divisor P of V'_E whose image in V' is dense. Its valuation is trivial both on $k = b(V')$ and on E .

For example, if $V' = \mathbf{P}_b^1$ with coordinate x and $b(I) = b(t)$, the divisor $x = t$ is moving, whereas $x = c$ for $c \in b$ is fixed. The next lemma applies to moving divisors over every finite extension of the parameter field.

Lemma 6.4. *At every moving test, including after any finite parameter extension as in Definition 6.3, the old coefficient B_P from Proposition 3.3 is zero. For the relative ordinary p -pluricanonical generator of the J -family, there is an actual source prime Q above P such that*

$$(6.13) \quad m_Q = 1, \quad \frac{1 + r_Q}{m_Q} = \inf_{Q' \mapsto P} \frac{1 + r_{Q'}}{m_{Q'}}.$$

Here $m_Q = \text{ord}_Q(x^*P)$ and $r_Q = p^{-1} \text{ord}_Q(\omega_J)$ are measured in the actual relative canonical bundle, and the infimum includes the divisorial valuations on higher smooth source models. The prime Q is inherited from the actual source in the rank-one construction after the indicated parameter extension, not inserted by a new reduced-exceptional boundary convention.

Proof. A moving test is trivial on k because its residue field contains the function field of its dense image in V' ; it is trivial on E because it is a divisor on an E -variety. The proper birational maps to the finite normalization used in Proposition 6.2 identify its generic point with a strict divisor of the old generic source H_0 . If that divisor had positive old B_0 coefficient, the boundary-image assertion of Proposition 6.2 would place its image in a fixed b -divisor on V' . Such a divisor, after any parameter extension, cannot dominate V' . Thus its old coefficient is zero.

We explain why these are indeed the old tests and orders. A finite separable extension of the parameter field is finite *étale* as a field morphism. Base change is therefore finite *étale* on both the base and the source of the generic family. Height-one primes lie over height-one primes, and actual source components persist, possibly splitting. Their pullback multiplicities and relative canonical orders are unchanged. Generic restriction and adjunction cancel the parameter-canonical factors in the relative canonical bundles. The same assertion for the all-valuation infimum follows by base changing a fixed log resolution computing the threshold. Extensions involved in reaching $b = \overline{\mathbf{C}(T_0)}$ are algebraic, and each prime and the finite data just used descend to a finite stage. Consequently the unchanged-divisor and *étale*-extension compatibility in Proposition 3.3 applies at this strict old prime. No new exceptional boundary coefficient enters the calculation.

Choose an actual component Q attaining the first minimum t_P of Proposition 3.3. With d_Q its old source boundary coefficient, that proposition gives

$$B_P = 1 - \lambda_P + t_P = 0, \quad 1 = \lambda_P - t_P \leq \frac{1 - d_Q}{m_Q} \leq 1.$$

Since m_Q is a positive integer and $0 \leq d_Q \leq 1$, all these inequalities are equalities. Hence $m_Q = 1$ and $d_Q = 0$. Equality in the first inequality says that this same actual component attains $\lambda_P = \inf_{Q' \mapsto P} (1 + r_{Q'})/m_{Q'}$. This proves (6.13). $\square$

7. DESCENT OF THE WHOLE FIBER

Let $b = \overline{\mathbf{C}(T_0)} \subset \Omega$ be the algebraically closed field from Section 6. Incorporate the finite parameter change of Proposition 6.2 into I , and put

$$(7.1) \quad L_0 = b(I), \quad k = b(V'), \quad K = L_0(V') = k(I).$$

The varieties I and V' are geometrically integral over b ; V' is normal and projective. In particular, K/k is regular. The generic x -fiber J/K is smooth, projective, and geometrically integral. Proposition 6.2 supplies a smooth projective geometrically integral reference variety J_0/k , with

$$(7.2) \quad J_{\overline{K}} \text{ is birational to } (J_0)_{\overline{k}}, \quad \kappa((J_0)_{\overline{k}}) = 0, \quad h^0((J_0)_{\overline{k}}, pK_{(J_0)_{\overline{k}}}) = 1,$$

for the ordinary pluricanonical index p used in the rank-one construction. Its one-dimensional degree- p space has a nonzero generator over k .

We will descend $K(J)$, after extending L_0 to its algebraic closure, to a field over b that still contains $k = b(V')$. Thus the coordinates of the relative Iitaka base remain part of the descended fiber. The proof has three steps: make the birational descent cocycle constant, use Lemma 6.4 to remove ramification at moving divisors, and descend the resulting finite *étale* cover.

Here constancy means independence of the parameter variety I . The reference variety J_0 is defined over $k = b(V')$, and the constant cocycle will be defined over a finite extension of k . This dependence on V' is retained when the resulting invariant field is descended to b .

7.1. Constant birational cocycles. We use birational groups with the group structure of their function-field automorphisms. More explicitly, a birational selfmap u is identified with $(u^{-1})^*$; products of the resulting automorphisms are composition. This convention fixes the order in the semilinear actions below.

Lemma 7.1 (Finite splitting with a constant cocycle). *There are a finite Galois extension k'/k in $\overline{k}$, a finite Galois extension $\widetilde{K}/K$ containing k' , and a birational identification*

$$J_{\widetilde{K}} \dashrightarrow (J_0)_{\widetilde{K}}$$

with the following properties. If $G = \text{Gal}(\widetilde{K}/K)$, the transported descent action on $\widetilde{K}(J_0)$ has the form

$$(7.3) \quad \rho_g = z_g s_g, \quad z_g \in \text{Aut}_{k'} k'(J_0),$$

where s_g transports coefficients by g . The map

$$(7.4) \quad G \longrightarrow \text{Aut}_{k'} k'(J_0) \rtimes \text{Gal}(k'/k), \quad g \longmapsto (z_g, g|_{k'})$$

is injective. In particular,

$$(7.5) \quad K(J) \simeq \widetilde{K}(J_0)^G$$

as extensions of K .

Proof. The birational group. Choose compatible algebraic closures $\bar{k} \subset \bar{K}$. A nonzero ordinary pluricanonical form on J_0 excludes uniruledness: if a smooth projective model of J_0 were dominated generically finitely by a ruled variety, its pullback would be a nonzero pluricanonical form on a resolution of that ruled variety, which has none.

Hanamura's structure theorem for non-uniruled projective varieties in characteristic zero provides a suitable smooth projective birational model T of $(J_0)_{\bar{k}}$ for which the reduced scheme of flat graphs of birational maps,

$$\mathcal{B} = \text{Bir}(T)_{\text{red}},$$

is a group scheme locally of finite type over $\bar{k}$. Its identity component

$$A_1 = \mathcal{B}^0 = \text{Aut}^0(T)$$

is an abelian variety. This group structure is intrinsic among such birational models, and represents their algebraic birational group actions. These are exactly the conclusions of [25, Theorems 2.1–2.2], also stated in [9, Theorems 3.8–3.9]. The theorem applies to a non-uniruled projective variety over an algebraically closed characteristic-zero field; it does not require a minimal model of J_0 .

The scheme of flat graphs is an open subscheme of the Hilbert scheme of $T \times T$; see [24, Definitions 1.6 and 1.8, Proposition 1.7] and [9, Proposition 3.7]. The Hilbert functor commutes with field extension, and the open graph condition is geometric. Every birational selfmap over the extended field is therefore represented by a point of the base-changed graph scheme. In characteristic zero a reduced locally finite type scheme is geometrically reduced, so passing to its reduction introduces no new field-valued points after extension. Every connected component of $\mathcal{B}$ contains a $\bar{k}$ -point, since $\mathcal{B}$ is locally of finite type over the algebraically closed field $\bar{k}$. Translation by such a point identifies that component with A_1 . Each of these abelian translates remains connected after extension to $\bar{K}$, and the disjoint union of the original open components exhausts the base-changed scheme. Thus, after identifying the birational groups of T and J_0 , we have

$$(7.6) \quad \begin{aligned} \text{Bir}((J_0)_{\bar{K}}) &= A_1(\bar{K}) \text{Bir}((J_0)_{\bar{k}}), \\ A_1(\bar{K}) \cap \text{Bir}((J_0)_{\bar{k}}) &= A_1(\bar{k}). \end{aligned}$$

This argument allows infinitely many components. The intrinsic group structure also shows that coefficient transport and conjugation preserve A_1 .

Removing the nonconstant abelian part. Set $\Gamma = \text{Gal}(\bar{K}/K)$, and choose an isomorphism of $\bar{K}$ -function fields

$$\theta : \bar{K}(J) \xrightarrow{\sim} \bar{K}(J_0).$$

Let s_σ^J and s_σ^0 denote the natural coefficient transports on these two fields. The automorphisms

$$u_\sigma = \theta s_\sigma^J \theta^{-1} (s_\sigma^0)^{-1} \in \text{Bir}((J_0)_{\bar{K}})$$

satisfy

$$(7.7) \quad u_{\sigma\tau} = u_\sigma \sigma(u_\tau), \quad \sigma(v) = s_\sigma^0 v (s_\sigma^0)^{-1}.$$

The map $\sigma \mapsto u_\sigma$ is continuous when the algebraic points have the discrete topology: the rational maps defining θ and its inverse use only finitely many algebraic coefficients over K . Its image is therefore finite.

For every component occurring among these values choose a representative over $\bar{k}$, choosing the identity in the identity component. By (7.6) this gives a locally constant choice h_σ and expressions

$$u_\sigma = a_\sigma h_\sigma, \quad a_\sigma \in A_1(\bar{K}).$$

The component cocycle law and the second equality in (7.6) imply

$$d_{\sigma,\tau} = h_\sigma \sigma(h_\tau) h_{\sigma\tau}^{-1} \in A_1(\bar{k}).$$

Consequently

$$\alpha_\sigma(a) = h_\sigma \sigma(a) h_\sigma^{-1}$$

defines an action on $A_1(\bar{K})$ preserving $A_1(\bar{k})$. Indeed the discrepancy between $\alpha_\sigma \alpha_\tau$ and $\alpha_{\sigma\tau}$ is conjugation by $d_{\sigma,\tau}$, which acts trivially on the commutative group A_1 .

Give

$$M_1 = A_1(\bar{K})/A_1(\bar{k})$$

the discrete topology. The induced action is continuous: every algebraic point has an open stabilizer for coefficient transport, and h_σ is locally constant. In additive notation the classes $c_\sigma = [a_\sigma]$ satisfy

$$(7.8) \quad c_{\sigma\tau} = c_\sigma + \alpha_\sigma(c_\tau).$$

The abelian group M_1 is uniquely divisible. Multiplication by an integer $n > 0$ is surjective on the points of an abelian variety over either algebraically closed field. Moreover,

$$A_1[n](\bar{K}) = A_1[n](\bar{k}),$$

since the finite torsion scheme is already defined over the algebraically closed smaller field. If na is constant, choose $a_0 \in A_1(\bar{k})$ with $na_0 = na$; then $a - a_0$ is constant torsion, so a is constant. This proves injectivity of multiplication on the quotient as well as its surjectivity. Thus M_1 is a $\mathbf{Q}$ -vector space.

Choose an open normal subgroup $N \subset \Gamma$ contained in the zero set of the continuous cocycle c . Equation (7.8) gives $c_{\sigma n} = c_\sigma$ for $n \in N$. It also gives

$$\alpha_n(c_\sigma) = c_{n\sigma} = c_{\sigma(\sigma^{-1}n\sigma)} = c_\sigma.$$

Thus N fixes every cocycle value. The finite sum

$$\beta = \frac{1}{[\Gamma : N]} \sum_{\tau \in \Gamma/N} c_\tau$$

is well-defined, and summing (7.8) over the cosets yields

$$(7.9) \quad c_\sigma = \beta - \alpha_\sigma(\beta).$$

Choose $t \in A_1(\bar{K})$ lifting β and replace θ by $\theta' = t^{-1}\theta$. Its cocycle is

$$z_\sigma = t^{-1} u_\sigma \sigma(t).$$

The class of its A_1 factor is $-\beta + c_\sigma + \alpha_\sigma(\beta) = 0$ by (7.9). Hence

$$z_\sigma \in \text{Bir}((J_0)_{\bar{k}}) \quad \text{for every } \sigma \in \Gamma.$$

A faithful finite splitting group. The modified cocycle is still continuous, because the algebraic point t has an open stabilizer. Its image is therefore finite. Choose a common finite Galois extension k'/k defining all these finitely many birational maps and their inverses. The map

$$\Gamma \longrightarrow \text{Aut}_{k'}(J_0) \rtimes \text{Gal}(k'/k), \quad \sigma \longmapsto (z_\sigma, \sigma|_{k'})$$

is a homomorphism, with the multiplication $(z, \gamma)(w, \delta) = (z\gamma(w), \gamma\delta)$. It has finite image and an open normal kernel N_0 . Put $\widetilde{K} = \bar{K}^{N_0}$. Then $\widetilde{K}/K$ is finite Galois and contains k' . For $\sigma \in N_0$ we have $z_\sigma = 1$, so θ' intertwines the natural coefficient actions of N_0 . Its rational maps descend to $\widetilde{K}$ by Galois descent. The quotient group $G = \Gamma/N_0$ acts by (7.3), the pairs are faithful by construction, and the fixed-field identity (7.5) follows. $\square$

7.2. Ramification at moving divisors. For a finite extension E/L_0 , a *moving divisor* is a prime divisor of V'_E whose image is dense in V' , as in Definition 6.3. Its valuation is trivial on both k and E . We recall the precise conclusion of Lemma 6.4. For the rational degree- p relative pluricanonical generator ω_J , and any such divisor P , there is an actual source divisor Q above P such that

$$(7.10) \quad m_Q = 1, \quad \frac{1 + r_Q}{m_Q} = \inf_{T \rightarrow P} \frac{1 + r_T}{m_T}, \quad m_T = \text{ord}_T(x^*P), \quad r_T = \frac{1}{p} \text{ord}_T(\omega_J).$$

Orders here are measured in the actual relative canonical bundle; the infimum includes divisors on higher smooth models. The same statement holds after every finite parameter extension.

Proposition 7.2 (Unramifiedness at moving divisors). *The extension $\widetilde{K}/K$ in Lemma 7.1 is unramified at every moving divisor. After a finite extension E/L_0 , each field component of*

$$\widetilde{K} \otimes_K E(V')$$

is likewise unramified at every moving divisor of V'_E .

Proof. We compare divisorial orders before and after splitting. On the constant model there is a unique minimizing divisor. The actual multiplicity-one minimizer in (7.10) will force equality of total and base inertia, and faithfulness will make both trivial.

Relative canonical orders. First work over L_0 . Put $L = K(J)$ and use (7.5) to identify

$$\widetilde{L} = L\widetilde{K} = \widetilde{K}(J_0).$$

Since J/K is geometrically integral, L/K is regular. Hence L and $\widetilde{K}$ are linearly disjoint over K , and restriction identifies

$$(7.11) \quad \text{Gal}(\widetilde{L}/L) \xrightarrow{\sim} \text{Gal}(\widetilde{K}/K) = G.$$

Fix a moving divisor P and a prolongation P' to $\widetilde{K}$. Write $e = e(P'/P)$. If a divisorial valuation Q of L above P prolongs to Q' above P' , write $e_Q = e(Q'/Q)$. On smooth models at these generic divisor points, the multiplicities and relative differential orders satisfy

$$(7.12) \quad m_{Q'} = \frac{e_Q m_Q}{e}, \quad r_{Q'} = e_Q r_Q + e_Q - 1 - (e - 1)m_{Q'}.$$

The first identity is the order of a base uniformizer. For the second, the total canonical differential acquires order $e_Q - 1$ under the finite extension, whereas the base canonical differential acquires order $e - 1$ and is pulled back with multiplicity $m_{Q'}$. These are the tame different orders of characteristic-zero divisorial DVRs. They can also be computed by differentiating uniformizers and completing them by separating residue parameters. Dividing the degree- p calculation by p gives the stated formula.

In particular,

$$(7.13) \quad \frac{1 + r_{Q'}}{m_{Q'}} = e \frac{1 + r_Q}{m_Q} - (e - 1).$$

This is an increasing affine transformation independent of Q .

The unique minimizing divisor. Let R' be the valuation ring of P' and let κ' be its residue field. The valuation is trivial on k , because P is moving. It is also trivial on the finite algebraic extension k'/k : the valuation of a nonzero element algebraic over a trivially valued field must be zero, as follows immediately from its minimal polynomial. Thus

$$(7.14) \quad k' \subset R', \quad k' \hookrightarrow \kappa'.$$

Consider the smooth proper constant model

$$\mathcal{J} = J_0 \times_{\text{Spec } k} \text{Spec } R'.$$

Its special fiber S' is smooth and geometrically integral over κ' . Choose the ordinary regular degree- p generator ω_0 on J_0 . The pulled-back generator ω_J is a scalar multiple $c\omega_0$ over $\widetilde{K}$, because the ordinary degree- p space has dimension one. Put $\ell = p^{-1} \operatorname{ord}_{P'}(c)$ and $D_0 = \operatorname{div}(\omega_0) \geq 0$.

Every divisorial valuation of $\widetilde{L}$ above P' has a center on $\mathcal{J}$, by properness, and that center lies in S' . For such a valuation T , let $A_{\mathcal{J}}(T)$ denote its log discrepancy over the smooth model $\mathcal{J}$. Removing the common scalar shift gives

$$(7.15) \quad \frac{1 + r_T}{m_T} = \ell + \frac{A_{\mathcal{J}}(T) + p^{-1} \operatorname{ord}_T(D_0)}{m_T}, \quad m_T = \operatorname{ord}_T(S').$$

The divisor D_0 is pulled back to the constant model in this formula. It has no special-fiber component. Hence S' itself has $m_{S'} = 1$ and ratio $\ell + 1$.

For every other divisorial valuation centered in S' ,

$$A_{\mathcal{J}}(T) > \operatorname{ord}_T(S') = m_T.$$

This is the pure log terminality of a smooth divisor on a smooth scheme. In local parameters it follows by including the equation of S' and at least one further parameter vanishing at the proper center: the log discrepancy is at least the sum of their positive orders. Since $\operatorname{ord}_T(D_0) \geq 0$, Equation (7.15) proves that S' is the unique minimizer of the ratio above P' .

Triviality of inertia. Take the multiplicity-one minimizer Q supplied by (7.10). It has a prolongation to $\widetilde{L}$; by Galois transport we may arrange that the restriction of this prolongation to $\widetilde{K}$ is the specified P' . Indeed G acts transitively on the prolongations of P , and its action on $\widetilde{L}$ fixes L by (7.11). Conversely, every divisorial valuation above P' restricts under the finite extension to a divisorial valuation above P . Equation (7.13) therefore shows that the chosen prolongation of Q attains the upstairs infimum. The uniqueness just proved identifies it with S' . Consequently $m_{Q'} = m_Q = 1$, and (7.12) gives

$$(7.16) \quad e_Q = e.$$

Let $\mathcal{I}_{Q'}$ and $\mathcal{I}_{P'}$ be the total and base inertia groups, viewed as subgroups of the common Galois group in (7.11). The embedding of base residue fields in total residue fields gives

$$\mathcal{I}_{Q'} \subset \mathcal{I}_{P'}.$$

All residue characteristics are zero, so the residue extensions are separable and the orders of these inertia groups are e_Q and e . Equation (7.16) forces equality of the two inertia groups. Thus every element of $\mathcal{I}_{P'}$ acts trivially on the special-fiber function field $\kappa'(J_0)$.

For $g \in \mathcal{I}_{P'}$, coefficient transport is already trivial on κ' . Equation (7.14) also gives $g|_{k'} = 1$. Its induced action on $\kappa'(J_0)$ is therefore the base extension of the constant map z_g . Such an action is faithful: a birational function-field automorphism over k' that becomes the identity after an injective extension of fields was the identity to begin with. Hence $z_g = 1$. The pair (7.4) is faithful, so $g = 1$. This proves $\mathcal{I}_{P'} = 1$, and hence unramifiedness at P .

Finite parameter extensions. Let E/L_0 be finite. The base change of the finite Galois splitting field is a finite product of fields, with a transitive G -action; each component is Galois over $E(V')$ with group its stabilizer in G . The constant field k' embeds in every component. The same constant maps and the restricted faithful pairs define its semilinear action. Geometric integrality of J and the multiplicity-one conclusion (7.10) both persist. Repeating the preceding argument for this component and its stabilizer proves the stated unramifiedness after E . In particular, no uniform choice of a finite parameter extension is being assumed for all divisors. $\square$

7.3. Descent of the cover and the fiber.

Proposition 7.3 (Descent of the whole fiber). *Let Λ be an algebraic closure of $L_0 = b(I)$. There is a finitely generated field F_b/b containing $k = b(V')$ and an isomorphism*

$$(7.17) \quad \text{Frac}(\Lambda \otimes_b F_b) \simeq \text{Frac}(\Lambda \otimes_{L_0} K(J)).$$

The right-hand side is the function field of the whole geometric generic fiber of the original fibration after identifying Λ with Ω . In particular,

$$(7.18) \quad \text{Var}(f) \leq \dim T_0.$$

Proof. **A fixed open set carrying the cover.** Extend the splitting algebra to $\Lambda(V')$:

$$\mathcal{B}_\Lambda = \widetilde{K} \otimes_K \Lambda(V').$$

It is a finite product of fields and a Galois algebra with group G . We first descend its normalization over a suitable fixed open subset of V' .

Every prime divisor of V'_Λ is either the base extension of a prime divisor of V' or is moving. On an affine chart with coordinate ring A , the map $A \rightarrow A \otimes_b \Lambda$ is faithfully flat. If a height-one prime $\mathfrak{q}$ contracts to $\mathfrak{p}$, flat going down gives $\text{ht } \mathfrak{p} \leq 1$. A height-zero contraction is zero and means that the divisor is moving. If $\text{ht } \mathfrak{p} = 1$, the corresponding integral b -divisor is geometrically integral, since b is algebraically closed. Hence $\mathfrak{p}(A \otimes_b \Lambda)$ is a height-one prime. Its inclusion in $\mathfrak{q}$ is equality. No finite-type assumption on the field extension Λ/b is used here.

A moving prime divisor of V'_Λ descends to a moving prime divisor after some finite extension E/L_0 . Indeed, on the above affine chart, choose finitely many generators of its prime ideal. Their coefficients lie in such an E , since Λ/L_0 is algebraic. The ideal generated by the same equations over E is prime by faithful-flat descent, and has height one by invariance of codimension under field extension. Its contraction to A is still zero. Thus the descended divisor remains moving, and Proposition 7.2 applies to every field component at this finite stage.

It follows that all divisorial branching of $\mathcal{B}_\Lambda$ lies on finitely many fixed divisors. The normalization is finite because the varieties over a field are excellent; its branch locus has only finitely many divisorial components. Remove those divisors and the singular locus of V' . Also remove the fixed closed locus over which the normalization in k'/k is not finite étale. This leaves a smooth nonempty open subset $O \subset V'$ over b . The normalization of O_Λ in $\mathcal{B}_\Lambda$ is finite, normal, and unramified at all height-one points. Purity of the branch locus makes it finite étale. Here the base is regular, the source is normal, the morphism is finite and generically separable, and all divisorial ramification has been removed, which are the hypotheses of Zariski–Nagata purity; see [22, Exposé X, Théorème 3.1]. Denote this finite étale cover by

$$\mathcal{C}_\Lambda \longrightarrow O_\Lambda.$$

Its generic G -action extends by normalization. It is a G -torsor, possibly disconnected, since the generic Galois algebra is a G -torsor and all these covers are finite étale over the connected base.

Let $\mathcal{C}_{1,\Lambda}$ be a connected component, and let $G_1 \subset G$ be its stabilizer. Then $\mathcal{C}_{1,\Lambda} \rightarrow O_\Lambda$ is a connected finite étale G_1 -torsor. Let $\mathcal{T} \rightarrow O$ be the finite étale cover obtained by normalization in k'/k . The inclusion $k' \subset \mathcal{B}_\Lambda$ gives a morphism

$$\mathcal{C}_{1,\Lambda} \longrightarrow \mathcal{T}_\Lambda$$

over O_Λ , compatible with the action of G_1 through its restriction to k' .

Descent with the coefficient field. Finite étale covers are invariant under an extension of algebraically closed fields in characteristic zero. In the needed form, base extension from b to Λ gives an equivalence between the categories of finite étale covers of the connected normal separated finite-type scheme O and of O_Λ . This follows from [33, Theorem 1.1], since every finite cover

is tame in characteristic zero; see also [33, Remark 1.5] and [22, Exposé XIII, Proposition 4.6]. Full faithfulness descends the morphisms as well as the covers.

Apply this equivalence simultaneously to $\mathcal{C}_{1,\Lambda}$, its G_1 -action, and its map to the fixed cover $\mathcal{T}_\Lambda$. We obtain a connected finite étale G_1 -torsor $\mathcal{C}_1 \rightarrow \mathcal{O}$ and a compatible map $\mathcal{C}_1 \rightarrow \mathcal{T}$. Let E_1 be the function field of $\mathcal{C}_1$. Then

$$(7.19) \quad E_1/k \text{ is finite Galois with group } G_1, \quad k' \subset E_1,$$

and the chosen component of $\mathcal{B}_\Lambda$ is $\text{Frac}(\Lambda \otimes_b E_1)$. The embedding of k' and the restriction of the G_1 -action to it are part of this descended diagram.

The invariant field. For $g \in G_1$, the birational map z_g is defined over k' , and $k' \subset E_1$. Hence the semilinear action $z_g s_g$ descends to the field $E_1(J_0)$. Its multiplication law follows from the cocycle law, because the action on k' in (7.19) is the descended one. Define

$$(7.20) \quad F_b = E_1(J_0)^{G_1}.$$

This is a finitely generated field over b . Indeed $E_1(J_0)$ is finitely generated over b and finite over its invariant field. Also

$$(7.21) \quad k = b(V') \subset F_b,$$

since both the coefficient action of G_1 and the constant birational maps fix k . This inclusion retains the coordinates of the entire x -base in the descended field.

The coefficient field and its action have both descended. We now verify (7.17), including its retained k -coordinates. Invariants on a finite product of fields permuted transitively by a finite group are identified with invariants on a chosen factor under its stabilizer: the other coordinates of an invariant element are its translates. Thus after extension to Λ , Equation (7.5) may be computed on the chosen component with group G_1 .

Invariants also commute with extension from b to Λ . For a ring with a finite G_1 -action this follows by applying the averaging projector $|G_1|^{-1} \sum_{g \in G_1} g$. Passage to fraction fields causes no change: if an invariant fraction has denominator a , multiplication by the invariant denominator $\prod_{g \in G_1} g(a)$ puts its numerator in the invariant ring. These arguments apply to the present tensor products, which are domains on the chosen component; the finitely generated fields over the algebraically closed field b are geometrically integral. Consequently

$$\text{Frac}(\Lambda \otimes_b F_b) \simeq (\text{Frac}(\Lambda \otimes_b E_1(J_0)))^{G_1} \simeq \text{Frac}(\Lambda \otimes_{L_0} K(J)).$$

Identification with the original fiber. Proposition 2.10 gives $\mathbf{C}(S) = \mathbf{C}(Y)$ in the original field tower

$$\mathbf{C}(Y) \subset \mathbf{C}(S) \subset \mathbf{C}(W) \subset \mathbf{C}(X).$$

In particular, $\mathbf{C}(T_0) \subset \mathbf{C}(Y)$, with its prescribed algebraic closure $b \subset \Omega$. Before the finite parameter changes the generic parameter field is

$$b(I) = b \mathbf{C}(Y) \subset \Omega.$$

It is algebraic over $\mathbf{C}(Y)$. Each subsequent finite parameter extension can be embedded in Ω , and an algebraic closure of the resulting L_0 is still Ω . We may therefore take $\Lambda = \Omega$.

By Proposition 6.2, after this parameter change the function field of the generic x -base is $L_0(V') = K$. The field $K(J)$ is its total field in the original fibration. Birational model changes preserve these fields, so

$$\text{Frac}(\Omega \otimes_{L_0} K(J)) \simeq \text{Frac}(\Omega \otimes_{\mathbf{C}(Y)} \mathbf{C}(X)).$$

The right-hand side is the function field of the whole smooth geometric generic fiber of f , not just that of J over an algebraic closure of k .

Finally, take a smooth projective b -model of F_b , by a projective compactification and resolution of singularities in characteristic zero. Equation (7.17) says that its base extension to Ω is birational to that whole fiber. Since

$$\mathrm{trdeg}_{\mathbb{C}} b = \dim T_0,$$

the definition of birational variation gives (7.18). $\square$

7.4. The variation inequality.

Proof of Theorem 1.1. If $\kappa(F) = -\infty$, the assertion holds by the stated convention. Otherwise set $d = \kappa(F) \geq 0$. The compactification and logarithmic subadditivity argument in Lemma 2.6, specifically (2.1), gives

$$\bar{\kappa}(U) \geq d + \bar{\kappa}(V).$$

Equations (2.4) and (2.10) identify the dimension of the absolute logarithmic Iitaka base and give

$$\bar{\kappa}(U) = \dim Z = d + \dim T_0.$$

Combining this identity with (7.18) gives $\bar{\kappa}(U) \geq d + \mathrm{Var}(f)$. Taking the larger of the two lower bounds proves

$$\boxed{\bar{\kappa}(U) \geq \kappa(F) + \max\{\bar{\kappa}(V), \mathrm{Var}(f)\}.$$

The dimension-zero cases fit the same field argument. If $d = 0$, then V' is a point over b , so $k = b$ and the geometric constancy already gives the required descent. If I is a point, there are no parameter directions and $\Lambda = b$. If J_0 is a point, its birational group is trivial and the descended field retains precisely the fixed V' coordinates. For a point base or a zero-dimensional fiber the same conventions give the asserted inequality. Throughout, the Kodaira dimension used for the original compact generic fiber is its ordinary Kodaira dimension, and every auxiliary marking has been removed from the final invariant field. $\square$

When U and V are projective, their logarithmic Kodaira dimensions are their ordinary Kodaira dimensions. The same inequality therefore gives the ordinary Iitaka–Viehweg C^+ statement.

8. A NUMERICAL SECOND IITAKA CONSTRUCTION

We record a numerical form of the second Iitaka construction for an arbitrary nef contribution pulled back from an adjoint-positive base. The original base may have nonnegative logarithmic Kodaira dimension; its reduced inverse boundary is retained throughout the section comparisons. The construction retains the dimension of the entire second Iitaka base and, with empty base boundary, yields the ordinary reduction in Section 9.

The argument separates the geometric construction from the numerical step. First, the Kodaira-zero fibers determine a covering family on the original base and hence a product diagram. Exact divisorial orders then transfer sections to the second base. Finally, a uniform volume estimate and interpolation show that this transfer retains all of its dimensions. The nef divisor in the theorem is arbitrary; its origin as a Hodge line is needed only in the subsequent ordinary application.

Theorem 8.1 (The dimension retained by interpolation). *Let $g : W \rightarrow Y$ and $h : W \rightarrow Z$ be surjective morphisms with connected fibers between smooth projective complex varieties. Let B be an effective rational SNC boundary, and let D_Y be a reduced SNC divisor, with the zero divisor allowed. Assume*

$$B \geq (g^* D_Y)_{\mathrm{red}}, \quad \kappa(Y, K_Y + D_Y) \geq 0.$$

Let A be a nef rational divisor on Z such that $K_Z + aA$ is big for all sufficiently large rational a . Put

$$M = h^*A, \quad L(c) = K_W + B + cM, \quad d = \dim(W/Y),$$

and suppose $L(1)$ is big over Y . On smooth birational models, with the boundary rule of Lemma 2.2, there are morphisms

$$W \xrightarrow{q} U \xrightarrow{j} Z, \quad U \xrightarrow{s} T_0, \quad S \xrightarrow{p} T_0, \quad \rho : S \rightarrow Y$$

with the following properties:

- (1) $h = jq$, and q is the relative Iitaka fibration of $K_W + B$ over Z . Its geometric general fiber G is integral and satisfies $\kappa(G, K_G + B_G) = 0$.
- (2) The map g factors through S ; ρ is birational, and $W \rightarrow (S \times_{T_0} U)_{\text{main}}$ is dominant and generically finite. The generic fibers of $S \rightarrow T_0$ and $U \rightarrow T_0$ are geometrically integral. With $D_S = (\rho^*D_Y)_{\text{red}}$ made SNC, one has $\kappa(S_t, K_{S_t} + D_S|_{S_t}) = 0$ for very general t .
- (3) One has

$$(8.1) \quad \kappa(W, L(1)) = \dim U = d + \dim T_0, \quad \kappa(Y, K_Y + D_Y) \leq \dim T_0.$$

Rational linear equivalence in the definition of M is sufficient. All section comparisons are with the fixed initial source model.

A divisor is big over Y when its restriction to the geometric generic fiber of g is big. All boundaries in the theorem have coefficients in $[0, 1]$. A point has Kodaira dimension zero, so the statement includes zero-dimensional bases and fibers.

Setting $D_Y = 0$ recovers the ordinary numerical construction, including $D_S = 0$ and $\kappa(S_t) = 0$. In the logarithmic setting the inverse boundary condition is needed both to contract the base pluriform and to apply the model-pair comparisons.

8.1. A covering family with Kodaira dimension zero. The first task is to determine the dimension of the family of images of a Kodaira-zero fiber. A logarithmic pluriform on Y gives both nonnegativity for each moving image and the differential-rank calculation for the family.

Fix $0 \neq \alpha \in H^0(Y, \ell(K_Y + D_Y))$, replacing it by a tensor power whenever needed so that ℓ clears the boundary denominators in the following comparisons. We will use two consequences of pulling it back to a family covering Y .

First, a general member S_b of any covering family of integral subvarieties of Y , resolved with reduced inverse boundary D_{S_b} , has $\kappa(S_b, K_{S_b} + D_{S_b}) \geq 0$. Indeed choose $\dim Y - \dim S_b$ parameter directions spanning its generic normal space. The family over a transverse parameter slice has dimension $\dim Y$ and maps generically finitely to Y . Pull back α to a resolution of this family and its inverse boundary, and use adjunction on the general fiber. This produces a nonzero logarithmic pluriform on a resolution of S_b . The assertion also holds after a finite cover with its reduced inverse boundary. These image boundaries consist only of the reduced inverse image of D_Y ; exceptional divisors outside that inverse image are not automatically added. Logarithmic pullback is regular away from that inverse image, and this convention ensures compatibility with the allowed boundary on the source. The construction uses the family of *images*; its domains need not map generically finitely to Y .

Lemma 8.2 (Rank of the image family). *Let $q : V \rightarrow U$ and $g : V \rightarrow Y$ be surjective morphisms between smooth connected projective complex varieties. Suppose the general fiber G of q is geometrically integral, B is an effective rational boundary, (V, B) is log smooth over a dense open of U , D_Y is reduced SNC, $B \geq (g^*D_Y)_{\text{red}}$, $\kappa(Y, K_Y + D_Y) \geq 0$, $\kappa(G, K_G + B_G) = 0$, and $G \rightarrow g(G)$ is generically finite. Then the family of reduced image cycles $g(G)$ has dimension $\dim Y - \dim G$.*

Proof. Put $r = \dim G$ and $k = \dim Y - r$. On a smooth parameter open, the image family is flat. At a general $y \in S = g(G)$ the normal evaluation map

$$e_y : T_u U \longrightarrow N_{S/Y, y}$$

has rank k , since the family covers Y . Its nonzero maximal minors are obtained from transverse k -dimensional parameter slices. Pullback of α to each sliced family and adjunction produce sections of $\ell(K_G + B_G)$. Their local coefficients are the ℓ -th powers of the minors, multiplied by the same tangential Jacobian and the coefficient of α . The reduced inverse-boundary inclusion makes every logarithmic pole allowed by B_G , proving global regularity in the indicated logarithmic bundle. Changing lifts of parameter vectors by vectors tangent to G does not change their wedge determinant.

Every nonzero section space of a divisor with Iitaka dimension zero is one-dimensional. Ratios of the resulting sections are therefore constant on G . The common factors cancel, so all nonzero minor ratios are constant as well: their ℓ -th powers are constant, and a rational function on an integral complex variety with a constant positive power is itself constant. Thus the Plücker point of the rank- k quotient e_y , and hence $\ker(e_y)$, is independent of general y . A parameter vector in that kernel induces an embedded deformation of S zero at its generic point. Such a deformation is a homomorphism from the conormal sheaf to $\mathcal{O}_S$; since S is integral, it is detected at the generic point and must be zero. The common kernel is therefore the kernel of the image-cycle map. In characteristic zero its generic differential rank is its image dimension, namely k . $\square$

8.2. Construction of the product. We first verify that a relative Iitaka fibration exists over Z . After constructing it, the preceding rank calculation will identify the parameter space of its images in Y .

For a general fiber W_z of h , let S_z be its resolved image in Y , with reduced inverse boundary D_{S_z} . The covering-family observation gives $\kappa(S_z, K_{S_z} + D_{S_z}) \geq 0$. The fibers of $W_z \rightarrow S_z$ have big log canonical divisor. To see this restriction carefully, write a multiple of $L(1)$, over a dense open of Y , as a relatively ample divisor plus an effective divisor. General intersections with h -fibers move in g -fibers and avoid the fixed effective support. The ample part remains ample on them; M is trivial there; and adjunction after resolving the combined map identifies the restriction of $K_W + B$ with their log canonical divisor. The reduced inverse boundary on S_z pulls back into B_{W_z} . The lc general-type bound following Equation (5.3) now gives

$$\kappa(W_z, K_{W_z} + B_{W_z}) \geq 0.$$

Take the relative Iitaka fibration $W \xrightarrow{q} U \xrightarrow{j} Z$. The relative Iitaka theorem [27, Section 6, Remark 1] gives

$$(8.2) \quad \kappa(G, K_G + B_G) = 0, \quad \dim U = \dim Z + \kappa(W_z, K_{W_z} + B_{W_z}).$$

One applies it to the full log divisor on every chosen model: Lemma 2.2 identifies the relative section sheaves as well as the global section spaces.

If $G \rightarrow g(G)$ had positive-dimensional fibers, the same ample-plus-effective restriction would make their log canonical divisors big. The image has nonnegative logarithmic Kodaira dimension by the covering-family observation. The source boundary on G contains the reduced inverse boundary of that image, so the lc general-type bound would then contradict $\kappa(G, K_G + B_G) = 0$. Hence $G \rightarrow g(G)$ is generically finite.

Let T_0 be a smooth model of the relative algebraic closure in $\mathbf{C}(U)$ of the image-cycle field. Pull the universal reduced image family to T_0 and resolve its dominating component to obtain $S \rightarrow T_0$. The regular extension $\mathbf{C}(U)/\mathbf{C}(T_0)$ ensures that $U \rightarrow T_0$ has geometrically integral generic fiber. The generic image member is geometrically integral too: after this extension it

is dominated by the geometrically integral generic fiber of q . Thus the fiber product has one component dominating T_0 . The maps fit into

$$(8.3) \quad \begin{array}{ccccccc} W & \longrightarrow & (S \times_{T_0} U)_{\text{main}} & \longrightarrow & S & \xrightarrow{\rho} & Y \\ q \downarrow & & \downarrow & & p \downarrow & & \\ U & \xlongequal{\quad} & U & \xrightarrow{s} & T_0 & & \end{array}$$

By Lemma 8.2, $\dim T_0 = \dim Y - \dim G$. Both the top-left map and ρ are consequently generically finite, and $\dim U - \dim T_0 = d$. Moreover $\mathbf{C}(Y) \subseteq \mathbf{C}(S) \subseteq \mathbf{C}(W)$, with the first extension finite. Connectedness of g makes $\mathbf{C}(Y)$ algebraically closed in $\mathbf{C}(W)$, so ρ is birational.

Put $D_S = (\rho^* D_Y)_{\text{red}}$ and resolve this support. The source boundary contains $(g_S^* D_S)_{\text{red}}$, since both reduced inverse boundaries are supported over D_Y . The pullback of α to S restricts, by adjunction, to a nonzero logarithmic pluriform on S_t . Logarithmic pullback to G , which generically finitely dominates S_t with the required inverse boundary, injects its systems into those of $K_G + B_G$. Therefore $\kappa(S_t, K_{S_t} + D_S|_{S_t}) = 0$. Because ρ is birational and D_S is exactly the reduced inverse boundary, logarithmic pullback and pushdown at every original prime identify the divisible section spaces of $K_Y + D_Y$ and $K_S + D_S$. Equivalently, the latter spaces lie between the pulled-back spaces and the identical spaces obtained by adding every reduced exceptional divisor, as in Lemma 2.2. The elementary upper addition bound gives

$$(8.4) \quad \kappa(Y, K_Y + D_Y) = \kappa(S, K_S + D_S) \leq \dim T_0.$$

It remains to prove that $L(1)$ retains all $\dim U$ directions.

8.3. The divisor measuring descending forms. The product diagram determines a rank-one family of forms along q . We now record exactly how much a coefficient from $\mathbf{C}(U)$ may have a pole while the associated form stays regular on the original source. These coefficients, rather than a limiting assertion about Kodaira dimensions, will give the transfer used in the last step.

Fix the original model W_0 whose sections are to be recovered. Apply Lemma 3.6 to q , and preserve its conclusion relative to W_0 under the further modifications below. Thus every divisor of a source model with codimension-two image on U is exceptional over W_0 .

Write $M_U = j^* A$. Take the ℓ -th tensor power of a rational relative top form on U/T_0 , wedge it tensorwise with the logarithmic ℓ -pluriform α on S , and pull back through the generically finite map in Equation (8.3). In each common degree m divisible by ℓ , the (m/ℓ) -th power of this combined form spans the entire fiber section space, since $\kappa(G, K_G + B_G) = 0$. After trivializing the bundles pulled back from U , division of a degree- m section by this power gives a ratio constant on the geometric generic q -fiber; connectedness of q puts that ratio in $\mathbf{C}(U)$.

For a prime $P \subset U$, let τ_P generate $\ell K_{U/T_0}$ at its generic point and let η_P be its combined rational ℓ -pluriform on W . Define

$$(8.5) \quad c_P = \min_{E \mapsto P} \frac{\ell^{-1} \text{ord}_E(\eta_P) + b_E}{\text{ord}_E(q^* P)}, \quad C = \sum_P c_P P, \quad P_U(c) = K_{U/T_0} + C + c M_U.$$

Here b_E is the coefficient of B and the order of η_P is measured in ℓK_W . Only finitely many coefficients are nonzero: outside the divisors of one fixed rational frame, its combined form, B , and the divisorial nonsmooth locus, every order is zero.

Fix a rational value of c and take common degrees divisible by ℓ and the indices of all divisors involved. Division by the fixed rank-one form, and multiplication by that form in the reverse direction, identify the full divisible section systems:

$$(8.6) \quad H^0(W, mL(c)) \xrightarrow{\sim} H^0(U, mP_U(c)).$$

For the forward map, a rational coefficient of order a at P has order

$$a \operatorname{ord}_E(q^*P) + (m/\ell) \operatorname{ord}_E(\eta_P) + mb_E$$

upstairs, so its allowed orders are exactly $a \geq -mc_P$. Conversely a section of $mP_U(c)$ satisfies every divisorial test on W_0 : divisors dominating U are handled by the generic rank-one form, divisors over base primes by this minimum, and all other divisors are exceptional over W_0 . Normality then gives a section on W_0 . The converse lands on the fixed original model W_0 , whose systems agree with those on the current model by Lemma 2.2. Consequently these are mutually inverse comparisons of the full divisible section systems. Tensor products use powers of the same generating form, so multiplication and section ratios are preserved, as are the dimensions of their rational-map images.

Lemma 8.3 (The horizontal and vertical orders). *On suitable fixed models, write $C = C_h + C_v$ relative to $U \rightarrow T_0$. The divisor C_h is an SNC boundary over the generic point of T_0 , $C_v \geq -R_s$, and*

$$(8.7) \quad P_U(c) \geq K_{V_s} + C_h + cM_U.$$

For very general t , the same minimum construction for $q_t : W_t \rightarrow U_t$, using the adjunction form on S_t , gives $C_h|_{U_t}$. In particular, for every rational divisor Q on U_t ,

$$(8.8) \quad H^0(W_t, m(K_{W_t} + B_{W_t} + q_t^*Q)) \hookrightarrow H^0(U_t, m(K_{U_t} + C_h|_{U_t} + Q)).$$

Proof. First check specialization. Fix rational frames and include their divisors, the combined form, B , and all divisorial nonsmooth data in one finite support. Record separately their codimension-two images on U . At very general t , vertical members are absent and horizontal strata meet the fiber transversely. Their coefficients, orders, and multiplicities are unchanged. Every minimizing divisor $E \rightarrow P$ remains surjective after base change, so every component of P_t receives an unchanged minimizing value. The complements of the order-computation loci have smaller fiber dimension and contain no such component. Thus the minimum remains c_P .

A listed divisor dominating U cannot create an extra vertical test: over the geometric generic point of T_0 , its components are conjugate, and one dominates the geometrically integral U_t , hence all do. This spreads after shrinking the base. A codimension-two image stays of codimension at least two, or is avoided. Outside the finite support the frames are units and the generic divisor fibers are smooth and reduced. The fiber minimum is therefore exactly $C_h|_{U_t}$. Division by the rank-one fiber form proves Equation (8.8); no extension of a fiber section to W is asserted.

For a horizontal P , work on such a fiber. A regular top form on U_t wedged with the logarithmic pluriform on S_t pulls back with only the poles allowed by B_{W_t} , so $c_P \geq 0$. In the finite normalization of $S_t \times U_t$, choose a divisor over $S_t \times P_t$. Its ramification index e gives the value $(e - 1 + b_E)/e \leq 1$, because the form from S_t is a unit at its generic point. Thus $c_P \leq 1$.

For a vertical prime P , work at its discrete valuation ring, with uniformizer x , and choose a local volume form θ on T_0 . If r_P is the coefficient of R_s , write $s^*\theta = x^{r_P}\beta$ with β primitive. The saturated subspace defined by this decomposable form is a direct summand over the discrete valuation ring. There is consequently a regular complementary d -form λ with $\beta \wedge \lambda$ a unit top form. As a relative form, λ represents $x^{r_P}\tau$, where τ generates K_{U/T_0} . Wedging $q^*\lambda$ with the form from S shows that $(q^*x)^{\ell_{r_P}}\eta_P$ is allowed by B : its only possible logarithmic poles lie in the reduced inverse boundary already contained in B . Therefore $c_P \geq -r_P$. This argument also applies when the image of P has codimension greater than one on T_0 .

Finally resolve the finite support on U , resolve the source map, and recompute the minimum. At an old prime, old minimizing valuations persist and pullback of a log-regular form stays regular with the boundary rule of Lemma 2.2. Thus old coefficients do not change. New horizontal

coefficients satisfy the same product calculation, giving SNC horizontal support and coefficients in $[0, 1]$. Equation (5.1) gives Equation (8.7). $\square$

8.4. Complete systems and finite group actions. Return to Theorem 8.1. Fix a rational $c \in (0, 1)$ close enough to one that $L(c)$ is still big over Y , and fix the models in Lemma 8.3. The next objective is to prove that, for very general $t \in T_0$,

$$(8.9) \quad D_t := K_{U_t} + C_h|_{U_t} + cM_U|_{U_t} \quad \text{is big.}$$

When $d = 0$, this is the zero-dimensional assertion. For $d > 0$ we will bound the volumes of small ample perturbations of D_t below by one positive constant. This bound must be independent of the perturbation: full Iitaka dimension for each positive perturbation alone would not prove bigness of D_t in the limit.

The zero-excess case of Proposition 5.3 will make the auxiliary general-type model pairs constant after finite extensions. The extensions may vary with the perturbation. The next two results retain complete model systems in all divisible degrees and bound the loss when taking invariants by the effective action on the section image. The first supplies one fixed normalization and one fixed open set for every required degree of each such model.

Proposition 8.4 (Complete systems in an isotrivial family). *Let $p : H \rightarrow I$ be a surjective morphism of smooth integral projective varieties over an algebraically closed field of characteristic zero. Let B be an effective rational SNC boundary on H , let D_I be a reduced SNC divisor on I , and assume $B \geq (p^*D_I)_{\text{red}}$ and $\kappa(I, K_I + D_I) \geq 0$. Put $r = \dim H - \dim I$. Choose a geometric generic fiber component after adjoining the Stein field. Suppose it is klt of log general type and its log canonical model pair is isotrivial; denote its fixed model by (F_c, Δ_c) . There are a finite Galois base extension, the normalization $\tau : H^a \rightarrow H$ in the corresponding compositum component, and an integer $m_0 > 0$ such that*

$$(8.10) \quad H^0(F_c, m(K_{F_c} + \Delta_c)) \hookrightarrow H^0(H^a, m\tau^*(K_H + B)) \quad (m \in m_0\mathbf{N}).$$

On one fixed dense open of a smooth altered base, restriction to a general fiber gives the pulled-back complete system of this model, times a common nonzero factor. Both the open and m_0 work for all these degrees. The isotriviality hypothesis holds in particular if $\kappa(H, K_H + B) \leq r$.

Proof. Write $C = B_{\text{hor}}$, and let $V = R_p + B_{\text{vert}} - p^*D_I \geq 0$ as in Equation (5.2). Prescribe the finite extension identifying the chosen generic model with (F_c, Δ_c) , including its Stein field, in Lemma 5.2. A model section defines a relative rational pluriform on H' . At horizontal primes it is allowed by the generic canonical-ring comparison. At a vertical prime tested by Lemma 5.2, work over $R = \mathcal{O}_{I', D}$ and compare with $(Z, \Delta_Z) = (F_c, \Delta_c) \times \text{Spec } R$. Its reduced special fiber Z_0 makes $(Z, \Delta_Z + Z_0)$ log canonical. On a common graph resolution, the discrepancy a_E of the tested valuation satisfies

$$a_E - \text{ord}_E(t) \geq -1, \quad \text{ord}_E(t) = 1, \quad a_E \geq 0.$$

Use the same base canonical divisor on both sides of the comparison, so its pullbacks cancel and these are also relative discrepancies. In a degree clearing the model index, the zero divisor of a model section is effective Cartier. Its pullback is effective, and the remaining relative-form order is $ma_E \geq 0$. Thus every model section is regular at every tested prime, including those centered on the boundary or singular locus of the fixed model.

Lemma 5.2 gives sections of $m\tau^*(K_{Y_p} + C)$. Fix $0 \neq \alpha \in H^0(I, \ell(K_I + D_I))$. Multiplication by its pulled-back m/ℓ -th power and by the section of $m\tau^*V$ gives Equation (8.10), using Equation (5.2). All factors are independent of the chosen model section and nonzero on a general fiber.

To fix one open for every degree, use finite generation of the generic log canonical ring [8]. Choose a divisible index m_1 for which its Veronese is generated in degree one and $A_c = m_1(K_{F_c} + \Delta_c)$ is basepoint-free Cartier. Write $L_\eta = K_{H'_\eta} + C'_\eta$ on the generic fiber of p' . Resolve the base ideal of $|m_1 L_\eta|$ and its model map on one common graph, with maps u to H'_η and v to the fixed model. The free and fixed parts give

$$u^*(m_1 L_\eta) = v^* A_c + E_{\text{fix}}, \quad E_{\text{fix}} \geq 0.$$

Generation in degree one implies, for every $a \geq 1$,

$$|au^*(m_1 L_\eta)| = v^* |aA_c| + aE_{\text{fix}}.$$

Spread these maps and this divisor identity over one dense open of I' . Remove the images of its finitely many vertical terms and the zeros of the base multiplier, as well as $p'(\text{Supp } \rho^* V)$. Increase m_0 to a common multiple of m_1 and the earlier indices. Every restricted comparison is then a tensor power of the same free-and-fixed-part identity, so this open and index work for all $m \in m_0 \mathbf{N}$. Injectivity is already visible on the chosen generic fiber. The last assertion follows from Proposition 5.3. $\square$

The alteration making a general-type model constant can have arbitrarily large degree. The next observation avoids dividing section growth by that degree. It bounds only the action on the image of the system.

Lemma 8.5 (A degree bound for invariant growth). *Let D be a rational Cartier divisor on a normal projective complex variety H , with $\kappa(H, D) = d > 0$. Let $H \dashrightarrow V$ be dominant, where $\dim V = d$, and let $\pi : H' \rightarrow H$ be a finite normal Galois cover. Suppose that in every sufficiently divisible degree m :*

- (1) *the image field of $|m\pi^* D|$ contains $\mathbf{C}(V)$;*
- (2) *one fixed integral subvariety $F' \subset H'$ dominates that image and maps generically finitely to V by the induced map, with degree at most e_0 ;*
- (3) *a subspace of the restriction image of $H^0(H', m\pi^* D)$ on F' is, up to a common rational factor nonzero at the generic point of F' , the pullback under a dominant rational map from F' of the complete degree- m system of a fixed ample rational divisor on a d -dimensional projective model, whose volume is at least $v > 0$.*

Then

$$(8.11) \quad \limsup_n \frac{d! h^0(H, nD)}{n^d} \geq \frac{v}{e_0}.$$

The cover, the ample model, and the degree index may vary in applications; v and e_0 alone control this bound.

Proof. Choose one sufficiently divisible m for which the ample-model system is an embedding, and let X_m be the complete-system image. Finite pullback preserves Iitaka dimension, so $\dim X_m = d$. Because F' dominates X_m , restriction of the global sections defining X_m is injective on their image: a section whose restriction vanishes gives a hyperplane containing the dense image of F' , and hence vanishes on X_m . The subspace in (3) therefore lifts to a linear subsystem on X_m . Its rational linear projection has as its image precisely the embedded ample model. A general linear section of the projected image lifts to a linear section of X_m ; the points in a general finite fiber are counted with their positive multiplicities. Hence $\deg X_m$ is at least the product of the projection degree and the degree of that model. In particular,

$$\deg X_m \geq vm^d.$$

The pullback linearization gives an action of the Galois group Γ on the complete system and on X_m . It fixes the embedded field $\mathbf{C}(V)$, since this field comes from H . The tower

$$\mathbf{C}(V) \subseteq \mathbf{C}(X_m) \subseteq \mathbf{C}(F'), \quad [\mathbf{C}(F') : \mathbf{C}(V)] \leq e_0$$

shows that its effective image on X_m has order $e \leq e_0$.

Let R be the homogeneous coordinate ring of X_m . Evaluation embeds R_k equivariantly into $H^0(H', km\pi^*D)$. Its invariants descend to $H^0(H, kmD)$, since $(\pi_*\mathcal{O}_{H'})^\Gamma = \mathcal{O}_H$. Choose a positive integer r divisible by the orders of the characters with which every stabilizer in the *full* group Γ acts on the fibers of $\mathcal{O}_{X_m}(1)$; for example, $|\Gamma|$ suffices. Thus the line bundle $\mathcal{O}_{X_m}(r)$ with its given linearization descends to an ample line bundle A_m on the finite quotient X_m/Γ . The quotient has generic degree e , the order of the effective action, even when the kernel acts nontrivially by scalars on the original line bundle. It follows that $A_m^d = r^d \deg(X_m)/e$.

For sufficiently large n , the coordinate-ring piece R_n agrees with $H^0(X_m, \mathcal{O}_{X_m}(n))$: this is the restriction sequence in projective space and Serre vanishing for the ideal sheaf of X_m . Finite quotient descent identifies the invariant sections in degree rk with $H^0(X_m/\Gamma, A_m^k)$. The Hilbert polynomial therefore gives

$$\dim R_{rk}^\Gamma = \frac{r^d \deg(X_m)}{e d!} k^d + O(k^{d-1}).$$

Normalize in degree mrk and let k tend to infinity. Cancelling r^d gives the lower bound

$$\frac{\deg(X_m)}{em^d} \geq \frac{v}{e_0}.$$

Thus even an arbitrarily large scalar kernel affects only the divisibility index, not the claimed lower bound. $\square$

8.5. Uniform volume on a fiber of the second base. We now prove Equation (8.9) with c and the models fixed as above. Choose a very general $t \in T_0$. The case $d = 0$ has already been noted, so assume $d > 0$ and abbreviate

$$H = W_t, \quad I = S_t, \quad V = U_t, \quad e_0 = \deg(H \rightarrow I \times V).$$

Put $D_I = D_S|_{S_t}$. We have

$$\kappa(I, K_I + D_I) = 0, \quad B_H \geq (p_H^* D_I)_{\text{red}}, \quad \dim V = d,$$

where $p_H : H \rightarrow I$ is the induced morphism, and $L(c)|_H$ is big over I . The fiber order test in Equation (8.8) bounds the Iitaka dimension of every divisor $L(c)|_H + q_t^* Q$ by d .

Fix an ample rational divisor H_V on V and, for rational $\delta > 0$, put

$$N_\delta = L(c)|_H + \delta q_t^* H_V.$$

Before choosing δ , decrease the coefficients of B_H horizontal over I slightly to obtain B_H^- , leaving all vertical coefficients fixed. Thus the coefficient-one inverse image of D_I is retained. Choose the decrease so small that $K_H + B_H^- + cM|_H$ remains big on every geometric generic fiber component over I . Its finitely many component volumes have a fixed positive lower bound v . Adding the nef divisor $\delta q_t^* H_V$ cannot lower this bound.

The divisor $cM_U|_V + \delta H_V$ is ample. A sufficiently divisible general member of its pulled-back system gives an snc rational boundary Θ_δ with small coefficients. Thus

$$E_\delta := K_H + B_H^- + \Theta_\delta \sim_{\mathbf{Q}} N_\delta - (B_H - B_H^-)$$

has klt general fibers over I , with volume at least v . The pair still contains the reduced inverse image of D_I , and its logarithmic base has Kodaira dimension zero. Equation (5.3), the inclusion

of these systems in those of N_δ , and the rank-one upper bound imply

$$d \leq \kappa(H, E_\delta) \leq \kappa(H, N_\delta) \leq d.$$

Their general-type model pairs consequently have variation zero. Proposition 8.4 supplies a finite normal Galois cover $\tau : H^a \rightarrow H$ induced by an extension of $\mathbf{C}(I)$, with complete ample-model subseries of volume at least v . Multiplying by the section of $B_H - B_H^-$ places them in the systems of τ^*N_δ . This common factor is nonzero on a geometric general fiber. The extension and the model may depend on δ .

We check the two remaining conditions of Lemma 8.5. The same argument for $\delta/2$ gives a nonzero section of a multiple of $N_{\delta/2}$. Its powers multiplied by the systems of $(\delta/2)q_t^*H_V$ have ratios generating $\mathbf{C}(V)$ in sufficiently divisible degrees. Hence the complete-system image field contains $\mathbf{C}(V)$, both downstairs and after finite pullback.

Next choose a smooth comparison model $\hat{H} \rightarrow H^a$ mapping to the altered smooth base I' and to its fixed canonical model. Use the fixed open and the fixed index in Proposition 8.4; these arise from one graph and one divisor identity, rather than degree-dependent open sets. Choose a very general fiber component $\hat{F}$ and let $F' \subset H^a$ be its reduced image. The birational map $\hat{H} \rightarrow H^a$ is birational on this component, since a geometric general fiber component is not contained in its exceptional locus. Its restricted model system has image dimension d and therefore dominates the d -dimensional complete-system image in every required degree. This fixes a single F' for the application of the lemma.

Finally, $[\mathbf{C}(F') : \mathbf{C}(V)] \leq e_0$, independently of the altered base. To see the precise bound, write $K_H = \mathbf{C}(H)$, $K_I = \mathbf{C}(I)$, and $K_{I'} = \mathbf{C}(I')$. For the selected compositum component of the base change, its degree over $I' \times V$ is

$$e' = [K_H K_{I'} : K_{I'}(V)] \leq [K_H : K_I(V)] = e_0.$$

Shrink one dense open of I' so that the corresponding finite map has this generic degree on its fibers. In characteristic zero the geometric generic fiber is reduced, and the degrees of its components over V sum to e' . The same holds for sufficiently general fibers after this shrinking. The chosen F' is birational to one such component, so its degree is at most e' . Intersect this open with the open already fixed for the model systems and choose F' there. The common multiplying sections are nonzero at its generic point, and the same F' works in every required degree. This argument includes any preliminary Stein extension and never divides by $[K_{I'} : K_I]$.

Lemma 8.5 now gives

$$\limsup_n \frac{d! h^0(H, nN_\delta)}{n^d} \geq v/e_0.$$

The fiber order test injects these spaces into those of $n(D_t + \delta H_V)$. Since $\dim V = d$, we obtain the ordinary volume bound

$$\text{vol}_V(D_t + \delta H_V) \geq v/e_0 \quad (\delta \in \mathbf{Q}_{>0}).$$

Continuity of volume [34, Section 2.2.C] gives $\text{vol}_V(D_t) \geq v/e_0 > 0$, proving Equation (8.9).

8.6. Interpolation on the second base. Only one positivity input remains: relative bigness gives a pseudoeffective saturated relative divisor, whereas a big divisor from the base supplies all base directions. We state both forms of weak positivity before interpolating between them.

Lemma 8.6 (Two consequences of weak positivity). *Let $p : H \rightarrow I$ be a surjective morphism between smooth connected projective complex varieties.*

- (i) *If Δ is a rational SNC boundary, Q is a big rational divisor on I , and the generic fiber is connected, then*

$$\kappa(H, K_{H/I} + \Delta + p^*Q) \geq \dim I + \kappa(H_i, K_{H_i} + \Delta_i).$$

- (ii) *If C is a rational horizontal boundary log smooth over the generic point of I , P is a nef real divisor, and $K_H + C + P$ is big over I , then $K_{\mathcal{V}_p} + C + P$ is pseudoeffective.*

Proof. Apply Lemma 5.2, including the Stein field when needed. For a log smooth pair with connected fibers, twisted weak positivity says that

$$\mathcal{E}_m = p'_* \mathcal{O}_{H'}(m(K_{H'/I'} + C'))$$

is weakly positive in divisible degrees [19, Theorem 1.1]. Its reflexive symmetric powers are generically generated after arbitrarily small positive ample twists.

Here is the extension step needed when p' is not equidimensional. Since $\mathcal{E}_m$ is torsion free, it is locally free on an open $J \subset I'$ containing every codimension-one point. On J , the reflexive symmetric power agrees with the ordinary symmetric power, and multiplication of sections defines the regular evaluation map

$$\mathrm{Sym}^b \mathcal{E}_m \longrightarrow p'_* \mathcal{O}_{H'}(bm(K_{H'/I'} + C')).$$

Thus a global section of the reflexive symmetric power with a base line-bundle twist evaluates to a rational pluriform regular over J . Every possible divisorial pole lies over $I' \setminus J$. Lemma 5.2 makes such a divisor exceptional over the original H ; its order test at all other primes and normality give a section on H^a . This argument is the replacement for a global reflexivity assertion about the direct image.

For (i), the assertion is automatic if the fiber Iitaka dimension is $-\infty$. Otherwise discard the vertical boundary and write the pulled-back big divisor as $Q' \sim_{\mathbf{Q}} 2\eta A + E$, with A ample, $\eta > 0$ rational, and $E \geq 0$. Choose a fiber degree m realizing the fiber Iitaka dimension k . Weak positivity with the first ηA supplies systems whose restrictions include all products of those degree- m fiber sections, so their fiber image has dimension k . Multiplying by the sections of the second ηA and by the fixed section of E makes the image field contain the base field. The resulting image has dimension $\dim I + k$. Lemma 5.2 descends these systems to the finite normalization of H ; adding R_p and the discarded vertical boundary embeds them in the required systems. Finite pullback preserves Iitaka dimension. This proves the big-base comparison by the weak-positivity method of [19, Proposition 9.1]; the divisorial tests above supply the extension step that equidimensionality supplies in that formulation.

For (ii), first assume P is rational, and take all perturbation parameters positive and rational. Fix ample divisors A' on H' and A_I on I' . Represent $\rho^*P + \delta A'$ by a general small-coefficient SNC boundary. Its addition preserves the big generic log divisor. The same weak-positivity argument, with a twist ϵA_I , produces a nonzero section of a sufficiently divisible positive multiple of

$$K_{H'/I'} + C' + \rho^*P + \delta A' + \epsilon p'^* A_I$$

regular at every prime tested over H . Push its divisor to H : any negative components are exceptional over H , so the pushforward is effective. Divide by $e = \deg \rho$. The tested lattice comparison and the norm for principal divisors show that

$$K_{\mathcal{V}_p} + C + P + \frac{\delta}{e} \rho_* A' + \frac{\epsilon}{e} \rho_* p'^* A_I$$

is pseudoeffective. These two error classes are fixed. Let δ, ϵ decrease to zero and use closedness of the pseudoeffective cone.

For a nef real divisor P , choose ample rational divisor classes Q_j tending to $[P]$ in $N^1(H)_{\mathbf{R}}$, by approximating the ample classes $P + \epsilon_j A_H$ with $\epsilon_j \downarrow 0$ and A_H ample. Their restrictions tend to $[P|_{H_i}]$ on each geometric generic fiber component. Bigness is open, so $K_H + C + Q_j$ is big over I for all sufficiently large j . The rational case makes $K_{\mathcal{V}_p} + C + Q_j$ pseudoeffective. Closedness of the pseudoeffective cone gives the assertion for P . $\square$

We now have every ingredient on the same fixed model U . By Lemma 8.6(ii) and Equation (8.9), $K_{\mathcal{V}_s} + C_h + cM_U$ is pseudoeffective. Thus Lemma 8.3 makes $P_U(c)$ pseudoeffective. Choose rational $a > 1$ with $K_Z + aA$ big. Applying Lemma 8.6(i) to h gives

$$\kappa(W, L(a)) \geq \dim Z + \kappa(W_z, K_{W_z} + B_{W_z}) = \dim U.$$

The section identification in Equation (8.6) makes $P_U(a)$ big. Both divisors lie on the same affine line, and

$$P_U(1) = \frac{a-1}{a-c}P_U(c) + \frac{1-c}{a-c}P_U(a).$$

The two coefficients are positive, so $P_U(1)$ is big. The inverse comparison in Equation (8.6) transfers its systems to $L(1)$ on the original W_0 , preserving their rational map dimensions. Therefore $\kappa(W, L(1)) \geq \dim U$. The opposite inequality is elementary upper addition for q , since $M|_G = 0$ and $\kappa(G, K_G + B_G) = 0$. Together with Equation (8.4) and the product dimension identity, this completes the proof of Theorem 8.1.

9. ORDINARY RELATIVE IITAKA REDUCTIONS

The absolute construction proves the main theorem directly. For projective fibrations it is also useful to start with the relative Iitaka fibration and retain all directions of a second Iitaka base. We derive that formulation from Theorem 8.1 and identify its parameter field inside the original $\mathbf{C}(Y)$. The additional companion inputs in this section are the relative rank-one section comparison and its reduced-boundary cyclic refinement; their exact outputs are specified in Proposition 9.1. The reduced boundary is essential to its entire-top-space assertion. The numerical construction itself used an arbitrary nef divisor; here the relative Kodaira-zero comparison identifies that divisor with a parabolic Hodge line. At the end, we identify the resulting fields inside $\mathbf{C}(X)$, so the conclusion concerns the original whole geometric generic fiber.

Proposition 9.1 (Ordinary relative reduction). *Let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective complex varieties. Assume $\kappa(Y) \geq 0$, and put $d = \kappa(F) \geq 0$ for the geometric generic fiber F . On smooth projective models there is a factorization*

$$X' \xrightarrow{x} W \xrightarrow{g} Y$$

with connected geometric generic fibers and $\dim(W/Y) = d$. There are an effective SNC rational boundary B on W , a nef rational divisor M , and a morphism $h_0 : W \rightarrow Z$ with connected fibers, with Z smooth projective, such that

- (i) $K_W + B + M$ is big over Y ;
- (ii) $M \sim_{\mathbf{Q}} h_0^* A$ for a nef rational divisor A on Z , and $K_Z + aA$ is big for every sufficiently large rational a ;
- (iii) the divisible section systems of $K_W + B + M$ identify with those of K_X , preserving their rational-map dimensions;
- (iv) the generic pair (J, D_J) of x , with the reduced exceptional boundary inherited from $X' \rightarrow X$, has logarithmic Kodaira dimension zero. Its minimal-index cyclic root cover is geometrically connected and has entire logarithmic top-form space of dimension one, with every positive logarithmic plurigenus at most one;
- (v) M is the rational parabolic extension of this entire top line, identified with the top line of the unique rational polarized pure weight grade carrying it.

On an initial reduction model B is the normalized boundary of [37, Lemma 7.4]. Subsequent source models of W carry its strict transform plus reduced exceptional boundary, and the pullback of M ; their section systems remain those of the original X .

Proof. Take the relative Iitaka fibration of K_X over Y , using the relative algebraic closure of its section field, and resolve its maps. The Iitaka fibration theorem gives relative dimension d and Kodaira dimension zero on its geometric generic fiber [27, Section 6]. Give X' the strict transform of the zero boundary plus the reduced exceptional divisor. The multiplicative section-ring identity in Lemma 2.2, also relative over Y , identifies its relative log Iitaka fibration with the one just chosen. Consequently (J, D_J) has logarithmic Kodaira dimension zero.

Apply the rank-one comparison of [37, Proposition 2.7], with the original X as its reference model. The relative section comparison gives $K_W + B + M$ the full section field of dimension d on the generic fiber over Y ; hence it is relatively big. It also gives (iii). The normalization lemma in [37, Lemma 7.4] identifies its initial boundary using two different order tests. The coefficient defining sections is the minimum over the actual components of a fiber, whereas the discriminant threshold also allows divisorial valuations on higher models. The stated normalization compares these quantities on the same SNC model; we do not replace one by the other. The reduced-boundary results [37, Lemma 7.1] and [37, Corollary 7.2] give (iv)–(v), including the normalization of M without an extra cyclic-index factor.

The entire highest line in (v), with its actual parabolic extension, satisfies the hypotheses of Proposition 3.9. That proposition gives h_0 , Z and A after smooth projective modifications. Its positive value of a suffices for all larger a , since A is nef. Use Lemma 2.2, with the pulled-back rational twist, for the boundary on the modified W . Its section comparison, and the original-reference convention in [37, Proposition 2.7], preserve (i) and (iii). The parabolic line pulls back functorially, so (v) is preserved as well. The modifications and the relative Iitaka construction do not change geometric generic integrality. $\square$

Corollary 9.2 (The ordinary second Iitaka construction). *For the data of Proposition 9.1, there are a morphism $q : W \rightarrow U$ with connected fibers to a smooth projective variety U and a morphism $j : U \rightarrow Z$ with $h_0 = j \circ q$. The general q -fiber G satisfies*

$$\kappa(G, K_G + B|_G) = 0, \quad G \longrightarrow g(G) \text{ is generically finite.}$$

There are smooth projective varieties T_0, S and a diagram

$$\begin{array}{ccccc} W & \xrightarrow{(g_S, q)} & (S \times_{T_0} U)_{\text{main}} & \longrightarrow & S & \xrightarrow{\rho} & Y \\ q \downarrow & & \downarrow & & \downarrow & & \\ U & \xlongequal{\quad} & U & \longrightarrow & T_0 & & \end{array}$$

with $g = \rho \circ g_S$, where ρ is birational and the map to the main product component is dominant and generically finite. The geometric generic fibers of $S \rightarrow T_0$ and $U \rightarrow T_0$ are integral, and

$$(9.1) \quad \kappa(S_t) = 0, \quad \kappa(X) = \kappa(W, K_W + B + M) = \dim U = d + \dim T_0.$$

Here t is a very general point of T_0 . All models use the section-preserving boundary convention of Proposition 9.1.

Proof. Take $M = h_0^* A$ in Theorem 8.1; rational linear equivalence suffices there. Its conclusions, with its product diagram (8.3) and dimension identity (8.1), give the asserted diagram and dimension formula. Its hypotheses are exactly (i)–(ii) of Proposition 9.1, the effective SNC boundary, and $\kappa(Y) \geq 0$, with $D_Y = 0$ in that theorem. The section-system identification in Proposition 9.1(iii) transfers the resulting dimension to X . This retains the actual image-cycle parameter space, before its dimension is compared with $\kappa(Y)$. $\square$

Corollary 9.3 (Ordinary cyclic covers along the second Iitaka fibers). *In the notation of Proposition 9.1 and Corollary 9.2, let J be the generic fiber of $x : X' \rightarrow W$ over $K = \mathbf{C}(W)$. The*

minimal logarithmic generator $\omega \in H^0(J, p(K_J + D_J))$ is an ordinary regular pluriform. Its resolved connected cyclic root cover $\tilde{J}$ satisfies

$$H^0(\tilde{J}, aK_{\tilde{J}}) = K\eta^a \quad (a \geq 1),$$

where η is its root top form. In particular, $\kappa(\tilde{J}) = 0$ and $h^0(\tilde{J}, K_{\tilde{J}}) = 1$.

On a dense smooth open of every very general fiber of $q : W \rightarrow U$, the family of these covers is geometrically birationally constant with its deck action, and the same holds for the J -family after taking the quotient. More precisely, over the geometric generic q -fiber these families become birational to varieties over $b_U = \overline{\mathbf{C}(U)}$ after a finite extension of that fiber's function field, equivariantly for the deck group in the cover case. These identifications persist under algebraically closed field extension.

Proof. Choose a nonzero ordinary pluricanonical form on the generic fiber of f , which has Kodaira dimension $d \geq 0$. Pull it to the induced smooth birational model and restrict to the generic fiber of its map to W_y . Adjunction, after dividing by a local determinant frame on W_y , gives a nonzero ordinary form $\theta \in H^0(J, mK_J)$ for some $m > 0$. The restriction is nonzero because a proper zero divisor cannot contain the generic fiber. The logarithmic section space in degree mp is one-dimensional, so

$$\theta^{\otimes p} = c\omega^{\otimes m} \quad (c \in K^*).$$

Testing divisorial orders shows that ω has no poles. Its index remains the original minimal p ; the common degree is used only to compare the forms.

In the reduced cyclic-cover lemma of [37, Lemma 7.1], this gives

$$E_J = \operatorname{div}(\omega) + pD_J, \quad \Delta_J = D_J - E_J/p = -p^{-1} \operatorname{div}(\omega) \leq 0.$$

Thus $\Delta_J^{\leq 1}$ and the specified pole boundary on the root cover are empty. That lemma proves the displayed equality of *ordinary* pluricanonical spaces. Moreover, [37, Corollary 7.2] now identifies M directly with the rational parabolic extension of the entire compact top Hodge line. There is no open-fiber boundary to compare.

Spread an equivariant smooth projective model of the cover over a nonempty smooth open $W^\circ \subset W$, shrinking it to remove any vertical remnants of the spread pole boundary. Let G be a very general smooth q -fiber meeting this open. Since $h_0 = j \circ q$ and $M \sim_{\mathbf{Q}} h_0^* A$, the degree of M on every complete test curve in G is zero. Lemma 3.2 and Corollary 3.13 therefore make the projective compact top line constant on each such curve. Complete-intersection curves through a general point with tangent directions spanning T_G show that the line is constant on $G \cap W^\circ$. Corollary 3.15, applied anew on this restricted locus, gives finite character.

Apply Corollary 4.9 to $W^\circ \rightarrow U$, using its relative assertion and the finite deck action. Its hypotheses on geometric generic integrality follow from Corollary 9.2. This proves the stated constancy of the covers, including over b_U . Taking invariant function fields gives the assertion for J : invariants under a finite group commute with field extension in characteristic zero. $\square$

Corollary 9.4 (The absolute Iitaka base and descent of the whole fiber). *The variety U in Corollary 9.2 is a birational model of the absolute Iitaka base of X . Inside $\mathbf{C}(X)$, the field $\mathbf{C}(W)$ is the relative algebraic closure of $\mathbf{C}(Y)\mathbf{C}(U)$. Consequently the field $\mathbf{C}(T_0) \subset \mathbf{C}(Y)$ in that Corollary agrees with the field obtained from Propositions 2.7 and 2.10 applied to the projective morphism f with empty original boundaries.*

In particular, inside the prescribed algebraic closure Ω of $\mathbf{C}(Y)$, put $b = \overline{\mathbf{C}(T_0)}$. There is a finitely generated field E_b/b such that

$$(9.2) \quad \operatorname{Frac}(\Omega \otimes_b E_b) \simeq \operatorname{Frac}(\Omega \otimes_{\mathbf{C}(Y)} \mathbf{C}(X)).$$

Thus the whole geometric generic fiber of f is birationally defined over this specific field b , and $\text{Var}(f) \leq \dim T_0$.

Proof. We identify first the absolute section field, then the intermediate field of the combined map, and finally the intersection of the original base field with the absolute section field. All these identifications take place inside $\mathbf{C}(X)$.

Put $L = K_W + B + M$, and let $E \subset \mathbf{C}(W)$ be its stabilized section field. Since $L|_G = K_G + B_G$ has Iitaka dimension zero, every section ratio is constant on the general q -fiber G . Connectedness of q gives $E \subseteq \mathbf{C}(U)$. The equality $\kappa(W, L) = \dim U$ makes $\mathbf{C}(U)/E$ a finite extension. The stabilized section field is relatively algebraically closed in $\mathbf{C}(W)$, by the Iitaka theorem as used in Proposition 2.7. Hence $E = \mathbf{C}(U)$. The multiplicative section comparison in Proposition 9.1 identifies this field with the absolute canonical section field of X .

The product diagram in Corollary 9.2, with $S \rightarrow Y$ birational, shows that $\mathbf{C}(W)$ is finite over $\mathbf{C}(Y)\mathbf{C}(U)$. It is relatively algebraically closed in $\mathbf{C}(X)$ because x has connected fibers. These two facts characterize the relative algebraic closure used in Proposition 2.7, so the intermediate fields coincide.

It remains to identify the parameter field with its embedding, rather than only its transcendence degree. In either construction, the product diagram and geometric generic integrality give an injection

$$\mathbf{C}(Y) \otimes_{\mathbf{C}(T_0)} \mathbf{C}(U) \hookrightarrow \mathbf{C}(W).$$

For $u \in \mathbf{C}(Y)$ and $v \in \mathbf{C}(U)$, linear independence over $\mathbf{C}(T_0)$ shows that $u \otimes 1 = 1 \otimes v$ forces $u = v \in \mathbf{C}(T_0)$. Consequently

$$\mathbf{C}(T_0) = \mathbf{C}(Y) \cap \mathbf{C}(U) \quad \text{inside } \mathbf{C}(X).$$

The two larger fields have already been identified, so the parameter fields coincide with their embeddings in $\mathbf{C}(Y)$. Proposition 7.3 now gives Equation (9.2) over this exact field b . The variation bound follows from its definition. $\square$

9.1. Comparing compact and open top lines. Corollary 9.3 used the empty pole boundary to identify the moduli line with the compact top line. The following comparison treats the additional case in which a relative SNC boundary is retained on a smooth cover. If the entire compact and open top-form spaces are both one-dimensional, it transfers flatness through the pure compact image and retains a prescribed finite group action.

Lemma 9.5 (Compact-to-open comparison). *Let $\pi : P \rightarrow T$ be a smooth projective morphism of relative dimension r with connected fibers, where T is a smooth connected complex algebraic variety. Let $D \subset P$ be a relative reduced SNC divisor, with every stratum smooth over T , and suppose*

$$h^0(P_t, K_{P_t}) = h^0(P_t, K_{P_t} + D_t) = 1 \quad (t \in T).$$

Write $\mathbb{H} = R^r \pi_ \mathbf{Q}$, let $\mathbb{V}$ be the degree- r cohomology variation of $P \setminus D$, and put $\mathbb{Q} = \text{im}(\mathbb{H} \rightarrow \mathbb{V})$. Then $\mathbb{Q} = W_r \mathbb{V}$ is a polarizable pure variation, and the compact-to-open map identifies its top line $F^r \mathbb{Q}$ with both $F^r \mathbb{H}$ and the entire top line $F^r \mathbb{V}$. If $F^r \mathbb{Q}$ is flat, the compact top line $F^r \mathbb{H}$ is flat. A prescribed finite group acting on (P, D) over T admits an equivariant horizontal Hodge splitting realizing this identification.*

Proof. The logarithmic Hodge description identifies the map on top pieces with the nonzero inclusion

$$H^0(P_t, K_{P_t}) \hookrightarrow H^0(P_t, K_{P_t} + D_t);$$

see [15, Corollaries 3.2.13(ii) and 3.2.14]. The dimensions make it an isomorphism. Degree- r open cohomology has weights at least r , and its compact image is precisely $W_r \mathbb{V}$ [15, Corollary 3.2.17].

Strictness therefore identifies the three top lines. In particular, the only weight grade carrying the logarithmic top line is this pure weight- r image.

The kernel of $\mathbb{H} \rightarrow \mathbb{Q}$ is a pure Hodge subvariation. Its orthogonal complement for a polarization of $\mathbb{H}$ is horizontal and is a Hodge subvariation, and maps isomorphically to $\mathbb{Q}$; this is the polarized splitting argument in [15, Lemma 4.2.3(i)]. It supplies a horizontal Hodge right inverse j . Since the kernel has zero F^r piece, $F^r \mathbb{H} = j(F^r \mathbb{Q})$. Thus flatness of the latter line gives flatness of the former.

For a finite group Γ , replace j by

$$j_\Gamma = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma_{\mathbb{H}} j \gamma_{\mathbb{Q}}^{-1}.$$

Every summand is a horizontal Hodge right inverse, so their average is one as well, and reindexing the sum proves equivariance. The rank-one top identification is unchanged. This proves the comparison with the finite action retained. $\square$

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